

The block below carries no module guard, so docstrip copies it into EVERY generated file. That is what we want for the TeX files and wrong for the latexmkrc, which is Perl and does not understand lines opening with two percent signs – hence the negated guard around it.

The guard lines themselves are hidden behind a false conditional. docstrip reads a guard line either way, but ltxdoc, which typesets this file, would otherwise set the literal text of the guard as a stray paragraph in the body of the manual.

Nothing in this area may name a TeX conditional, not even in prose: outside the meta-comment at the top of the file, every documentation line here IS LaTeX source and is executed, and a conditional named in passing opens or closes a real one. That is not hypothetical – an earlier draft of this very comment swallowed the rest of the manual.

Three conditional primitives are deliberately absent from the lists below: the ones that close or branch a conditional. `\DoNotIndex` scans its argument with TeX’s own conditional machinery live, so a bare one of them closes the wrong conditional. The manual ended its run with “Too many }’s” and exited with status 1, which failed the docs step of the harness even though the PDF had been written. Leaving them out of the lists means they get indexed, which is harmless. Note that they cannot be named here either, not even inside `...` shorthand split across a line.

The coppe document class

Version 4.0

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a WhatsApp group where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Name}{Surname}{Degree}{Institution}
\advisor{Second Advisor's Name}{Surname}{Degree}{Institution}
\advisor{Third Advisor's Name}{Surname}{Degree}{Institution}
```

`\coadvisor` takes exactly the same four arguments.

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently – given name and surname go in one argument:

```
\examiner{First Examiner's Name Surname}{Degree}{Institution}
\examiner{Second Examiner's Name Surname}{Degree}{Institution}
...
\examiner{N-th Examiner's Name Surname}{Degree}{Institution}
```

The argument order changed in v4.1. The treatment (“Prof.”, “Prof.^a”) used to be the first mandatory argument and the institution an optional one; now the treatment is the optional argument and is EMPTY by default, and the institution is the last mandatory argument. 3.1.2.1.3(e) asks for name, titulacao and institution, and asks for no treatment at all, so that is what the folha de aprovacao now prints. The institution may be left empty – `\advisor{Ana}{Lima}{D.Sc.}{}{}` compiles and simply prints nothing in its place.

To print a treatment anyway, pass it in the optional argument:

```
\advisor[Prof.]{Ana}{Lima}{D.Sc.}{UFRJ}
\examiner[Prof.\textsuperscript{a}]{Bia Sousa}{Ph.D.}{UFF}
```

A document written for v4.0 or earlier has to be rewritten. See `MIGRATION_v3_to_v4.md` for the mechanical rule.

Who appears on the folha de aprovacao. Since v4.1, only the EXAMINERS. The advisors stay on the capa and the folha de rosto, where they belong, but the approval sheet records who examined the work, and the advisor conducted it. Where the Programa does seat the advisor on the board, the `orientadorexamina` class option puts the advisors and coadvisors back, ahead of the examiners.

`\dataaprovacao` The date the board approved the work, printed on the folha de aprovacao after “Aprovada em”. Give it in words, the way the sheet should read: `\dataaprovacao{15 de setembro de 2026}`. **With no date given the sheet says “a ser determinada”**, in the main language of the work – there is no blank rule to fill in by hand, because the deposit is digital and nobody writes on a PDF.

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Folha adicional: Coleta CAPES

Since August 2026 every thesis and dissertation must carry a *folha adicional* immediately after the folha de rosto, holding the fields the CAPES/MEC collection requires plus the ficha catalografica (SiBI manual, 9th rev. ed. 2026, 3.1.2.1.2 and Annex H). The sheet is filled in by the graduate programme, and is neither counted nor numbered.

The macros below feed that sheet. They are institutional data, so the sheet is always typeset in Portuguese regardless of the main language of the work, as the COPPE norm requires of institutional identity.

Since v4.1 the sheet is drawn as Annex H shows it: a CLOSED FRAME the width of the text block, one cell per field, cells separated by a horizontal rule, and **no rules to write on** – the space to fill in is the blank space of the cell. A field left empty simply leaves its cell blank. The ficha catalografica sits centred below the frame, on the same sheet.

`\areaconcentracao` The area of concentration, e.g. `\areaconcentracao{Engenharia de Sistemas e Computacao}`. It appears on the folha de rosto, on the folha de aprovacao and on the folha adicional.

`\linhapesquisa` The research line, required by the CAPES model of the folha de rosto, where it is printed at the left margin just above the advisors. Its label follows the MAIN LANGUAGE of the work (“Linha de pesquisa”, “Research line”, “Línea de investigación”, “Axe de recherche”, “Linea di ricerca”). Leave it out, or give it an empty argument, and the folha de rosto comes out without the line and without any error.

`\tipoproducao` The kind of intellectual output, one of `bibliografica`, `artistica`, `tecnologica` or `tecnica`. Anything else raises a warning.

`\projetovinculado` Whether the work belongs to a research project (`sim` or `nao`) and, if so, its name.

`\agenciafomento` A funding agency, given as full name and acronym; repeat once per agency:

```

\agenciafomento{Conselho Nacional de Desenvolvimento
                Cientifico e Tecnologico}{CNPq}
\agenciafomento{Fundacao Carlos Chagas Filho de Amparo a
                Pesquisa do Estado do Rio de Janeiro}{FAPERJ}

```

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.
`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which
`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is
`\printloabbreviations` provided, it will used as sort key. This was later, in 2024, implemented also for
`\printlosymbols` `\abbrev`, otherwise `Symbol`, or `Abreivation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```

makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx

```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

Illustrations: caption on top, source below

`\source` 2.10 of the Manual puts the CAPTION ABOVE every illustration – figure, table, quadro, code listing, algorithm – with the designating word, the number in arabic figures, an em dash and the title; and it makes the SOURCE mandatory, BELOW the illustration, in the smaller size. The class does that for you: the caption goes on top wherever you write `\caption`, and `\source{...}` (or `\fonte{...}`, the same command) sets the source line. The source advances no counter and appears in no list.

```
\begin{figure}[ht]
\centering
\caption{0 que a figura mostra}    % em cima
\label{fig:algo}
\includegraphics[width=0.5\textwidth]{arquivo}
\fonte{Elaboracao propria.}        % embaixo
\end{figure}
```

If another package has already defined `\source` or `\fonte`, use `\cpsource` and `\cpfonte`, which are always available.

A **quadro** is an illustration whose data is bordered on all sides, unlike a *tabela*, which is ruled only at the top and the bottom. The `quadro` environment behaves like `figure` and `table` and has its own list, `\listofquadros`.

For an illustration the Manual does not name – a map, a photograph, a score – declare your own float in the preamble:

```
\newcoppefloat{mapa}{Mapa}{Lista de Mapas}
```

You get the `mapa` environment, numbered within the chapter like every other illustration, the caption on top, `\source` below, `\autoref` support, and a `\listofmapas` command to put with the other pre-textual lists. The example does exactly this.

Long quotations, footnotes, captions and the folio all come out in the smaller uniform size 2.2(b) prescribes; you do not set that yourself.

References

COPPE_{TEX} produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no Bib_{TEX} `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` Bib_{TEX} styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib}    % in the preamble
...
\printbibliography              % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic `@book` (`@livro`); `@article` (`@artigo`); `@artigodejornal` (newspaper); `@incollection` (`@partedelivro`); `@inbook` (`@emlivro`); `@inreference` (`@verbete`); `@proceedings` (`@anais`); `@inproceedings` (`@trabalhodeevento`); `@thesis` (`@tese/@dissertacao`, with the presets `@dissertacaomestrado`, `@tesedoutorado`, `@tesephD`, `@tcc`); `@report` (`@relatorio`); `@periodical` (`@periodico`); `@manual`; `@misc` (`@diverso`); `@online`.

Special (NBR 6023 §4.2.5–4.2.14) `@patent` (`@patente`); `@legislation` (`@legislacao/@atonormat`); `@jurisdiction` (`@jurisprudencia`, with a `relator` field); `@legal` (`@documentojuridico`); `@standard` (`@norma`); `@music` (`@partitura`); `@map` (`@mapa`); `@image` (`@iconografico`); `@artwork` (`@obradearte`); `@video` (`@audiovisual`); `@movie` (`@filme`); `@letter` (`@correspondencia`); `@dataset` (`@conjuntodedados`).

Games and TV `@game` (`@jogo`); `@boardgame` (`@jogodetabuleiro`); `@videogame` (`@jogoeletronico`); `@tvshow` (`@programadetv`).

When an entry has no `author` or `editor`, it is entered by title, with the first significant word (and a leading article) in capitals. A periodical is the exception: 4.2.3 sets its title in capitals throughout.

Fields the ABNT categories need

Beyond the usual `biblatex` fields, each with an unaccented Portuguese synonym:

Events `eventtitle` (`evento`) — the name of the event, set in capitals ahead of the title of the proceedings, as 4.2.4 requires; `number` (`numero`) the edition; `eventdate` (`dataevento`) the year it was held; `venue` (`localevento`) the city. The title (`@anais`) or `booktitle` (`@trabalhodeevento`) then carries only “Anais [...]”, which is the element in bold.

Patents `filingdate` (`deposito`) and `grantdate` (`concessao`), printed as “Depósito: Concessão:” (4.2.5).

Theses `course` (`curso`), `institution` (`instituicao`), `pagetotal` — rendered as “1997. 203 f. Dissertação (Mestrado em ...) — Instituição, Local, 1997.” (4.2.1.1).

Legal documents `relator`, plus `volume`, `number` and `pages` of the vehicle in which the act was published (4.2.6).

Anything electronic `url` and `urldate` (`dataacesso`), closing the entry as “Disponível em: Acesso em:”

Every one of these is exercised, with the manual’s own example data, by `adversativa/referencias-manual.bib` — one entry per category of 4.2 — and checked against the manual’s printed form by `tools/conferir-referencias.py`. Neither is distributed: they prove the styles, they are not part of them.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard L^AT_EX command used before all appendices. COPPETEX introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (`\coppetexfinalpage`)

`\coppetexfinalpage` Optional closing page modelled on the printer’s *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before `\end{document}`. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (`\vfill`);
- renders the text in Portuguese unconditionally (Brazilian `babel`), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overridable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the T_EX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Latin Modern* (`lmodern`), which is what the class loads. Redefine if you loaded `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPET_EX class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, <https://github.com/COPPE-UFRJ/CoppeTeX>.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsceexam, msceexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msceexam`, and `dsceexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is **brazilian**.

- **brazilian** and **english** are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- **spanish, french, italian** are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.
- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein / \brazilianabstract` API, and how to create a new language pack.

pdfa Produce the deposited file as **PDF/A-2b**, which 2.2(d) of the manual requires of the digital version. The class loads `pdfx` at level **a-2b** and writes the XMP metadata from the document’s own `\title`, `\author`, `\keyword` and `\department`. Level a-2b rather than a-1b: a-1b forbids transparency, which `tcolorbox` and several graphics packages produce. The metadata reaches the PDF on the SECOND run, because `pdfx` reads it from a file the class writes at the end of the first one – which is no burden, since three runs are needed for cross-references anyway.

semlinks Drop the colour and the frame from every hyperlink. The class loads `hyperref` and colours links; this option calls `hidelinks` after everything else, so the page comes out sober while the links stay clickable and the bookmarks stay in the reader’s panel. Neither the Manual nor the COPPE norm has an opinion: 2.2(b) fixes the colour of the TEXT, and a link is not text. Use it if you are going to print.

orientadorexamina Put the advisors and coadvisors back on the folha de aprovacao, ahead of the examiners, as president of the board (3.1.2.1.3(e)). **Off by default since v4.1**: the folha de aprovacao records who EXAMINED the work, and the advisor conducted it rather than examined it, so only the examiners are listed unless the Programa says otherwise. Before v4.1 there was no choice – the advisors always appeared, and the only way out was not to declare them, which also removed them from the capa and the folha de rosto, where they belong.

assinaturas **Gone in v4.1, and now a no-op that only warns.** It used to draw a signature rule above each board member. The deposit has been digital only since Resolucao CEPG n. 246/2023, so there is nothing to sign by hand; the folha de aprovacao now always uses the compact list. The option is still accepted so a document written before v4.1 still compiles.

coorientador List the coadvisors on the three abstract pages, under the advisors, in the same shape the folha de rosto uses. Off by default: the abstract page listing only the advisors is a COPPE tradition, and 3.1.2.1.4 does not require either form.

rascunhoficha While writing, fill the folha adicional with a ficha catalografica typeset by the class itself, so the page is not an empty frame. **Never valid for deposit**: the real ficha comes from the SiBI generator or the Programa’s library, and goes in with `\fichacatalografica`.

twoside, oneside The 2026 edition dropped the verso margins and the deposit is digital, so the class sets one-sided pages with a 3 cm left margin on every folha. **twoside** restores the mirrored layout for anyone who is going to PRINT and bind the work; **oneside** is the default and is there to be explicit.

semmorewrites, morewrites TeX has sixteen output streams and a work that asks for every list runs out of them; the failure, “No room for a new `\write`”, lands on whichever list came last and explains nothing. Under pdfTeX the

class therefore loads the `morewrites` package by itself when it is installed, which removes the ceiling. `semmorewrites` turns that off, for the rare case where the package disagrees with something else in the document – it buffers what is written until shipout, so a package that writes a file and reads it back in the same run can misbehave. `morewrites` means “I insist”: a missing package becomes an error instead of a warning. Under LuaLaTeX, which has 128 streams, none of this is loaded and none of it is needed.

`comserifa`, `semserifa` Since v4.1 the class sets the whole document in the sans-serif family by default; `comserifa` goes back to the serif one. Neither the manual nor the COPPE norm prescribes a typeface: 2.2(b) fixes the colour, the body size 12 and the smaller uniform size of the four items it lists, and says nothing about the family. The choice is the author’s, and the SiBI manual that carries the rule is itself set in Arial – which is why sans is now the default. The class loads `lmodern`, which brings both families as vector fonts, so either way the document is PDF/A material. `semserifa` is still accepted and now does nothing, so a document written before v4.1 still compiles. For a DIFFERENT sans face – Helvetica, say – load its package in the preamble and redefine `\familydefault` there; the default is for the common case of an author who does not want to choose a font at all. Redefine `\coppefinalfont` if you do load another face, so the colophon does not announce Latin Modern.

`listasnosumario` Put the pre-textual lists – figures, tables, quadros, programas, abreviaturas, simbolos – back into the sumario. **Contrary to 3.1.2.1.6**, which tells the author to follow the manual’s own sumario, and that one opens at “1 INTRODUCAO”: the lists are pre-textual elements and come BEFORE the sumario, so listing them inside it is listing what the reader has already passed. The option exists because every CoppeTeX document written before this release had them there, and a work in its final weeks should not have its sumario change under it.

`numeraisromanos` Number the pre-textual folhas in lowercase roman numerals. **Contrary to 2.7**, which has the count running from the folha de rosto with nothing printed until the textual part. Kept only so that documents written under the older norm still compile; do not use it for a new work.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the **abstract** environment. Picked by the class option (**brazilian** by default; one of **english**, **spanish**, **french**, **italian** also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the **foreignabstract** environment. **english** by default for every main language; switches automatically to **brazilian** when **main = english**.

third optional. Loaded on demand with `\usecoppelanguage{<lang>}` in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

`\titlein`
brazilianabstract (*env.*)
`\usecoppelanguage`

Two pieces of user-facing API come with the multilingual model:

- `\titlein{<lang>}{<text>}` records the title in any language. `\title{...}` still writes the Brazilian-Portuguese title and `\foreigntitle{...}` still writes the English one (back-compat); for a main language outside the **brazilian/english** pair, also call `\titlein{<lang>}{...}`. The cover and folha de rosto print whichever title matches `\coppe@mainlang`.
- The **brazilianabstract** environment is the optional third abstract used when the main language is neither **brazilian** nor **english** – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in `\begin{otherlanguage}{brazilian}` and uses the Portuguese template strings. A document with **main = brazilian** can simply keep using **abstract**; **brazilianabstract** is redundant then.
- `\usecoppelanguage{<lang>}` loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings **universityname**, **cityname**, **statename**, **countryname**, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to **coppe.cls** (or anywhere on the TEXINPUTS path):

coppe-lang-<lang>.def class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for **brazilian/english** in **coppe.cls** (**appendix**, **annex**, **frame**, **listframe**, **source**, **program**, **listprogram**, **references**, **in:**, **art**, **platform**, **directedby**, **producedby**, **listabbreviation**, **listsymbol**, **glossary**, **advisor**, **advisors**, **dept**, **approvedmale**, **approvedfemale**, **universityname**, **cityname**,

statename, countryname, monthname1...monthname12, algorithm2eopt, babelname). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```

so the biblatex mapping is registered after the bibliography engine is loaded by coppe.cls (which happens *after* the language pack is read).

<lang>-coppe.lbx biblatex localization strings. \InheritBibliographyExtras{<lang>} then \DeclareBibliographyStrings for the keys availablefrom, urlseen, in, editor, editors, depositor, coppescale, msdiss, dscthis, phdthesis, tcc, coppemscdeg, coppedscdeg, coppephddeg, coppetccdeg, coppefiling, coppegrant. The shipped spanish-coppe.lbx, french-coppe.lbx and italian-coppe.lbx are the reference templates to copy from.

Tip: the algorithm2eopt key holds whatever algorithm2e calls the language. Mind the spelling: algorithm2e ships portuguese, english, spanish, french and italiano – not italian.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T'\i tulo em portugu\~es} % Brazilian abstract slot
  \foreigntitle{English title} % English abstract slot
  \titlein{spanish}{T'\i tulo en espa\~nol} % cover/folha
  \author{Juan}{P\erez}
  \advisor{Mar'\i a}{Garc'\i a}{D.Sc.}{UFRJ}
  \department{PESC}\date{01}{2026}
  \keyword{multilenguaje}
  \maketitle
  \frontmatter
  \begin{abstract} Resumen principal en espa\~nol. \end{abstract}
  \begin{foreignabstract} English foreign abstract. \end{foreignabstract}
  \begin{brazilianabstract} Resumo em portugu\~es para a banca. \end{brazilianabstract}
  \tableofcontents
  \mainmatter
  \chapter{Introducci\on} Cuerpo en espa\~nol\dots
\end{document}
```

6 Files in your project folder

In a normal install where CoppeTeX is part of your TEXINPUTS path (e.g. inside the local texmf tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or

- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom biblatex datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` biblatex bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` biblatex localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```
\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}
```

unconditionally, so biblatex tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, biblatex never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one biblatex-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
<code>brazilian</code> (default)	— (built into <code>coppe.cls</code>)
<code>english</code>	— (built into <code>coppe.cls</code>)
<code>spanish</code>	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
<code>french</code>	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
<code>italian</code>	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Required L^AT_EX packages

The class is built on the standard `book` class (`12pt`, `a4paper`, `oneside`) and brings in the stock CTAN packages listed below with `\RequirePackage`. On a full T_EX Live or MiK_TE_X install they are all present already, so this list matters only for a minimal install — or when you simply want to know which commands are already in scope. You do *not* need to `\usepackage` any of these yourself; every command and environment they provide is available in your manuscript. Options in brackets are the ones the class passes.

Engine and encoding

`iftex` detects the running engine, so the class can branch its setup between pdfT_EX and the Unicode engines (LuaT_EX/XeT_EX).

`inputenc` (`utf8`) UTF-8 input; loaded *only under pdfT_EX* (the Unicode engines read UTF-8 natively).

`fontenc` (`T1`) T1 font encoding, for correctly composed accented glyphs and hyphenation.

Mathematics and symbols

`amsmath` the AMS mathematics environments.

`amssymb` the AMS symbol fonts; loaded *only under pdfT_EX*. Under a Unicode engine load `unicode-math` (or `amssymb`) yourself — the class warns at end of document if neither is present.

Page layout and spacing

`geometry` A4 one-sided margins (3/3/2/2 cm) and header height.

`fancyhdr` running headers and footers.

`titlesec` chapter and section title formatting.

`setspace` one-and-a-half line spacing (double with the `doublespacing` option).

Lists, tables and rules

`enumitem` customisable list spacing and labels.

`tabularx` tables with automatic-width columns.

`booktabs` professional horizontal rules (`\toprule`, `\midrule`, `\bottomrule`).

`longtable` tables that break across pages.

Floats, graphics and captions

`graphicx` image inclusion.

`float` float placement control (e.g. the `[H]` specifier).

`caption` caption typography (the ABNT caption-on-top layout).

`subcaption` subfigures and subtables.

Code, verbatim and algorithms

`listings` source-code listings.

`fancyvrb` extended verbatim.

`algorithm2e` (`linesnumbered`, `lined`, `algochapter`, `ruled`) typeset algorithms;
the natural-language option tracks the main language.

Bibliography, quotations and language

`babel` multilingual typesetting; the language options are computed from the main
and foreign language slots.

`csquotes` context-sensitive quotation marks (the recommended companion of
`biblatex`).

`biblatex` (`backend=biber`) the bibliography engine and the ABNT `coppe` styles
(`coppe` or, with the `numbers` option, `coppe-numeric`). The backend is
`biber`, not `bibTeX`.

Hyperlinks

`hyperref` clickable cross-references, PDF bookmarks and document metadata.

Low-level utilities

`etoolbox` e-TeX programming toolkit.

`ltxcmds` low-level command helpers (`\ltx@ifpackageloaded`, ...).

`ifthen` the `\ifthenelse` conditional.

`hyphenat` hyphenation control (`\nohyphens`, used on the cover).

`lastpage` `\pageref{LastPage}` for the cataloging page count.

`hologo` T_EX-family logos (`\hologo{LaTeX}, \dots`).

`xcolor` colour support (used by `listings` and `hyperref`).

8 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and `pdflatex` will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (`amssymb` vs. `unicode-math`). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under pdfL^AT_EX: it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (LuaL^AT_EX, XeL^AT_EX). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math` nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with `unicode-math`:** it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

`longquote` **Quotation** A quotation longer than three lines is INDENTED BY 4 cm from the left margin – not set with a 4 cm left margin, which is a different and much narrower thing. 2.2(b) of the manual says “recuo de 4 cm da margem esquerda”, so on an A4 page with a 3 cm margin the quoted text starts 7 cm from the edge of the paper. It carries no quotation marks and is set in the smaller font, single spaced. The `coppe` class provides the `longquote` environment, which does all of that; the indentation is `\recuolongquote`, should a Programa ever need to change it.

9 A simple example

```

1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating}          % rodando coisas, como tabelas
5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz}             % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 \usepackage{unicode-math}
11 \setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
19 \addbibresource{example.bib}% base de referências (biblatex/biber)
20
21 \makelosymbols
22 \makeloabbreviations
23 % Flutuante criado pelo autor: mesma regra dos demais (legenda acima, fonte
24 % abaixo) e uma lista própria, \listofmaps.
25 \newcoppefloat{mapa}{Mapa}{Lista de Mapas}
26 % Fonte: a norma fixa o corpo 12, e não a família. A classe compõe sem serifa
27 % por padrão; para compor com serifa, passe a opção de classe ‘comserifa’ lá
28 % em cima -- NÃO redefine \familydefault aqui, porque isso venceria a opção.
29 % Para uma sem serifa diferente da Latin Modern, carregue o pacote da fonte e
30 % troque a família:
31 \usepackage{helvet}\renewcommand{\familydefault}{\sfdefault}
32
33
34 \begin{document}
35   \title{Título da Tese}
36   \foreigntitle{Thesis Title}
37   \author{Nome do Autor}{Sobrenome}
38   % Orientador: nome, sobrenome, titulacao e instituicao. O tratamento
39   % ("Prof.") e o argumento OPCIONAL entre colchetes e vem vazio por padrao --
40   % a norma nao pede tratamento nenhum. A instituicao e obrigatoria, mas pode
41   % ficar vazia: \advisor{Ana}{Lima}{D.Sc.}{} compila sem erro.
42   \advisor{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}{UFRJ}
43   % O segundo orientador vai com tratamento, so para mostrar o argumento

```

```

44 % opcional funcionando; a norma nao pede tratamento nenhum.
45 \advisor[Prof.]{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}{UFRJ}
46 % Coorientador: mesmos quatro argumentos de \advisor. Aparece na folha de
47 % rosto e na folha de aprovacao; para ve-lo tambem nas paginas de resumo,
48 % passe a opcao de classe 'coorientador'.
49 \coadvisor{Nome do Coorientador}{Sobrenome}{D.Sc.}{UFF}
50 % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
51 % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
52 % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
53 % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
54
55 \examiner{Nome do Primeiro Examinador Sobrenome}{D.Sc.}{UFRJ}
56 \examiner{Nome do Segundo Examinador Sobrenome}{Ph.D.}{UFRJ}
57 \examiner{Nome do Terceiro Examinador Sobrenome}{D.Sc.}{UFF}
58 \examiner{Nome do Quarto Examinador Sobrenome}{Ph.D.}{UNICAMP}
59 % Instituicao vazia: o argumento e obrigatorio, mas aceita ficar em branco e
60 % entao nada e impresso no lugar dela.
61 \examiner{Nome do Quinto Examinador Sobrenome}{Ph.D.}{}
62 \department{PESC}
63 \date{06}{2026}
64 % Data da defesa, na folha de aprovacao. Sem esta linha a folha sai com
65 % "a ser determinada", que e o que se quer enquanto a data nao existe.
66 % \dataaprovacao{15 de setembro de 2026}
67 %
68 % A folha de aprovacao lista so os EXAMINADORES. Se no seu Programa o
69 % orientador integra a banca, passe a opcao de classe 'orientadorexamina'
70 % la em cima e ele volta para a lista, na frente dos examinadores.
71
72 \keyword{Primeira palavra-chave}
73 \keyword{Segunda palavra-chave}
74 \keyword{Terceira palavra-chave}
75
76 % Dados da folha adicional (Coleta CAPES), obrigatoria desde agosto de 2026.
77 % A folha e preenchida pelo Programa; estes campos apenas a compoem.
78 \areaconcentracao{Engenharia de Sistemas e Computa\c c\~ao}
79 \linhapesquisa{Engenharia de Dados e Conhecimento}
80 \tipoproducao{bibliografica} % bibliografica | artistica | tecnologica | tecnica
81 \projetovinculado{sim} % sim | nao
82 \nomeprojeto{Nome do Projeto de Pesquisa ao qual o trabalho se vincula}
83 \agenciafomento{Conselho Nacional de Desenvolvimento Cientifico e
84 Tecnologico}{CNPq}
85 \agenciafomento{Fundacao Carlos Chagas Filho de Amparo a Pesquisa do
86 Estado do Rio de Janeiro}{FAPERJ}
87 % A ficha oficial vem de fichacatalografica.sibi.ufrj.br; tendo o arquivo,
88 % descomente a linha abaixo e a moldura vazia da folha adicional some.
89 % \fichacatalografica{ficha.pdf}
90
91 \maketitle
92
93 \frontmatter
94 \dedication{A alguem cujo valor e digno desta dedicatória.}
95
96 \chapter*{Agradecimentos}
97

```

```

98  Gostaria de agradecer a todos os que contribu\iram para este trabalho:
99  orientadores, colegas, familiares e \as ag\encias de fomento. Lorem ipsum
100 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
101 ut labore et dolore magna aliqua.
102
103  \begin{abstract}
104
105  Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
106  consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
107  dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
108  ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
109  in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
110  pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
111  officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
112  iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
113  aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
114  vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
115  aspernatur aut odit aut fugit.
116
117  \end{abstract}
118
119  \begin{foreignabstract}
120
121  In this work, we present an example abstract. Lorem ipsum dolor sit amet,
122  consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
123  dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
124  ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
125  in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
126  pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
127  officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
128  iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
129  aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
130  vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
131  aspernatur aut odit aut fugit.
132
133  \end{foreignabstract}
134
135  % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
136  % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
137  \listoffigures
138  \listoftables
139  \listofquadros
140  \listofmapas
141  \listofprogramas
142  \listofalgorithms
143  \printloabbreviations
144  \printlosymbols
145  \tableofcontents
146
147  \mainmatter
148  \chapter{Introdução}
149
150  Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos do
151

```

152 A classe `\verb|coppe|` foi criada originalmente por Vicente Helano e George Ainsworth, poré
 153 Em 2026, Geraldo Xexéo, com forte ajuda das ferramentas Claude 4.7 Opus e Max e 4.8 Opus M
 154
 155 Provavelmente Geraldo Xexéo, professor do Programa de Engenharia de Sistemas e Computação
 156
 157 A versão mais atual dessa classe é mantida no GitHub, no repositório `\url{https://github.c`
 158
 159 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
 160
 161 Há um grupo de WhatsApp para perguntas e discussões do estilo: `\href{https://github.com/CO`
 162
 163 Bugs devem ser reportados na `\href{https://github.com/COPPE-UFRJ/CoppeTeX/issues}{página d`
 164
 165 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
 166
 167 É importante de notar que essa classe não foi construída sobre a classe `\LaTeX \` para a A
 168
 169 Essa classe implementa citações ABNT modernas e vários tipos de entrada descritos na norma
 170
 171 Este documento não substitui, mas complementa, o documento que descreve a classe.
 172
 173
 174
 175 `\chapter{Configurações Iniciais}`
 176
 177 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c
 178
 179 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
 180
 181 Recomendo ler o documento “The `\verb*|coppe|` document class” para entender melhor todas
 182
 183 `\section{Linguagem principal do texto}`
 184
 185 Essa classe considera que o texto principal está em português e algumas partes específicas
 186 `\begin{itemize}`
 187 `\item` A opção `\verb|english|` deve ser usada no comando `\verb|\documentclass|`.
 188 `\item` A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
 189 `\end{itemize}`
 190
 191 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe `\CoppeT`
 192
 193
 194 `\section{Por que usar o \LaTeX}`
 195
 196 Há uma grande discussão entre usuário de Word e `\LaTeX`, principalmente, quanto ao uso dess
 197
 198 Nós escolhemos o `\LaTeX` por alguns motivos: grande facilidade de seguir um estilo sem se p
 199
 200 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob
 201
 202 `\section{Como e onde usar o \LaTeX}`
 203
 204 Existem muitos tutoriais de `\LaTeX`, mas basicamente, em 2024, ele é usado em dois ambiente
 205 `\begin{enumerate}`

206 \item Na sua máquina, instalando uma versão completa como o \hologo{MiKTeX}, típico do Win
207 \item Usar um ambiente na rede, como o Overleaf.
208 \end{enumerate}
209
210 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b
211
212
213 \chapter{Algumas Regras da COPPE}
214
215 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei
216
217 É imprescindível definir os símbolos, tal como o
218 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
219 \syml{\mathbb{R}}{Conjunto dos números reais}
220 \syml{\emptyset}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como
221
222 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \ve
223
224 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice
225
226 \section{Citações}
227 \label{sec:citacoes}
228
229 Citações curtas podem ser feitas \enquote{o comando quote} ou direto com ‘‘duas crases e do
230
231 \begin{longquote}
232 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
233 feita com o ambiente \verb*|longquote| The primary objective of this
234 investigation was to determine the feasibility of detecting corrosion in
235 aluminum Naval aircraft components with neutron radiographic interrogation
236 and the use of standard corrosion penetrameters. Secondary objectives
237 included the determination of the effect of object thickness on image quality,
238 the defining of minimum levels of detectability and a preliminary investigation
239 of a means whereby the degree of corrosion could be quantified with neutron
240 radiographic data. \citep{article-example}
241 \end{longquote}
242
243 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \verb*|na
244
245 Em todo caso, \textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em
246
247 Quando se cita uma obra por meio de outra (\textit{apud}), pela ABNT entra na lista de ref
248
249 \chapter{Floats}
250
251 Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \texti
252
253 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm
254
255 \textbf{Pela norma atual (manual2024), a legenda} \verb|\caption| \textbf{de toda ilustração
256
257 Quadros são ilustrações cujos dados são delimitados por linhas em \emph{todas} as bordas (ao
258
259

```

260 \section{Tabelas e Figuras Padrão}
261
262 Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.
263
264 \begin{table}[ht]
265 \centering % Centraliza a tabela
266 \caption{Exemplo de Tabela de Números}
267 \label{tab:exemplo_numeros}
268 \begin{tabular}{ccc} % Define a quantidade de colunas
269 \hline % Linha superior
270 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos
271 \hline % Linha média
272 1 & 2 & 3 \\ % Primeira linha de dados
273 \hline
274 4 & 5 & 6 \\ % Segunda linha de dados
275 \hline
276 7 & 8 & 9 \\ % Terceira linha de dados
277 \hline
278 10 & 11 & 12 \\ % Quarta linha de dados
279 \hline % Linha inferior
280 \end{tabular}
281 \fonte{Elabora\c{c}\~ao pr'\opria.} % fonte obrigatoria, abaixo do float
282 \end{table}
283
284
285
286 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
287
288 \begin{figure}[ht]
289 \centering % Centraliza a figura
290 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
291 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
292 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
293 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
294 \end{figure}
295
296 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
297
298 \begin{quadro}[ht]
299 \centering
300 \caption{Exemplo de Quadro}
301 \label{qua:exemplo}
302 \begin{tabular}{|l|l|}
303 \hline
304 \textbf{Ilustração} & \textbf{Delimitação} \\
305 \hline
306 Tabela & Linhas apenas no topo e na base. \\
307 \hline
308 Quadro & Linhas em todas as bordas. \\
309 \hline
310 \end{tabular}
311 \fonte{Elabora\c{c}\~ao pr'\opria.}
312 \end{quadro}
313

```



```

314 Um segundo quadro, no \autoref{qua:segundo}, prova que o contador anda e que a lista de quad
315
316 \begin{quadro}[ht]
317 \centering
318 \caption{Segundo Quadro, para provar a numeração}
319 \label{qua:segundo}
320 \begin{tabular}{|l|l|}
321 \hline
322 \textbf{Elemento} & \textbf{Onde fica} \\
323 \hline
324 Legenda & Acima da ilustração. \\
325 \hline
326 Fonte & Abaixo da ilustração. \\
327 \hline
328 \end{tabular}
329 \fonte{Adaptado do manual do SiBI, 9.\textsuperscript{a} ed. rev., 2026.}
330 \end{quadro}
331
332 \section{Criando um float novo}
333
334 \verb|\newcoppefloat{nome}{Nome}{Titulo da Lista}| cria um flutuante que segue a mesma regra
335
336 \begin{mapa}[ht]
337 \centering
338 \caption{Primeiro mapa, num float criado pelo autor}
339 \label{mapa:primeiro}
340 \begin{tikzpicture}[scale=0.8]
341 \draw[thick] (0,0) rectangle (4,2.4);
342 \draw[fill=gray!20] (0.4,0.4) rectangle (1.8,2);
343 \draw[fill=gray!40] (2.2,0.4) rectangle (3.6,1.2);
344 \node at (2,-0.4) {\footnotesize esquema de ocupação};
345 \end{tikzpicture}
346 \fonte{Elabora\c{c}\~ao pr\'opria.}
347 \end{mapa}
348
349 \begin{mapa}[ht]
350 \centering
351 \caption{Segundo mapa, para provar o contador e a lista}
352 \label{mapa:segundo}
353 \begin{tikzpicture}[scale=0.8]
354 \draw[thick] (0,0) circle (1.2);
355 \draw[->] (-1.6,0) -- (1.6,0);
356 \draw[->] (0,-1.6) -- (0,1.6);
357 \node at (0,-2) {\footnotesize esquema de alcance};
358 \end{tikzpicture}
359 \fonte{Elabora\c{c}\~ao pr\'opria.}
360 \end{mapa}
361
362
363 \section{Tabelas mais elegantes}
364
365 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
366
367 \begin{table}[ht]

```

```

368 \centering % Centraliza a tabela
369 \caption{Exemplo de Tabela de Números mais elegantes}
370 \label{tab:exemplo_numerosbom}
371 \begin{tabular}{ccc} % Define a quantidade de colunas
372 \toprule % Linha superior
373 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos
374 \midrule % Linha média
375 1 & 2 & 3 \\ % Primeira linha de dados
376 4 & 5 & 6 \\ % Segunda linha de dados
377 7 & 8 & 9 \\ % Terceira linha de dados
378 10 & 11 & 12 \\ % Quarta linha de dados
379 \bottomrule % Linha inferior
380 \end{tabular}
381 \fonte{Elabora\c{c}\~ao pr\'opria.}
382 \end{table}
383
384 \section{Tabelas Longas ou Largas}
385
386 Se sua tabela é muito longa ou larga, existem várias opções.
387 \begin{itemize}
388 \item alterar o tamanho da letra
389 \item Usar o longtable
390 \item rodar a tabela, fazendo ela em \textit{landscape}
391 \item fazer a tabela dentro de um minibox
392 \end{itemize}
393
394
395 \subsection{Tabelas largas demais}
396
397 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
398
399 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
400
401 \begin{table}[ht]
402 \centering % Centraliza a tabela
403 \caption{Exemplo de Tabela Larga com Fonte Menor}
404 \label{tab:tabela_largafns}
405 \footnotesize % Aplica uma fonte menor para a tabela
406 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
407 \toprule
408 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} & \textbf{Col. 4} & \textbf{Col. 5} & \textbf{Col. 6} & \textbf{Col. 7} \\
409 \midrule
410 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
411 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
412 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
413 \bottomrule
414 \end{tabular}
415 \fonte{Elabora\c{c}\~ao pr\'opria.}
416 \end{table}
417
418 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
419
420 \begin{table}[ht]
421 \centering

```

```

422 \caption{Exemplo de Tabela Redimensionada}
423 \label{tab:examplerb}
424 \resizebox{\textwidth}{!}{%
425 \begin{tabular}{llll}
426 \toprule
427 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
428 \midrule
429 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
430 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
431 \bottomrule
432 \end{tabular}}%
433 }
434 \fonte{Elabora\c{c}\~ao pr\'opria.}
435 \end{table}
436
437
438 Para rodar uma tabela muito larga em 90 graus no LaTeX, voc\ea pode usar o pacote \verb*|rotat
439
440 Aqui est\aa um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
441
442 \begin{sidewaystable}
443 \centering
444 \caption{Sua Legenda Aqui}
445 \label{tab:sua_tabela}
446 \begin{tabular}{llll}
447 \toprule
448 Coluna 1 & Coluna 2 & Coluna 3 \\
449 \midrule
450 Item 1 & Item 2 & Item 3 \\
451 Item 4 & Item 5 & Item 6 \\
452 \bottomrule
453 \end{tabular}
454 \fonte{Elabora\c{c}\~ao pr\'opria.}
455 \end{sidewaystable}
456
457 Se a tabela for muito longa, o ambiente \verb|longtable| \e o ideal. Ele fornece comandos par
458
459 % Exemplo de tabela longa que se estende por v\arias p\aginas
460 \begin{longtable}{|c|c|c|}
461 % primeiro cabe\cacho (e o caption)
462 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
463 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
464 \endfirsthead
465 % cabe\cacho normal
466 \multicolumn{3}{c}%
467 {\table\ -- continua\c\o da p\agina anterior} \\
468 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
469 \endhead
470 % p\ea normal
471 \hline \multicolumn{3}{|r|}{Continua na pr\oxima p\agina} \\ \hline
472 \endfoot
473 \hline
474 % \u ltimo p\ea
475 \multicolumn{3}{|r|}{Continua na pr\oxima p\agina} \\

```

```

476 \hline \hline
477 \endlastfoot
478
479 % Conteúdo da tabela
480 1 & 2 & 3 \\
481 4 & 5 & 6 \\
482 1 & 2 & 3 \\
483 4 & 5 & 6 \\
484 1 & 2 & 3 \\
485 4 & 5 & 6 \\
486 1 & 2 & 3 \\
487 4 & 5 & 6 \\
488 1 & 2 & 3 \\
489 4 & 5 & 6 \\
490 1 & 2 & 3 \\
491 4 & 5 & 6 \\
492 1 & 2 & 3 \\
493 4 & 5 & 6 \\
494 1 & 2 & 3 \\
495 4 & 5 & 6 \\
496 1 & 2 & 3 \\
497 1 & 2 & 3 \\
498 4 & 5 & 6 \\
499 1 & 2 & 3 \\
500 4 & 5 & 6 \\
501 1 & 2 & 3 \\
502 4 & 5 & 6 \\
503 1 & 2 & 3 \\
504 4 & 5 & 6 \\
505 1 & 2 & 3 \\
506 4 & 5 & 6 \\
507 1 & 2 & 3 \\
508 4 & 5 & 6 \\
509 1 & 2 & 3 \\
510 4 & 5 & 6 \\
511 1 & 2 & 3 \\
512 4 & 5 & 6 \\
513 1 & 2 & 3 \\
514 4 & 5 & 6 \\
515 1 & 2 & 3 \\
516 4 & 5 & 6 \\
517 1 & 2 & 3 \\
518 4 & 5 & 6 \\
519 1 & 2 & 3 \\
520 4 & 5 & 6 \\1 & 2 & 3 \\
521 4 & 5 & 6 \\
522 1 & 2 & 3 \\
523 4 & 5 & 6 \\
524 1 & 2 & 3 \\
525 1 & 2 & 3 \\
526 4 & 5 & 6 \\
527 1 & 2 & 3 \\
528 4 & 5 & 6 \\
529 1 & 2 & 3 \\

```

530 4 & 5 & 6 \\
531 1 & 2 & 3 \\
532 4 & 5 & 6 \\
533 1 & 2 & 3 \\
534 4 & 5 & 6 \\
535 1 & 2 & 3 \\
536 4 & 5 & 6 \\
537 1 & 2 & 3 \\
538 4 & 5 & 6 \\
539 1 & 2 & 3 \\
540 4 & 5 & 6 \\
541 1 & 2 & 3 \\
542 4 & 5 & 6 \\
543 1 & 2 & 3 \\
544 4 & 5 & 6 \\
545 1 & 2 & 3 \\
546 4 & 5 & 6 \\
547 1 & 2 & 3 \\
548 4 & 5 & 6 \\
549 1 & 2 & 3 \\
550 4 & 5 & 6 \\
551 1 & 2 & 3 \\
552 4 & 5 & 6 \\
553 1 & 2 & 3 \\
554 4 & 5 & 6 \\
555 1 & 2 & 3 \\
556 4 & 5 & 6 \\
557 1 & 2 & 3 \\
558 4 & 5 & 6 \\
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566 4 & 5 & 6 \\
567 1 & 2 & 3 \\
568 4 & 5 & 6 \\
569 1 & 2 & 3 \\
570 4 & 5 & 6 \\
571 1 & 2 & 3 \\
572 4 & 5 & 6 \\
573 1 & 2 & 3 \\
574 4 & 5 & 6 \\
575 1 & 2 & 3 \\
576 4 & 5 & 6 \\
577 1 & 2 & 3 \\
578 4 & 5 & 6 \\
579 1 & 2 & 3 \\
580 4 & 5 & 6 \\
581 1 & 2 & 3 \\
582 4 & 5 & 6 \\
583 1 & 2 & 3 \\

```

584 4 & 5 & 6 \\
585 1 & 2 & 3 \\
586 4 & 5 & 6 \\
587
588 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
589 \end{longtable}
590 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
591
592 \section{Subfiguras (subcaption)}
593
594 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
595
596 \begin{figure}[ht]
597   \centering
598   \begin{subfigure}[b]{0.42\textwidth}
599     \centering\includegraphics[width=\linewidth]{example-image-a}
600     \caption{Primeira vista}\label{fig:sub-a}
601   \end{subfigure}
602   \hfill
603   \begin{subfigure}[b]{0.42\textwidth}
604     \centering\includegraphics[width=\linewidth]{example-image-b}
605     \caption{Segunda vista}\label{fig:sub-b}
606   \end{subfigure}
607   \caption{Figura com duas subfiguras}
608   \label{fig:subs}
609   \fonte{Elabora\c{c}\~ao pr\'opria.}
610 \end{figure}
611
612 Referencie a figura inteira com \verb|\autoref{fig:subs}|. Cada parte tem
613 rótulo próprio: \verb|\autoref{fig:sub-a}| leva à \autoref{fig:sub-a}.
614
615 \section{Imagens (graphicx)}
616
617 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
618 \begin{itemize}
619   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
620   \item \verb|[angle=90]| --- rotação;
621   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, direi}
622 \end{itemize}
623 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt{t}
624
625 \section{Introdução ao tikz}
626
627 0 \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
628
629 \begin{figure}[ht]
630   \centering
631   \begin{tikzpicture}
632     \node[draw, rounded corners] (a) at (0,0) {Início};
633     \node[draw, rounded corners] (b) at (4,0) {Fim};
634     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
635   \end{tikzpicture}
636   \caption{Diagrama simples em tikz}
637   \label{fig:tikz}

```

```

638 \fonte{Elabora\c{c}\~ao pr\'opria.}
639 \end{figure}
640
641 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|tikz|
642
643 \section{Float ou direto no texto?}
644
645 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \verb|caption|
646
647 \chapter{Using BibLaTeX}
648 \label{cap:biblatex}
649
650 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
651
652 \textbf{É} obrigatório compilar com \texttt{biber} (e não com \hologo{BibTeX}). A sequência
653
654 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque
655
656 \begin{table}[h]
657 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao}
658 \texttt{\textbackslash cite} do \LaTeX\ e
659 o comando \texttt{\textbackslash citep},
660 disponível pela opção \texttt{natbib} do BibLaTeX.
661 \label{tab:citation}
662 \centering
663 {\footnotesize
664 \begin{tabular}{|c|c|c|}
665 \hline
666 Tipo da Publicação & \verb|\cite| & \verb|\citep|\\
667 \hline
668 Livro & \cite{book-example} & \citep{book-example}\\
669 Artigo & \cite{article-example} & \citep{article-example}\\
670 Relatório & \cite{techreport-example} & \citep{techreport-example}\\
671 Relatório & \cite{techreport-exampleIn} & \citep{techreport-exampleIn}\\
672 Anais de Congresso & \cite{inproceedings-example} &
673 \citep{inproceedings-example}\\
674 Séries & \cite{incollection-example} & \citep{incollection-example}\\
675 Em Livro & \cite{inbook-example} & \citep{inbook-example}\\
676 Dissertação de mestrado & \cite{mastersthesis-example} &
677 \citep{mastersthesis-example}\\
678 Tese de doutorado & \cite{phdthesis-example} & \citep{phdthesis-example}\\
679 \hline
680 \end{tabular}}
681 \fonte{Elabora\c{c}\~ao pr\'opria.}
682 \end{table}
683
684 \begin{table}[h]
685 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao}
686 \texttt{\textbackslash cite} do \LaTeX\ e
687 o comando \texttt{\textbackslash citet},
688 disponível pela opção \texttt{natbib} do BibLaTeX. Além disso, usando o booktabs.}
689 \label{tab:citation1}
690 \centering
691 {\footnotesize

```

```

692 \begin{tabular}{ccc}
693   \toprule
694   Tipo da Publicação & \verb|\cite| & \verb|\citet|\
695   \midrule
696   Livro & \cite{book-example} & \citet{book-example}\
697   Artigo & \cite{article-example} & \citet{article-example}\
698   Relatório & \cite{techreport-example} & \citet{techreport-example}\
699   Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\
700   Anais de Congresso & \cite{inproceedings-example} &
701     \citet{inproceedings-example}\
702   Séries & \cite{incollection-example} & \citet{incollection-example}\
703   Em Livro & \cite{inbook-example} & \citet{inbook-example}\
704   Dissertação de mestrado & \cite{mastersthesis-example} &
705     \citet{mastersthesis-example}\
706   Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\
707   \bottomrule
708 \end{tabular}}
709 \fonte{Elabora\c{c}\~ao pr\'opria.}
710 \end{table}
711
712 \section{Comandos de citação}
713
714 Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex
715
716 \subsection{Parâmetro opcional: página e “tradução nossa”}
717
718 Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser
719
720 \subsection{Citação de citação (\emph{apud})}
721
722 Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citepapud| (demonstra
723
724 \subsection{Múltiplas bibliografias}
725
726 Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|refsection|
727
728 \section{Tipos de entrada bibliográfica}
729
730 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinôni
731
732 \subsection{Tipos acadêmicos principais}
733 \begin{itemize}
734   \item Livro / monografia --- \verb|@book| (\verb|@livro|)
735   \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \v
736   \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
737   \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento
738   \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo
739   \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); periód
740   \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/sit
741 \end{itemize}
742
743 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
744 \begin{itemize}
745   \item Patente --- \verb|@patent| (\verb|@patente|)

```



```

746 \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato
747 \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\
748 \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
749 \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb
750 \end{itemize}
751
752 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
753 \begin{itemize}
754 \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@j
755 \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadetv|, \verb|@serietv|)
756 \end{itemize}
757
758 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
759
760 \chapter{Alguns outros exemplo úteis}
761
762 \begin{tcolorbox}[title=Meu Textbox]
763 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
764 \end{tcolorbox}
765
766 \begin{tcolorbox}
767 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
768 \end{tcolorbox}
769
770 \begin{figure}[ht]
771 \centering
772 \begin{tikzpicture}
773 \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\te
774 \begin{scope}[x={(image.south east)},y={(image.north west)}]
775 % Definindo o textbox dentro da figura
776 \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text
777 \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
778 Este textbox fala sobre como inserir um textbox dentro de uma figura usa
779 \end{tcolorbox}
780 };
781 \end{scope}
782 \end{tikzpicture}
783 \caption{Figura com Textbox}
784 \label{fig:figura_com_textbox1}
785 \fonte{Elabora\c{c}\~ao pr\'opria.}
786 \end{figure}
787
788
789 \begin{figure}[ht]
790 \centering
791 \begin{tcolorbox}
792 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
793 \end{tcolorbox}
794 \caption{Figura com Textbox simples}
795 \label{fig:figura_com_textbox}
796 \fonte{Elabora\c{c}\~ao pr\'opria.}
797 \end{figure}
798
799 \section{Listagens de código e algoritmos}

```

```

800
801 0 CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
802
803 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
804 def saudacao(nome):
805     # imprime uma saudação acentuada
806     print(f"Olá, {nome}!")
807
808 saudacao("COPPE")
809 \end{python}
810 \source{Elaboração própria.}
811
812 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
813
814 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
815 Olá, COPPE!
816 \end{Verbatim}
817
818 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|. 0 \auto
819
820 \begin{java}[caption={Exemplo de programa em Java},label=prog:java]
821 public class Saudacao {
822     public static void main(String[] args) {
823         System.out.println("Olá, COPPE!");
824     }
825 }
826 \end{java}
827 \fonte{Elaboração própria.}
828
829 Há também \verb|\pythoninline| e \verb|\javainline| para código no meio do
830 texto, como em \javainline{new Saudacao()}, e o ambiente \verb|Verbatim| (com as opções acim
831
832 \begin{algorithm}[H]
833 \caption{Soma dos elementos de um vetor}
834 \label{alg:soma}
835 \Dados{vetor  $v$  com  $n$  elementos}
836 \Resultado{soma  $s$  dos elementos de  $v$ }
837  $s \leftarrow 0$ ;
838 \Para{ $i \leftarrow 1$  \Ate{ }  $n$ }{
839      $s \leftarrow s + v[i]$ ;
840 }
841 \Retorna  $s$ ;
842 \end{algorithm}
843
844 0 \autoref{alg:busca} é o segundo algoritmo, e usa \verb|\Enquanto|. Como no caso das listag
845
846 \begin{algorithm}[H]
847 \caption{Busca sequencial em um vetor}
848 \label{alg:busca}
849 \Dados{vetor  $v$  com  $n$  elementos e uma chave  $k$ }
850 \Resultado{posição de  $k$  em  $v$ , ou  $0$  se  $k$  não estiver em  $v$ }
851  $i \leftarrow 1$ ;
852 \Enquanto{ $i \leq n$  \textbf{e}  $v[i] \neq k$ }{
853      $i \leftarrow i + 1$ ;

```

```

854 }
855 \Se{$i > n$}{\Retorna $0$};
856 \Retorna $i$;
857 \end{algorithm}
858
859 \chapter{Escrevendo matemática}
860
861 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carrega
862
863 \section{Alinhamento (align)}
864 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebra a linha)
865 \begin{align}
866 f(x) &= (x+1)^2 \\
867 &= x^2 + 2x + 1 .
868 \end{align}
869
870 \section{Matrizes (matrix)}
871 O \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
872 \begin{equation}
873 A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}
874 \quad
875 \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
876 \end{equation}
877
878 \section{Subequações (subequations)}
879 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
880 \begin{subequations}
881 \begin{align}
882 a &= b + c , \\
883 d &= e - f .
884 \end{align}
885 \end{subequations}
886
887 \section{Estruturas alinhadas (array)}
888 O \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
889 \begin{equation}
890 |x| = \left\{ \begin{array}{l} x, \text{ se } x \geq 0, \\ -x, \text{ caso contrário.} \end{array} \right.
891 \end{equation}
892
893 \section{Operadores (\textbackslash DeclareMathOperator)}
894 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no preâmbulo
895
896 \begin{verbatim}
897 \usepackage{amsmath}
898 \DeclareMathOperator{\RMSE}{RMSE}
899 \DeclareMathOperator{\diag}{diag}
900 \end{verbatim}
901 Assim:
902 \begin{itemize}
903 \item \verb|\RMSE(y,\hat{y})| produz  $\text{RMSE}(y,\hat{y})$ ;
904 \item \verb|\diag(\lambda_1,\lambda_2)| produz  $\text{diag}(\lambda_1,\lambda_2)$ .
905 \end{itemize}

```

```

908 \end{itemize}
909
910 \section{Unicode e LuaLaTeX (unicode-math)}
911 \label{sec:unicode-math}
912 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss}
913 \begin{verbatim}
914 \usepackage{amsmath}      % a classe já carrega
915 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
916 \setmathfont{Latin Modern Math} % ou STIX Two Math
917 \end{verbatim}
918 O código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x)dx$ ,  $\alpha$ ) e você ai
919
920 \chapter{Referências cruzadas, notas e índices}
921 \label{cap:refs}
922
923 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
924
925 \section{Rótulos e referências cruzadas}
926
927 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
928
929 \section{Notas de rodapé}
930
931 Use \verb|\footnote{texto}|\footnote{Como esta nota, que a classe imprime em espaçamento s
932
933 \section{hyperref e marcadores}
934
935 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref
936
937 \section{URLs}
938
939 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
940 use \verb|\href{https://exemplo.org}{texto}|. Evite digitar URLs como texto comum --- elas n
941
942 \section{Índice remissivo}
943
944 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre
945
946 \section{Traduções (NBR 10520)}
947
948 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
949
950 \chapter{Truques de LaTeX}
951
952 \section{Comentários mágicos (\texttt{\% !TeX})}
953 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
954 \begin{verbatim}
955 % !TeX program = lualatex
956 % !TeX encoding = UTF-8
957 % !TeX spellcheck = pt_BR
958 % !BIB program = biber
959 \end{verbatim}
960 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
961

```

```

962 \section{Espaço engolido}
963 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como “\LaTeX é
964
965 \section{Escopo (grupos)}
966 Chaves \verb|\{...\}| (ou \verb|\begingroup|\dots\verb|\endgroup|) criam um escopo: mudanças
967
968 \section{Descarregar figuras: \texttt{\textbackslash clearpage} × \texttt{\textbackslash m
969 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
970
971 \section{\texttt{latexmkrc}, perl e Overleaf}
972 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
973
974 \chapter{Método Proposto}
975 \chapter{Resultados e Discussões}
976
977 \section{Algumas Demonstrações}
978
979 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha
980 \syml[beta]{Beta}{A palavra Beta more e corrigida}
981 \syml[zzbeta]{\beta}{A letra $\beta$ corrigida}
982 \syml[beta]{A palavra beta}
983 \syml[alpha]{A palavra alpha}
984 \syml[alpha]{Alpha}{A palavra Alpha}
985 \syml[zzalpha]{\alpha}{A letra $\alpha$ corrigida}
986 \syml[marco]{Marco}{A palavra Marco corrigida}
987
988 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
989 \abbrev{GoT}{Game of Thrones}
990 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
991 \abbrev[iot]{IoT}{IoT ordenado como iot}
992 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
993 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
994 \abbrev{IoT}{IoT com ordenação default}
995 \abbrev[ITU]{ITU}{ITU mesmo}
996
997 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
998 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
999
1000 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas auto
1001
1002
1003
1004
1005 \chapter{Conclusões}
1006
1007 \backmatter
1008
1009 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
1010 % partitura --- declarados via sinônimos em exemplo.bib.
1011 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
1012 \nocite{jurisprudencia-example}
1013 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
1014 \nocite{jornalcaderno-example,jornal-example}
1015 \printbibliography

```

```

1016
1017
1018
1019 \appendix
1020
1021 \chapter{Um apêndice}
1022
1023 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
1024
1025 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo
1026
1027
1028 \chapter{Outro apêndice}
1029
1030 \annex
1031
1032
1033 \chapter{Um Anexo}
1034 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
1035
1036
1037
1038 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
1039
1040 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comando
1041
1042
1043 \chapter{Outro Anexo}
1044
1045
1046 \coppetexfinalpage% colofão institucional (página par, face do contracapa)
1047
1048 \end{document}
1049
1050 \example

```

10 Implementation

10.1 The ‘coppe.cls’ file

```

1051 \*class
1052 \def\filename{coppe.dtx}
1053 \def\fileversion{v4.1}
1054 \def\filedate{2026/09/10}
1055 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
1056 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
1057 % Save the class version and date into private macros immediately, before any
1058 % required package can redefine the very common \fileversion / \filedate
1059 % global names (LaTeX has no convention reserving them for the host class).
1060 % \coppe@version / \coppe@date are what user-facing macros (e.g.
1061 % \coppetexfinalpage) should reference.
1062 \let\coppe@version\fileversion
1063 \let\coppe@date\filedate
1064 % Document date stored in private slots so \date can set them without ever

```

```

1065 % touching TeX's \month/\year primitives (which would break \today). They
1066 % default to the current system month/year, so a document without \date
1067 % still renders sensibly on the cover and folha de rosto.
1068 \edef\coppe@docmonth{\the\month}
1069 \edef\coppe@docyear{\the\year}
1070 \LoadClass[12pt,a4paper,oneside]{book}
1071 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
1072 % the END of the class (after every class hook is defined), in that order:
1073 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
1074 % it patches (TOC, refs, lists) are already defined, and biblatex must come
1075 % after hyperref so its citation links and back-references work.
1076 \RequirePackage{hyphenat}
1077 \RequirePackage{lastpage}
1078 \RequirePackage{ifthen}
1079 \RequirePackage{graphicx}
1080 % Os logotipos da capa. Nao ha \graphicspath aqui, e nao adianta haver: quando a
1081 % classe e instalada na arvore do TeX, o kpathsea a encontra sem dizer a
1082 % ninguem ONDE a encontrou -- \CurrentFilePath vem vazio e \@filef@und devolve
1083 % so o nome. Nao ha como a classe deduzir a propria pasta.
1084 %
1085 % Nem e preciso. O \includegraphics procura pelo kpathsea, o mesmo mecanismo que
1086 % achou o .cls, de modo que um logotipo instalado AO LADO da classe e achado sem
1087 % mais nada -- conferido em MiKTeX, compilando de outra pasta com so o TEXINPUTS
1088 % apontando para a da classe.
1089 %
1090 % O que quebrava, e e o que esta corrigido aqui, e o outro caso: quem copia so o
1091 % coppe.cls para a pasta do trabalho e deixa os logotipos para tras. Ai o TeX
1092 % dizia "File 'coppe-logo' not found" e mandava digitar outro nome, sem nenhuma
1093 % pista de que o arquivo vem junto com a classe. \coppe@logo diz.
1094 \newcommand\coppe@logo[2][2cm]{%
1095   \IfFileExists{#2.pdf}%
1096     {\includegraphics[height=#1]{#2}}
1097     {\ClassError{coppe}{Logotipo ' #2.pdf ' nao encontrado}%
1098       {Ele vem junto com a classe, na mesma pasta do coppe.cls.\MessageBreak
1099         Copie coppe-logo.pdf e ufrj-logo.pdf para a pasta do seu\MessageBreak
1100         trabalho, ou instale-os ao lado do coppe.cls na arvore do TeX.\MessageBreak
1101         Os dois estao em dist/ no repositorio da CoppeTeX.}%
1102       \framebox[3cm][c]{\rule{0pt}{#1}\ttfamily #2?}}}
1103 \RequirePackage{tabularx}
1104 \RequirePackage{booktabs} % regras de tabela profissionais (toprule/midrule/bottomrule)
1105 \RequirePackage{longtable} % tabelas que quebram entre páginas
1106 \RequirePackage{etoolbox}
1107 \RequirePackage{ltxcmds}
1108 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
1109 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
1110 \def\TeX{\hologo{TeX}}
1111 \def\LaTeX{\hologo{LaTeX}}
1112 \def\LaTeXe{\hologo{LaTeXe}}
1113 \def\BibTeX{\hologo{BibTeX}}
1114 \def\pdfLaTeX{\hologo{pdfLaTeX}}
1115 \def\pdfTeX{\hologo{pdfTeX}}
1116 \def\LuaLaTeX{\hologo{LuaLaTeX}}
1117 \def\XeLaTeX{\hologo{XeLaTeX}}
1118 \def\MiKTeX{\hologo{MiKTeX}}

```

```

1119 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
1120 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
1121 % files may use accented Latin characters (áéíóú) directly. Under pdfLaTeX we also
1122 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
1123 \RequirePackage{iftex}
1124 \ifPDFTeX
1125   \RequirePackage[utf8]{inputenc}
1126   \AtEndOfClass{\ClassWarningNoLine{coppe}{%
1127     Although slower, you should try compiling with LuaLaTeX}}
1128 \fi
1129 % lmodern MUST come with the T1 encoding. Without it, T1 falls back to the EC
1130 % fonts, which exist only as METAFONT bitmaps in most installations, and the
1131 % whole document is typeset in Type3 bitmap fonts. Measured before this change:
1132 % example.pdf carried 87 Type3 fonts against 6 Type1, and coppe.pdf 133 against
1133 % 2. The consequences are not cosmetic:
1134 %
1135 %   * the text renders blurred at any zoom and prints poorly;
1136 %   * copying and searching text in the PDF is unreliable;
1137 %   * PDF/A -- required for deposit by 2.2(d) of the 2026 manual -- is not
1138 %     attainable with bitmap text.
1139 %
1140 % Latin Modern is the vector successor of Computer Modern, metrically and
1141 % visually near-identical, and ships with every current distribution, so the
1142 % page layout does not move.
1143 \RequirePackage{lmodern}
1144 \RequirePackage[T1]{fontenc}
1145 % Math support. amsmath is universal and works on every engine, so the class
1146 % brings it in unconditionally (it must precede hyperref, which is loaded at
1147 % the end of the class). amssymb is the legacy AMS symbol package, built
1148 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
1149 % XeLaTeX) it clashes with unicode-math, which already provides all of its
1150 % symbols through Unicode math codes. So amssymb is loaded ONLY under
1151 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
1152 % unicode-math (the class does not auto-load unicode-math because choosing
1153 % an OpenType math font is a typographic decision that belongs to the author).
1154 \RequirePackage{amsmath}
1155 \ifPDFTeX
1156   \RequirePackage{amssymb}
1157 \else
1158   % Unicode engine: defer the check to end of document. Only warn if the
1159   % user did NOT add either unicode-math (the modern OpenType-math route)
1160   % or amssymb (the legacy route, which they would have loaded manually).
1161   % This avoids nagging authors who already followed the documented
1162   % \usepackage{unicode-math} workflow.
1163   \AtEndDocument{%
1164     \@ifpackageloaded{unicode-math}{\}%
1165     \@ifpackageloaded{amssymb}{\}%
1166     \ClassWarningNoLine{coppe}{%
1167       You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
1168       neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
1169       (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\MessageBreak
1170       Add to your preamble:\MessageBreak
1171       \space\space\string\usepackage{unicode-math}\MessageBreak
1172       \space\space\string\setmathfont{Latin Modern Math}\MessageBreak

```



```

1173          Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
1174          with the OpenType math font setup}}}}
1175 \fi
1176 % Margins per the SiBI manual, 9th rev. ed. (2026), 2.3: left 3cm, top 3cm,
1177 % right 2cm, bottom 2cm -- ONE-SIDED. The 2025 edition also specified verso
1178 % margins (right 3cm / left 2cm) and recommended printing the body on both
1179 % sides; the 2026 revision DELETED the verso margins entirely, following CEPG
1180 % Resolution 246/2023, which ended the printed deposit. A digital-only document
1181 % has no gutter, so the mirrored layout (previously inner=outer=2cm plus a 1cm
1182 % bindingoffset) no longer has any normative basis and is now opt-in via the
1183 % \texttt{twoside} class option, handled after \ProcessOptions below.
1184 %
1185 % headheight=15pt is given here (not patched later) so geometry's own page
1186 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
1187 % height to host the running head, and the standard 12pt default triggers
1188 % its "headheight is too small" warning. Letting geometry know up front
1189 % keeps \topmargin a derived value -- we never touch it by hand.
1190 % (geometry has no \texttt{oneside} key -- only \texttt{twoside}; with the key
1191 % absent it follows the class, which \LoadClass set to one-sided above.)
1192 % headsep is given explicitly to place the folio. 2.7 measures it the way it
1193 % measures the right margin -- to the digits themselves: "a 2cm da borda
1194 % superior, ficando o ultimo algarismo a 2cm da borda direita". The right edge
1195 % already landed at 2.02cm; the top of the digits sat at 1.87cm, too high.
1196 % geometry derives \topmargin from top - 1in - headheight - headsep, so
1197 % shortening headsep lowers the header and leaves the body top untouched:
1198 % topmargin + headheight + headsep is 13.088pt either way, and 3cm is where the
1199 % body still starts. The value is CALIBRATED, not computed -- the folio sits a
1200 % little lower than the arithmetic alone predicts, because fancyhdr's box does
1201 % not put the digits flush with the top of \headheight. Two measured points,
1202 % taken from the rendered page rather than from the PDF's font metrics:
1203 %   headsep 18.07pt -> digits at 1.8669cm   headsep 14.29pt -> 2.0616cm
1204 % which put 2.0000cm at 15.49pt while the folio was still set at the body size.
1205 % It is \footnotesize now (2.2(b) puts pagination in the smaller font), the
1206 % digits are shorter, and their top therefore sits lower for the same baseline:
1207 % re-measured, 15.49pt gave 2.0659cm and the value moved to 16.87pt. That is
1208 % precisely why this is a measured constant and not a computed one -- change
1209 % the folio's size or headheight and it has to be measured again, not scaled.
1210 % Verified on the rendered page: the top of the digits and their right edge come
1211 % out at the same 2.019cm, and the right margin is 2cm by construction, so that
1212 % shared 0.019cm is the ruler's own bias and not a residual error.
1213 \RequirePackage[a4paper,headheight=15pt,headsep=16.87pt,%
1214 top=3cm,bottom=2cm,left=3cm,right=2cm]{geometry}
1215 % The first paragraph of every section must be indented like any other. LaTeX's
1216 % default is the Anglo-Saxon convention -- no indent after a heading -- which is
1217 % not what Brazilian typography (nor the SiBI manual's own examples) uses.
1218 % indentfirst was mentioned only as a commented suggestion in the example
1219 % preamble, so every document built with this class came out with the first
1220 % paragraph of each section flush left. Loading it in the class fixes it for
1221 % everybody instead of asking each author to remember.
1222 %
1223 % Loaded BEFORE titlesec so titlesec's \@afterindent handling sees the patched
1224 % \@afterindentfalse.
1225 % Long unbreakable tokens -- \verb|...| in running prose, command names, URLs --
1226 % cannot be hyphenated, so TeX overfills the line rather than stretching the

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1227 % spaces. example.tex alone produced 17 overfull boxes this way, the worst 86pt,
1228 % which is the "text does not always respect the margins" a recent review
1229 % reported. \emergencystretch gives TeX a final pass in which it may stretch
1230 % interword spaces instead of running into the margin. It cannot save a single
1231 % token longer than the measure, but it clears the ordinary cases.
1232 \setlength\emergencystretch{3em}
1233 \RequirePackage[indentfirst]
1234 \RequirePackage{titlesec}
1235 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1236 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
1237 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
1238 % keep "N Title". \appendix and \annex raise the flag (see below).
1239 \newif\if@coppeappendix \@coppeappendixfalse
1240 \titleformat{\chapter}[hang]
1241   {\normalfont\Large\bfseries}
1242   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1243   {1ex}}{\MakeUppercase}
1244 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1245 % 2.6 says the numeric indicative of a section precedes its title flush left,
1246 % and that titles WITHOUT a numeric indicative are centred -- and then lists
1247 % them, "apendice(s)" and "anexo(s)" among them. A letter is not a numeric
1248 % indicative, and the manual sets its own Annex A centred. The chapter format is
1249 % therefore re-issued when \appendix or \annex is called: same label, same
1250 % travessao, but a centred block instead of a flush-left hang. Numbered chapters
1251 % are untouched -- they DO carry a numeric indicative and stay flush left.
1252 \newcommand\coppe@appendixheadings{%
1253   \titleformat{\chapter}[block]
1254     {\normalfont\Large\bfseries\filcenter}
1255     {\MakeUppercase{\appendixname}\ \thechapter\ \textendash}
1256     {1ex}{\MakeUppercase}%
1257   \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}}
1258 \let\coppe@orig@appendix\appendix
1259 \renewcommand\appendix{%
1260   \coppe@orig@appendix\@coppeappendixtrue\coppe@appendixheadings}
1261 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1262 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1263 % which macro is written (so it is correct at .toc-render time even though the
1264 % appendixname has changed back); the chapter letter is baked in at write time.
1265 \newif\if@coppeannex \@coppeannexfalse
1266 \newcommand\coppe@tocapp[1]{%
1267   \MakeUppercase{\coppestring{appendix}}%
1268   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1269 \newcommand\coppe@tocanx[1]{%
1270   \MakeUppercase{\coppestring{annex}}%
1271   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1272 \def\@chapter[#1]#2{%
1273   \ifnum \c@secnumdepth >\m@ne
1274     \if@mainmatter
1275       \refstepcounter{chapter}%
1276       \typeout{\@chapapp\space\thechapter.}%
1277       \if@coppeappendix
1278         \if@coppeannex
1279           \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
1280         \else

```

```

1281         \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1282     \fi
1283     \else
1284         \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1285     \fi
1286     \else
1287         \addcontentsline{toc}{chapter}{#1}%
1288     \fi
1289 \else
1290     \addcontentsline{toc}{chapter}{#1}%
1291 \fi
1292 \chaptermark{#1}%
1293 \addtocontents{lof}{\protect\addvspace{10\p@}}%
1294 \addtocontents{lot}{\protect\addvspace{10\p@}}%
1295 \if@twocolumn
1296     \@topnewpage[\@makechapterhead{#2}]%
1297 \else
1298     \@makechapterhead{#2}%
1299     \@afterheading
1300 \fi}
1301 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1302 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1303 \titleformat{name=chapter,numberless}[hang]
1304     {\normalfont\Large\bfseries\filcenter}{\Opt}{\MakeUppercase}
1305 \titlespacing*{name=chapter,numberless}{Opt}{10pt}{1.5\baselineskip}
1306 % section: ALL CAPS, normal weight -- the \bfseries here contradicted this very
1307 % comment until v4.1, and made the secondary level as heavy as the primary one,
1308 % defeating the "destaque gradativo" of 2.6. The gradation is now
1309 % chapter = ALL CAPS bold, section = ALL CAPS regular, subsection = Title Case
1310 % bold, subsubsection = Title Case bold italic, as read from the manual.
1311 \titleformat{section}
1312     {\normalfont\large}{\thesection}{1ex}{\MakeUppercase}
1313 % subsection: bold (author writes Title Case).
1314 \titleformat{subsection}
1315     {\normalfont\bfseries}{\thesubsection}{1ex}{\}
1316 % subsubsection: bold italic.
1317 \titleformat{subsubsection}
1318     {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{\}
1319 % paragraph = quinary section: italic, the author capitalising only the first
1320 % word. 2.6 recommends going as deep as the quinary, and the annotation on the
1321 % 2025 manual (specs/ANOTACOES.md) describes exactly this shape.
1322 %
1323 % book makes \paragraph a RUN-IN heading, so the text continued on the same
1324 % line -- 2.6 requires "o texto deve ser iniciado em outra linha". The positive
1325 % "after" spacing in \titlespacing turns it into a displayed heading.
1326 \titleformat{paragraph}[hang]
1327     {\normalfont\itshape}{\theparagraph}{1ex}{\}
1328 \titlespacing*{paragraph}{Opt}{2.5ex plus 1ex minus .2ex}{1.5ex plus .2ex}
1329 % book defaults secnumdepth to 2, so \subsubsection and \paragraph came out
1330 % UNNUMBERED however carefully \titleformat named \thesubsubsection. 2.6 has
1331 % the numbering following the whole chain down to the quinary, and the manual's
1332 % own sumario is four levels deep, so tocdepth follows.
1333 \setcounter{secnumdepth}{4}
1334 \setcounter{tocdepth}{4}

```

```

1335 % --- 0 sumario (2.6 e 3.1.2.1.6) -----
1336 %
1337 % Duas exigencias o governam, e a segunda tem um exemplo obrigatorio.
1338 %
1339 % A 2.6: "no sumario, as secoes devem ser grafadas conforme apresentadas no
1340 % corpo do trabalho". A gradacao do corpo -- primaria em CAIXA ALTA negrito,
1341 % secundaria em CAIXA ALTA, terciaria em negrito, quaternaria em negrito
1342 % italico, quinaria em italico -- tem de reaparecer aqui. So as duas primeiras
1343 % reapareciam: da terciaria em diante o sumario saia em redondo, contradizendo
1344 % o corpo.
1345 %
1346 % A 3.1.2.1.6: os titulos "devem ser alinhados a esquerda... pela margem do
1347 % titulo do indicativo mais extenso", e "utilize como exemplo o sumario deste
1348 % trabalho". Medido no PDF do manual: o indicativo comeca em 3,00 cm nos cinco
1349 % niveis -- a margem -- e o titulo em 4,75 cm nos cinco. Uma coluna so.
1350 %
1351 % O \@dottedtocline do LaTeX faz o contrario. Recua progressivamente (3,00 /
1352 % 3,62 / 4,57 / 5,89 / 7,13 cm nesta classe) e da a cada nivel uma caixa de
1353 % indicativo de largura fixa. Isso empurrava o titulo quinario para 9,20 cm --
1354 % 6,2 dos 16 cm de mancha -- e, pior, a caixa fixa ESTOURA: com o indicativo
1355 % "10.10.10.10.10", que uma tese de dez capitulos alcanca, o numero entrava
1356 % 4,6 mm POR CIMA do titulo. A 2.6 quer o titulo "separado por um espaco" do
1357 % indicativo; ali nao havia espaco nenhum.
1358 %
1359 % A coluna abaixo e medida, nao arbitrada: \coppe@tocnumberline mede cada
1360 % indicativo enquanto compoe o sumario e guarda o maior; o valor vai para o
1361 % .aux e governa a passada seguinte. E literalmente "a margem do titulo do
1362 % indicativo mais extenso". Custa uma passada -- a mesma que as referencias
1363 % cruzadas ja custam.
1364 \newlength\coppe@tocnumwidth
1365 \newlength\coppe@tocnummax
1366 \newlength\coppe@tocnumsep
1367 % 0 espaco entre o indicativo e o titulo (2.6, "separado por um espaco").
1368 \setlength\coppe@tocnumsep{0.5em}
1369 % Piso: a coluna do proprio manual. Pela regra literal, um trabalho so com
1370 % capitulos teria uma coluna de meio centimetro e o titulo quase colado na
1371 % margem; o piso evita isso. A coluna nunca fica MENOR que o indicativo mais
1372 % extenso mais o espaco -- o piso so a impede de ficar estreita demais.
1373 \setlength\coppe@tocnumwidth{1.75cm}
1374 \setlength\coppe@tocnummax{\z@}
1375 \newcommand\coppe@tocnumberline[1]{%
1376   \settowidth\@tempdima{#1}%
1377   \ifdim\@tempdima>\coppe@tocnummax \global\coppe@tocnummax\@tempdima \fi
1378   \hb@xt@\coppe@tocnumwidth{#1\hfil}}
1379 % A largura medida na passada anterior chega pelo .aux, que o \document le
1380 % antes dos ganchos de \AtBeginDocument.
1381 \AtBeginDocument{%
1382   \ifundefined{coppe@tocnumsaved}{\fi}%
1383   \setlength\@tempdima{\coppe@tocnumsaved}%
1384   \addtolength\@tempdima\coppe@tocnumsep
1385   \ifdim\@tempdima>\coppe@tocnumwidth
1386     \setlength\coppe@tocnumwidth\@tempdima
1387   \fi}}
1388 \AtEndDocument{%

```

```

1389 \ifdim\coppe@tocnummax>\z@
1390 \if@filesw
1391 \immediate\write\@auxout{\string\gdef\string\coppe@tocnumsaved
1392 {\the\coppe@tocnummax}}%
1393 \fi
1394 \setlength\@tempdima\coppe@tocnummax
1395 \addtolength\@tempdima\coppe@tocnumsep
1396 \ifdim\@tempdima>\coppe@tocnumwidth
1397 \ClassWarning{coppe}{A coluna do sumario cresceu; rode LaTeX de novo}%
1398 \fi
1399 \fi}
1400 % Uma linha do sumario. #1 nivel, #2 fonte, #3 entrada, #4 folha. A entrada
1401 % traz \numberline{N} seguido do titulo; o \hskip-\leftskip puxa a primeira
1402 % linha de volta a margem, e o \leftskip deixa as linhas seguintes de um
1403 % titulo longo alinhadas pela coluna do titulo -- nunca debaixo do numero.
1404 % A folha herda a fonte do nivel, como no sumario do manual.
1405 \newcommand\coppe@tocline[4]{%
1406 \ifnum #1>\c@tocdepth\else
1407 \vskip\z@\@plus.2\p@
1408 {\leftskip\z@ \rightskip\@tocrmarg \parfillskip-\rightskip
1409 \parindent\z@ \interlinepenalty\@M
1410 \leavevmode
1411 #2%
1412 \advance\leftskip\coppe@tocnumwidth
1413 \null\nobreak\hskip-\leftskip
1414 #3\nobreak
1415 \leaders\hbox{$\m@th\mkern\@dotsep mu\hbox{.}\mkern\@dotsep mu$}%
1416 \hfill\nobreak
1417 \hb@xt@\@pnumwidth{\hss #4}\par}%
1418 \fi}
1419 % O pontilhado do manual e mais fechado que o do book; \@dotsep e a distancia
1420 % entre os pontos, em unidades matematicas.
1421 \renewcommand\@dotsep{2}
1422 % Os cinco niveis. O \MakeUppercase vai so nos dois primeiros porque so eles
1423 % saem em caixa alta no corpo; ele atravessa os algarismos sem toca-los e
1424 % preserva comandos robustos, inclusive os do hyperref.
1425 %
1426 % O nivel 1 nao ganha mais o \addvspace{1.0em} do book: o sumario do manual
1427 % tem espacamento uniforme, sem folga extra antes da secao primaria. As
1428 % penalidades ficam, para que uma entrada de capitulo nao seja abandonada
1429 % sozinha no pe da folha.
1430 \renewcommand*\l@chapter[2]{%
1431 \addpenalty{-\@highpenalty}%
1432 \coppe@tocline{0}{\bfseries}{\MakeUppercase{#1}}{#2}%
1433 \penalty\@highpenalty}
1434 \renewcommand*\l@section[2]{%
1435 \coppe@tocline{1}{\normalfont}{\MakeUppercase{#1}}{#2}}
1436 \renewcommand*\l@subsection[2]{%
1437 \coppe@tocline{2}{\normalfont\bfseries}{#1}{#2}}
1438 \renewcommand*\l@subsubsection[2]{%
1439 \coppe@tocline{3}{\normalfont\bfseries\itshape}{#1}{#2}}
1440 \renewcommand*\l@paragraph[2]{%
1441 \coppe@tocline{4}{\normalfont\itshape}{#1}{#2}}
1442 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT

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```

1443 %      "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1444 \RequirePackage{caption}
1445 % justification=centering makes MULTI-LINE captions centre like the one-line
1446 % ones. Without it, caption's singlelinecheck centres a caption that fits on one
1447 % line and justifies one that wraps, so in a document with mixed caption lengths
1448 % a couple of tables looked left-aligned while the rest looked centred -- which
1449 % is what a recent review noticed about tables 5.1 and 5.2. 2.10 does not
1450 % prescribe the alignment; it does not tolerate two alignments either.
1451 \captionsetup{font=footnotesize,labelsep=endash,justification=centering}
1452 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1453 % caption so it picks up the ABNT captionsetup above.
1454 \RequirePackage{subcaption}
1455 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1456 %      manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1457 %      at the bottom is mandatory). The float package's plaintop style forces the
1458 %      caption above the body for figure, table and the new quadro float. -----
1459 \RequirePackage{float}
1460 \floatstyle{plaintop}
1461 \restylefloat{figure}
1462 \restylefloat{table}
1463 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1464 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1465 % unlike a tabela (rules on top and bottom only).
1466 \providecommand{\quadroname}{\coppestring{frame}}
1467 \providecommand{\listquadroname}{\coppestring{listframe}}
1468 \newfloat{quadro}{htbp}{loq}[chapter]
1469 \floatname{quadro}{\quadroname}
1470 \providecommand{\quadroautorefname}{\quadroname}
1471 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1472 % (float's own \l@quadro lays the number/title out incorrectly).
1473 \let\l@quadro\l@figure
1474 \newcommand\listofquadros{%
1475   \chapter*{\listquadroname}%
1476   \coppe@listtoc{\listquadroname}%
1477   \@mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1478   \@starttoc{loq}}
1479 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1480 % float as a smaller, single-spaced, centred line; it does NOT advance any
1481 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1482 % the label is "Fonte" (pt) / "Source" (en).
1483 \newcommand{\cpsourcenam}{\coppestring{source}}
1484 \newcommand{\cpsource}[1]{%
1485   \par\addvspace{\smallskipamount}%
1486   {\footnotesize\singlespacing\centering\cpsourcenam: #1\par}}
1487 \let\cpfonte\cpsource
1488 \@ifundefined{source}{\let\source\cpsource}{}%
1489 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1490 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1491 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1492 % \floatstyle is NOT wrapped in a group here: \newfloat defines the environment
1493 % locally, so a group around it threw the new environment away as soon as it
1494 % closed and the document then failed with "Environment undefined". plaintop is
1495 % the class-wide style anyway, set once above and never changed, so leaving it
1496 % selected costs nothing.

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1497 % The new float is numbered WITHIN THE CHAPTER, like figure, table and quadro,
1498 % and its list entries are laid out by \l@figure -- float's own \l@<env> runs
1499 % the number into the title and drops the leaders, which is what the Lista de
1500 % Mapas of the example showed the first time.
1501 \newcommand{\newcoppefloat}[3]{%
1502   \floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}[chapter]%
1503   \floatname{#1}{#2}%
1504   \expandafter\let\csname l@#1\endcsname\l@figure
1505   \expandafter\providecommand\csname #1autorefname\endcsname{#2}%
1506   \expandafter\newcommand\csname listof#1s\endcsname{%
1507     \chapter*{#3}\coppe@listtoc{#3}%
1508     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1509     \@starttoc{lo#1}}}%
1510 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1511 %       (filete) from the left margin; footnotesize is the book default. -----
1512 \let\coppe@orig@footnotetext\@footnotetext
1513 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1514 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1515 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1516 %       (manual2024 2.6 / proposta P24). -----
1517 \RequirePackage{enumitem}
1518 \newlist{alineas}{enumerate}{2}
1519 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1520 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
1521 % --- Page numbering: arabic in the top OUTER corner on textual pages
1522 %       (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1523 %       \frontmatter (unnumbered by default; roman with the numeraisromanos
1524 %       class option). -----
1525 % Every list goes through \@starttoc, which allocates a stream the first time
1526 % it sees an extension. That is where the ceiling is hit in practice, so that
1527 % is where the class checks -- one \ifnum before the allocation, and an error
1528 % that names the three ways out instead of the kernel's bare "No room".
1529 %
1530 % \count17 is the write allocation counter, and 15 is the last usable stream.
1531 % Under LuaTeX there are 128 and the counter is managed differently, so the
1532 % check is skipped there; the same is true when morewrites is in charge.
1533 % The wrap waits for \begin{document} on purpose: hyperref redefines
1534 % \@starttoc as well, and it is loaded at the very end of this class, so a
1535 % \let taken here would be thrown away.
1536 \newcommand\coppe@checkwrites[1]{%
1537   \ifdefined\directlua\else
1538     \if@coppemorewrites\else
1539       \ifundefined{tf@#1}{%
1540         \ifnum\count17>14
1541           \ClassError{coppe}{Acabaram os 16 fluxos de escrita do TeX}{%
1542             Este trabalho pede mais listas do que o pdfTeX consegue manter
1543             abertas ao mesmo tempo.\MessageBreak
1544             Tres saidas, nesta ordem de preferencia:\MessageBreak
1545             (1) compile com lualatex, que tem 128 fluxos;\MessageBreak
1546             (2) passe a opcao 'morewrites' a classe coppe;\MessageBreak
1547             (3) remova uma lista que o trabalho nao usa.\MessageBreak
1548             A lista que faltou agora foi a de extensao '#1', mas o problema
1549             nao e dela: e a soma de todas.}%
1550           \fi}{}%

```

```

1551     \fi
1552 \fi}
1553 % The wrapping itself lives in a macro of its own rather than inside the hook:
1554 % \AtBeginDocument stores its argument by expansion, and a parameter written
1555 % there does not survive it -- the document died with "Parameters must be
1556 % numbered consecutively" at \begin{document}. One level of nesting, one
1557 % doubling, and the hook only has to call it.
1558 \newcommand\coppe@wrapstarttoc{%
1559   \let\coppe@orig@starttoc\@starttoc
1560   % \numberline mede o indicativo (ver \coppe@tocnumberline) apenas no sumario:
1561   % a lista de figuras e a de tabelas numeram ilustracoes, nao secoes, e nao tem
1562   % por que compartilhar a coluna.
1563   \def\@starttoc##1{%
1564     \coppe@checkwrites{##1}%
1565     \begingroup
1566       \def\coppe@@ext{##1}\def\coppe@@toc{toc}%
1567       \ifx\coppe@@ext\coppe@@toc \let\numberline\coppe@tocnumberline \fi
1568       \coppe@orig@starttoc{##1}%
1569     \endgroup}}
1570 \AtBeginDocument{\coppe@wrapstarttoc}
1571 \RequirePackage{fancyhdr}
1572 % \headheight is set to 15pt by geometry up above (see the geometry call),
1573 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1574 % 2.2(b) lists what is set in "fonte menor e uniforme": long quotations,
1575 % footnotes, PAGINATION, and the captions of illustrations and tables. The
1576 % folio was the one item on that list still coming out at the body size of 12pt.
1577 % It is \footnotesize now, which is the same smaller size the other three
1578 % already use -- "uniforme" is part of the requirement, not just "menor".
1579 % Pre-textual lists reach the sumario only under the listasnosumario option
1580 % (see the option's declaration for the reason). Post-textual material calls
1581 % \addcontentsline directly and is unaffected.
1582 \newcommand\coppe@listtoc[1]{%
1583   \if@coppe@listastoc\addcontentsline{toc}{chapter}{#1}\fi}
1584 % \coppe@glosspre marks the two PRE-textual uses of the glossary environment --
1585 % the list of abbreviations and the list of symbols -- apart from the genuine
1586 % post-textual Glossario, which shares the same environment but must appear in
1587 % the sumario.
1588 \newif\if@coppeglosspre \@coppeglossprefalse
1589 \newcommand\coppe@folio{{\footnotesize\thepage}}
1590 \fancypagestyle{coppe}{{\fancyhf{}}\fancyhead[R0,LE]{\coppe@folio}}%
1591 \renewcommand{\headrulewidth}{0pt}}
1592 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1593 % number may leak onto them (e.g. the verso right after the cover).
1594 \renewcommand{\cleardoublepage}{%
1595   \clearpage
1596   \if@twoside
1597     \ifodd\c@page\else
1598       \hbox{} \thispagestyle{empty} \newpage
1599     \if@twocolumn\hbox{} \newpage \fi
1600   \fi
1601   \fi}
1602 \def\CoppeTeX{{\rm C}\kern-.05em{\sc o}\kern-.025em p\kern-.025em
1603 p\kern-.025em e}\kern-.08em
1604 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}

```



```

1605 \newboolean{maledoc}
1606 \setboolean{maledoc}{false}
1607 % maledoc diz apenas que o substantivo do tipo de documento e masculino
1608 % ("Exame de Qualificacao ... apresentado", "SUBMETIDO"): serve a concordancia,
1609 % e nada mais. Quem NAO vai a deposito -- e portanto nao tem ficha
1610 % catalografica nem folha da Coleta CAPES -- e o exame de qualificacao, que e
1611 % outra coisa. As duas condicoes coincidem hoje por acaso, e por isso ganham
1612 % cada uma o seu booleano.
1613 \newif\if@coppeexame \@coppeexamefalse
1614 %
1615 % Class options.
1616 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1617 % fixed language slots:
1618 % * main      -- the language of the body, captions, and the main abstract.
1619 %              Set by a class option: brazilian (default), english,
1620 %              spanish, french, italian, or any user-supplied language
1621 %              that ships a coppe-lang-<name>.def file.
1622 % * foreign   -- the language used by the foreignabstract environment and
1623 %              by \foreign@... registers. Defaults to english (or to
1624 %              brazilian when main=english) for back-compat.
1625 % * third     -- optional, loaded on demand with \usecoppelanguage{<name>}
1626 %              in the preamble (for citations or quotations in a third
1627 %              language). Has no class option of its own.
1628 % Note: the legacy \if@english boolean (defined and toggled here in
1629 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1630 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.ltx)
1631 % tests it. Code that needs to branch on the main language should test
1632 % \ifx\coppe@mainlang\coppe@englishstr instead.
1633 \def\coppe@mainlang{brazilian}
1634 \def\coppe@foreignlang{english}
1635 \DeclareOption{brazilian}{%
1636   \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1637 \DeclareOption{english}{%
1638   \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1639 \DeclareOption{spanish}{%
1640   \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1641 \DeclareOption{french}{%
1642   \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1643 \DeclareOption{italian}{%
1644   \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1645 \DeclareOption{msc}{%
1646   \newcommand{\@degree}{M.Sc.}
1647   \newcommand{\@degreename}{Mestrado}
1648   \newcommand{\local@degname}{Mestre}
1649   \newcommand{\foreign@degname}{Master}
1650   \newcommand\local@doctype{Disserta{c c}{\~ a}o}
1651   \newcommand\foreign@doctype{Dissertation}
1652 }
1653 \DeclareOption{dsccexam}{%
1654   \newcommand{\@degree}{D.Sc.}
1655   \newcommand{\@degreename}{Doutorado}
1656   \newcommand{\local@degname}{Doutor}
1657   \newcommand{\foreign@degname}{Doctor}
1658   \setboolean{maledoc}{true}\@coppeexamtrue

```

```

1659 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1660 \newcommand\foreign@doctype{Qualifying Exam}
1661 }
1662 \DeclareOption{mscexam}{%
1663 \newcommand{\@degree}{M.Sc.}
1664 \newcommand{\@degreename}{Mestrado}
1665 \newcommand{\local@degname}{Mestre}
1666 \newcommand{\foreign@degname}{Master}
1667 \setboolean{maledoc}{true}\@coppeexamtrue
1668 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1669 \newcommand\foreign@doctype{Qualifying Exam}
1670 }
1671 \DeclareOption{dsc}{%
1672 \newcommand{\@degree}{D.Sc.}
1673 \newcommand{\@degreename}{Doutorado}
1674 \newcommand{\local@degname}{Doutor}
1675 \newcommand{\foreign@degname}{Doctor}
1676 \newcommand\local@doctype{Tese}
1677 \newcommand\foreign@doctype{Thesis}
1678 }
1679 \newif\if@coppenumeric\@coppenumericfalse
1680 \DeclareOption{numbers}{\@coppenumerictrue}
1681 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1682 % pre-textual pages are counted but not numbered (manual2024 2.7).
1683 \newif\if@copperoman\@copperomanfalse
1684 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Default line spacing is one-and-a-half; the `doublespacing` option switches to double spacing. Since `setspace` is loaded at the end of the class (after `hyperref`), the actual `\onehalfspacing`/`\doublespacing` call is deferred: the option only flips a flag, which the late-load block reads.

```

1685 \newif\if@coppedoublespace\@coppedoublespacefalse
1686 \DeclareOption{doublespacing}{\@coppedoublespacetrue}
1687 % Two-sided layout is opt-in since v4.1: the 2026 manual dropped the verso
1688 % margins and the deposit is digital (CEPG 246/2023). Given \texttt{twoside},
1689 % the mirrored 3cm inner / 2cm outer layout is restored after \ProcessOptions.
1690 % With no \fichacatalografica file given, the folha adicional leaves a framed
1691 % blank where the ficha goes. The rascunhoficha option fills that blank with the
1692 % class's own LaTeX-composed ficha instead -- useful while writing, never valid
1693 % for deposit: the official ficha comes from the SiBI generator.
1694 % PDF/A output. Opt-in for now: 2.2(d) of the 2026 manual requires the deposited
1695 % file to be PDF/A, but the constraints (embedded fonts, colour profile, no
1696 % encryption, restricted transparency) can trip up a document that pulls in
1697 % arbitrary images or packages, and breaking existing theses to enforce it would
1698 % be worse than letting the author switch it on when depositing.
1699 \newif\if@coppepdfa\@coppepdfafalse
1700 \DeclareOption{pdfa}{\@coppepdfatrue}
1701 % A opcao 'assinaturas' punha uma regua para assinatura a mao acima do nome de
1702 % cada membro da banca. Com a entrega so digital (Resolucao CEPG n. 246/2023)
1703 % nao ha o que assinar a mao, e a v4.1 tirou as reguas. A opcao continua aceita,
1704 % sem efeito, para nao quebrar documento escrito antes disso.
1705 % Quem examina o trabalho e a BANCA EXAMINADORA, e a folha de aprovacao lista
1706 % quem examinou. O orientador nao e, por definicao, examinador: ele conduziu o
1707 % trabalho. Por isso a folha lista, por padrao, so os examinadores.

```

```

1708 %
1709 % Ha Programas em que o orientador integra a banca e assina a aprovacao. Para
1710 % esses existe 'orientadorexamina': com ela, orientadores e coorientadores
1711 % voltam para a lista, na frente dos examinadores, na ordem que a 3.1.2.1.3(e)
1712 % prescreve para o presidente da banca.
1713 %
1714 % Ate a v4.1 nao havia escolha: os orientadores saiam sempre, e um Programa que
1715 % nao os quisesse ali tinha de nao os declarar -- o que os tirava tambem da capa
1716 % e da folha de rosto, onde eles tem de estar.
1717 \newif\if@coppeorientexam \@coppeorientexamfalse
1718 \DeclareOption{orientadorexamina}{\@coppeorientexamtrue}
1719 % A classe carrega o hyperref e colore os links. Quem vai IMPRIMIR, ou quem
1720 % simplesmente prefere a pagina sobria, pedia isso a mao com \hypersetup no
1721 % preambulo -- que funciona, mas exige saber que a classe carrega o hyperref e
1722 % em que ordem. 'semlinks' faz o mesmo sem exigir nada disso. Os links continuam
1723 % existindo e clicaveis; some so a cor e a moldura.
1724 \newif\if@coppesemlinks \@coppesemlinksfalse
1725 \DeclareOption{semlinks}{\@coppesemlinkstrue}
1726 \DeclareOption{assinaturas}{\ClassWarningNoLine{coppe}{A opcao 'assinaturas'
1727   nao faz mais nada: a folha de aprovacao nao tem mais linhas de assinatura,
1728   porque a entrega e digital desde a Resolucao CEPG n. 246/2023}}
1729 % As paginas de resumo listam os ORIENTADORES abaixo do titulo. O coorientador
1730 % nao aparece ali, e isso e tradicao da COPPE, nao exigencia do manual: a
1731 % 3.1.2.1.4 descreve o resumo e nao lista quem assina a orientacao. Como ha
1732 % Programas que preferem ve-lo, a opcao acrescenta os coorientadores nas tres
1733 % paginas de resumo, no mesmo formato da folha de rosto. Sem a opcao nada muda.
1734 \newif\if@coppecoadvresumo \@coppecoadvresumofalse
1735 \DeclareOption{coorientador}{\@coppecoadvresumotrue}
1736 % 3.1.2.1.6 says the sumario enumerates the divisions "na mesma ordem e grafia
1737 % em que a materia nele se sucede" and adds "inclusive os elementos
1738 % pos-textuais" -- and then settles the question by pointing at its own sumario
1739 % as the model. That sumario opens at "1 INTRODUCAO": the lists of figures, of
1740 % tables, of abbreviations and of symbols come BEFORE the sumario and are not
1741 % listed in it, which is also the rule of NBR 6027 for pre-textual elements.
1742 % The class was adding all of them, so every thesis carried six entries the
1743 % manual's own example does not have. They come out. Post-textual elements --
1744 % Referencias, Glossario, Apendices, Anexos, Indice -- stay, as required.
1745 %
1746 % Programas that want the old behaviour ask for it by name.
1747 \newif\if@coppelistastoc \@coppelistastocfalse
1748 \DeclareOption{listasnosumario}{\@coppelistastoctrue}
1749 \newif\if@copperascunho \@copperascunhofalse
1750 \DeclareOption{rascunhoficha}{\@copperascunhotrue}
1751 \newif\if@coppetwoaside \@coppetwoasidefalse
1752 \DeclareOption{twoaside}{\@coppetwoasidetrue}
1753 \DeclareOption{oneside}{\@coppetwoasidefalse}
1754 % --- Fluxos de escrita -----
1755 % TeX has SIXTEEN \write streams and no more. Every list costs one for the
1756 % whole run: the .toc, the .lof, the .lot, and one for each list this class
1757 % adds (.loq for quadros, .lol for programas, .loa for algoritmos), plus the
1758 % .abx and .syx behind the lists of abbreviations and of symbols, the .idx of
1759 % the index, hyperref's .out, biblalex's .bcf and the kernel's .aux and friends.
1760 % A work that asks for all of them sits exactly at the ceiling.
1761 %

```

```

1762 % The kernel's failure in that situation -- "No room for a new \write" -- names
1763 % neither the cause nor the way out, and lands on whichever list happens to be
1764 % requested last, which is usually \tableofcontents and has nothing to do with
1765 % it. The class therefore checks before the allocation and says what is
1766 % happening (see \coppe@starttoc below).
1767 %
1768 % There are three ways out and the class chooses none of them for the author:
1769 %
1770 % * compile with LuaLaTeX. It has 128 streams; nothing else has to change,
1771 % and the class supports it.
1772 % * let the class load 'morewrites', which multiplexes the streams so the
1773 % ceiling disappears under pdfLaTeX.
1774 % * drop a list the work does not use. Each one returns a stream.
1775 %
1776 % The DEFAULT is the second one, and it is on. An error telling the author to
1777 % pass an option is advice for someone who already knows: a student two days
1778 % from the deposit reads "No room for a new \write" and concludes the class is
1779 % broken. So under pdfTeX the class loads morewrites by itself.
1780 %
1781 % Three things keep that from becoming a burden:
1782 %
1783 % * it is loaded ONLY IF INSTALLED. Missing, the class carries on without it
1784 % and warns; morewrites does not become a hard dependency of a work that
1785 % would never have reached the ceiling anyway.
1786 % * 'semmorewrites' turns it off, for whoever hits a conflict. morewrites
1787 % patches \newwrite, \openout and \write and buffers what is written until
1788 % shipout, so a package that writes a file and reads it back in the SAME run
1789 % can misbehave. That is the real cost of this default, and it is why the
1790 % escape exists.
1791 % * 'morewrites' still exists and now means "I insist": with it, a missing
1792 % package is an error rather than a warning.
1793 %
1794 % Under LuaTeX none of this applies and nothing is loaded.
1795 \newif\if@coppemorewrites \@coppemorewritetrue
1796 \newif\if@coppemwforced \@coppemwforcedfalse
1797 \DeclareOption{morewrites}{\@coppemorewritetrue\@coppemwforcedtrue}
1798 \DeclareOption{semmorewrites}{\@coppemorewritetrue\@coppemwforcedfalse}
1799 % A 2.2(b) fixa a cor, o corpo 12 e o corpo menor uniforme dos quatro itens
1800 % listados nela, e nao diz uma palavra sobre familia de fonte -- nem a Norma da
1801 % COPPE. A escolha e do autor. Desde a v4.1 o padrao da classe e a SEM SERIFA,
1802 % que e como o proprio manual do SiBI, que fixa a regra, esta composto;
1803 % 'comserifa' volta para a serifada, que e a do LaTeX. 'semserifa' continua
1804 % aceita, agora sem efeito, para nao quebrar documento escrito antes da v4.1.
1805 \newif\if@coppesansserif \@coppesansseriftrue
1806 \DeclareOption{semserifa}{\@coppesansseriftrue}
1807 \DeclareOption{comserifa}{\@coppesansseriffalse}
1808 \ProcessOptions\relax
1809 % A troca da familia padrao vai aqui porque so agora se sabe se a opcao foi
1810 % pedida; o lmodern ja esta carregado la em cima e traz as duas familias, a
1811 % serifada e a sem serifa, ambas vetoriais -- que e o que o PDF/A exige.
1812 % Quem quiser uma sem serifa DIFERENTE (Helvetica, por exemplo) carrega o
1813 % pacote da fonte no preambulo e refaz \familydefault ali; o padrao existe
1814 % para o caso comum, em que ninguem quer escolher fonte.
1815 \if@coppesansserif

```

```

1816 \renewcommand{\familydefault}{\sfdefault}%
1817 \fi
1818 % morewrites has to come before anything else allocates, so it goes first
1819 % thing after the options are read.
1820 \if@coppemorewrites
1821 \ifdefined\directlua
1822 % Under LuaTeX there is nothing to solve: 128 streams already. The flag is
1823 % cleared too, so the \@starttoc guard further down knows the ceiling is not
1824 % in play here either.
1825 \ClassInfo{coppe}{morewrites dispensado: o LuaTeX ja tem 128 fluxos}%
1826 \@coppemorewritesfalse
1827 \else
1828 \IfFileExists{morewrites.sty}
1829 {\RequirePackage{morewrites}}
1830 {\if@coppemwforced
1831 \ClassError{coppe}{Opcao 'morewrites' pedida, mas morewrites.sty nao
1832 esta instalado}{Instale o pacote morewrites, ou compile com lualatex,
1833 que nao precisa dele.}%
1834 \else
1835 \ClassWarningNoLine{coppe}{O pacote morewrites nao esta instalado.\MessageBreak
1836 Sem ele o pdfTeX so aguenta 16 fluxos de escrita, e um trabalho\MessageBreak
1837 com todas as listas passa disso. Se a compilacao parar por falta\MessageBreak
1838 de fluxo, instale morewrites ou compile com lualatex}%
1839 \@coppemorewritesfalse
1840 \fi}%
1841 \fi
1842 \fi
1843 % \LoadClass and geometry ran before \ProcessOptions (see above), so the
1844 % two-sided layout can only be applied here. geometry reads left/right as
1845 % inner/outer once twoside is on, so the 3cm/2cm pair set above becomes the
1846 % mirrored gutter with no further arithmetic. \@twosidetrue is what book's own
1847 % twoside option sets, and is what \cleardoublepage and fancyhdr test.
1848 % In one-sided mode fancyhdr treats every page as odd, so the \fancyhead[R0,LE]
1849 % declarations elsewhere already put the folio in the top right corner of every
1850 % folha, as 2.7 requires.
1851 \if@coppetwoside
1852 \@twosidetrue
1853 \geometry{twoside}%
1854 \fi
1855 % Built-in language names (used by the loader and the dispatcher below).
1856 \def\coppe@@brazilianstr{brazilian}
1857 \def\coppe@@englishstr{english}
1858 % Load Babel with the foreign language first and the main language last
1859 % (Babel treats the LAST option in the list as the main language). With
1860 % pt-main this produces [english,brazilian]{babel} and with en-main
1861 % [brazilian,english] -- identical to the historical \if@english calls.
1862 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1863 % Some Babel modules change LaTeX beyond translating captions, and one of
1864 % those changes breaks the norm. With Spanish as the MAIN language,
1865 % babel-spanish redefines \numberline to put a period after the section
1866 % number, so the sumario reads "2.1. SECCION SECUNDARIA" while the section
1867 % heading itself -- which the class formats -- reads "2.1 SECCION SECUNDARIA".
1868 % The 2.6 separates the indicative from the title by a space (NBR 6024) and
1869 % the 3.1.2.1.6 wants the sumario to reproduce the text; the period fails

```

```

1870 % both. 'es-nosectiondot' turns off that single adjustment and leaves the rest
1871 % of babel-spanish (quoting, shorthands, roman numerals, captions) alone.
1872 % \coppe@babelextra@<lang> is the general hook: it applies to the MAIN
1873 % language only, which is where these layout modules act, and it is kept out
1874 % of \coppe@babelopts because the language-pack loader below iterates that
1875 % list and would look for coppe-lang-es-nosectiondot.def.
1876 \@namedef{coppe@babelextra@spanish}{,es-nosectiondot}
1877 \let\coppe@babelload\coppe@babelopts
1878 \@ifundefined{coppe@babelextra@\coppe@mainlang}{%
1879   \edef\coppe@babelload{\coppe@babelload\@nameuse{coppe@babelextra\coppe@mainlang}}
1880 \expandafter\RequirePackage\expandafter[\coppe@babelload]{babel}
1881 % --- Multilingual string dispatcher -----
1882 % \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1883 % language <lang>. The three retrievers below look strings up either at the
1884 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1885 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1886 \providecommand{\copperdefstring}[3]{%
1887   \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1888 \providecommand{\coppestring}[1]{\@nameuse{coppe@str\language @#1}}
1889 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str\coppe@mainlang @#1}}
1890 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str\coppe@foreignlang @#1}}
1891 % Load an extra language pack at any point in the preamble (third-language
1892 % slot for citations or quotations beyond the main+foreign pair).
1893 \providecommand{\usecoppelanguage}[1]{%
1894   \def\coppe@@cur{#1}%
1895   \ifx\coppe@@cur\coppe@@brazilianstr\else
1896     \ifx\coppe@@cur\coppe@@englishstr\else
1897       \InputIfFileExists{coppe-lang-#1.def}{}
1898       {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1899     \fi\fi}
1900 % \titlein{<lang>}{<text>}: record the title in any language. Will be
1901 % consulted by the cover/folha renderers once they are migrated in step 2.
1902 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1903 % still drive every visible site, so behavior for pt/en is unchanged.
1904 \providecommand{\titlein}[2]{%
1905   \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
1906 % --- Built-in string tables: brazilian + english. The values mirror the
1907 % literals currently scattered through the class; step-1 only populates the
1908 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1909 \copperdefstring{brazilian}{appendix}{Ap\^endice}
1910 \copperdefstring{english}{appendix}{Appendix}
1911 \copperdefstring{brazilian}{annex}{Anexo}
1912 \copperdefstring{english}{annex}{Annex}
1913 \copperdefstring{brazilian}{frame}{Quadro}
1914 \copperdefstring{english}{frame}{Frame}
1915 \copperdefstring{brazilian}{listframe}{Lista de Quadros}
1916 \copperdefstring{english}{listframe}{List of Frames}
1917 \copperdefstring{brazilian}{source}{Fonte}
1918 \copperdefstring{english}{source}{Source}
1919 \copperdefstring{brazilian}{program}{Programa}
1920 \copperdefstring{english}{program}{Program}
1921 \copperdefstring{brazilian}{listprogram}{Lista de Programas}
1922 \copperdefstring{english}{listprogram}{List of Programs}
1923 \copperdefstring{brazilian}{references}{Refer\^encias}

```

1924 \copperdefstring{english} {references} {References}
1925 \copperdefstring{brazilian}{in:} {em}
1926 \copperdefstring{english} {in:} {in}
1927 \copperdefstring{brazilian}{art} {Ilustra\c{c}\~ao}
1928 \copperdefstring{english} {art} {Art}
1929 \copperdefstring{brazilian}{platform} {Plataforma}
1930 \copperdefstring{english} {platform} {Platform}
1931 \copperdefstring{brazilian}{directedby} {Dire\c{c}\~ao de}
1932 \copperdefstring{english} {directedby} {Directed by}
1933 \copperdefstring{brazilian}{producedby} {Produ\c{c}\~ao de}
1934 \copperdefstring{english} {producedby} {Produced by}
1935 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
1936 \copperdefstring{english} {listabbreviation}{List of Abbreviations and Acronyms}
1937 \copperdefstring{brazilian}{listsymbol} {Lista de S{\~ i}mbolos}
1938 \copperdefstring{english} {listsymbol} {List of Symbols}
1939 \copperdefstring{brazilian}{glossary} {Gloss{\~ a}rio}
1940 \copperdefstring{english} {glossary} {Glossary}
1941 \copperdefstring{brazilian}{advisor} {Orientador}
1942 \copperdefstring{english} {advisor} {Advisor}
1943 \copperdefstring{brazilian}{advisors} {Orientadores}
1944 \copperdefstring{english} {advisors} {Advisors}
1945 \copperdefstring{brazilian}{dept} {Programa}
1946 \copperdefstring{english} {dept} {Department}
1947 \copperdefstring{brazilian}{approvedmale} {Aprovado por}
1948 \copperdefstring{brazilian}{approvedfemale}{Aprovada por}
1949 \copperdefstring{brazilian}{approvedonmale} {Aprovado em}
1950 \copperdefstring{brazilian}{approvedonfemale}{Aprovada em}
1951 \copperdefstring{brazilian}{keywordsname} {Palavras-chave}
1952 \copperdefstring{brazilian}{volumename} {Volume}
1953 \copperdefstring{brazilian}{coadvisor} {Coorientador}
1954 \copperdefstring{brazilian}{coadvisors} {Coorientadores}
1955 \copperdefstring{brazilian}{ofname} {de}
1956 \copperdefstring{brazilian}{researchline}{Linha de pesquisa}
1957 \copperdefstring{brazilian}{tbddate} {a ser determinada}
1958 \copperdefstring{english} {approvedmale} {Approved by}
1959 \copperdefstring{english} {approvedfemale}{Approved by}
1960 \copperdefstring{english} {approvedonmale} {Approved on}
1961 \copperdefstring{english} {approvedonfemale}{Approved on}
1962 \copperdefstring{english} {keywordsname} {Keywords}
1963 \copperdefstring{english} {volumename} {Volume}
1964 \copperdefstring{english} {coadvisor} {Co-advisor}
1965 \copperdefstring{english} {coadvisors} {Co-advisors}
1966 \copperdefstring{english} {ofname} {of}
1967 \copperdefstring{english} {researchline}{Research line}
1968 \copperdefstring{english} {tbddate} {to be determined}
1969 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1970 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1971 \copperdefstring{english} {universityname}{Universidade Federal do Rio de Janeiro}
1972 \copperdefstring{brazilian}{cityname} {Rio de Janeiro}
1973 \copperdefstring{english} {cityname} {Rio de Janeiro}
1974 \copperdefstring{brazilian}{statename} {RJ}
1975 \copperdefstring{english} {statename} {RJ}
1976 \copperdefstring{brazilian}{countryname} {Brasil}
1977 \copperdefstring{english} {countryname} {Brasil}

```

1978 % Month names.
1979 \copperdefstring{brazilian}{monthname1} {Janeiro}
1980 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1981 \copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
1982 \copperdefstring{brazilian}{monthname4} {Abril}
1983 \copperdefstring{brazilian}{monthname5} {Maio}
1984 \copperdefstring{brazilian}{monthname6} {Junho}
1985 \copperdefstring{brazilian}{monthname7} {Julho}
1986 \copperdefstring{brazilian}{monthname8} {Agosto}
1987 \copperdefstring{brazilian}{monthname9} {Setembro}
1988 \copperdefstring{brazilian}{monthname10}{Outubro}
1989 \copperdefstring{brazilian}{monthname11}{Novembro}
1990 \copperdefstring{brazilian}{monthname12}{Dezembro}
1991 \copperdefstring{english}{monthname1} {January}
1992 \copperdefstring{english}{monthname2} {February}
1993 \copperdefstring{english}{monthname3} {March}
1994 \copperdefstring{english}{monthname4} {April}
1995 \copperdefstring{english}{monthname5} {May}
1996 \copperdefstring{english}{monthname6} {June}
1997 \copperdefstring{english}{monthname7} {July}
1998 \copperdefstring{english}{monthname8} {August}
1999 \copperdefstring{english}{monthname9} {September}
2000 \copperdefstring{english}{monthname10}{October}
2001 \copperdefstring{english}{monthname11}{November}
2002 \copperdefstring{english}{monthname12}{December}
2003 % algorithm2e package option for this language (empty -> english default).
2004 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
2005 \copperdefstring{english} {algorithm2eopt}{}
2006 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
2007 \copperdefstring{brazilian}{babelname}{brazilian}
2008 \copperdefstring{english} {babelname}{english}
2009 % Auto-load language packs for any non-built-in language listed in babelopts.
2010 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
2011 % (defined just above) and need no extra file; others (spanish/french/italian
2012 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
2013 % must already be defined when these run, so the loader sits AFTER the tables.
2014 \@for\coppe@@lang:=\coppe@babelopts\do{
2015   \edef\coppe@@cur{\coppe@@lang}%
2016   \ifx\coppe@@cur\coppe@@brazilianstr\else
2017     \ifx\coppe@@cur\coppe@@englishstr\else
2018       \InputIfFileExists{coppe-lang-\coppe@@cur.def}
2019       {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
2020       {\ClassError{coppe}
2021         {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
2022         {Install it next to coppe.cls, or pick a built-in language.}}}%
2023   \fi\fi}
2024 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
2025 % Portuguese alias for short, inline quotations (proposta P3).
2026 \RequirePackage{csquotes}
2027 \newcommand{\citacao}[1]{\enquote{#1}}
2028 % --- Code listings & algorithms (TOD03 "tratar listagens"): the listings
2029 % package gives the "Programa" listing (caption on top, ABNT R46) with
2030 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
2031 % the language styles below are based on the user's listagens.tex. -----

```



```

2032 \RequirePackage{xcolor}
2033 \RequirePackage{listings}
2034 \RequirePackage{fancyvrb}
2035 \DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
2036 \DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
2037 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
2038   literate=%
2039   {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{e}}1%
2040   {\~}{\~{e}}1 {\~}{\~{e}}1 {\~}{\~{i}}1 {\~}{\~{i}}1 {\~}{\~{o}}1%
2041   {\~}{\~{o}}1 {\~}{\~{o}}1 {\~}{\~{u}}1 {\~}{\~{u}}1 {\~}{\~{c}}1%
2042   {\~}{\~{c}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{E}}1%
2043   {\~}{\~{E}}1 {\~}{\~{I}}1 {\~}{\~{O}}1 {\~}{\~{O}}1 {\~}{\~{O}}1%
2044   {\~}{\~{U}}1%
2045 }
2046 % Babel-aware listing names, set after listings' own babel setup.
2047 \AtBeginDocument{%
2048   \renewcommand\lstlistingname{\coppestring{program}}%
2049   \renewcommand\lstlistlistingname{\coppestring{listprogram}}%
2050 % \listofprogramas in the coppe list style (heading + ToC + running head).
2051 \newcommand\listofprogramas{%
2052   \chapter*{\lstlistlistingname}%
2053   \coppe@listtoc{\lstlistlistingname}%
2054   \@mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
2055   \@starttoc{lol}}
2056 % Per-language styles + environments (python, java, xml, html, prolog).
2057 % The red hook that marks a wrapped line is built ONCE, into a box, and the
2058 % styles below insert the box. Writing the colour inline -- postbreak=\mbox{%
2059 % \textcolor{red}{\hookrightarrow}\space} -- worked for years and broke the
2060 % moment the pdfa option was switched on: pdfx puts xcolor into a converting
2061 % mode (\selectcolormodel plus \convertcolorsU/D), and a colour command run
2062 % from inside listings' line-breaking machinery then fails with "Missing
2063 % number, treated as zero" at the first wrapped line. Every listing in the
2064 % document dies, so the class's own example could not be compiled as PDF/A at
2065 % all. A box is immune: it is built here, in ordinary horizontal mode where
2066 % xcolor works normally, and listings only ever copies it.
2067 %
2068 % The \sbox waits for \begin{document} because xcolor's colours are not
2069 % defined while the class is still loading.
2070 \newsavebox\coppe@lstarrow
2071 \AtBeginDocument{\sbox\coppe@lstarrow{\textcolor{red}{\hookrightarrow}\space}}
2072 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
2073   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
2074   morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
2075   emph={MyClass,__init__},emphstyle=\ttb\color{rgb}{0.6,0,0},%
2076   stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
2077   extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
2078   escapechar=',postbreak=\usebox\coppe@lstarrow}}
2079 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}%
2080   \csname\@lst @SetFirstNumber\endcsname]{\csname\@lst @SaveFirstNumber\endcsname}
2081 \newcommand\pythoninline[1][\pythonstyle\lstinline!#1!}%
2082 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}]%
2083 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
2084   numberstyle=\small\ttfamily,numbersep=0.5em,%
2085   morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%

```

```

2086 keywordstyle=\ttb\color[rgb]{0,0,0.5},%
2087 emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
2088 emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
2089 showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
2090 postbreak=\usebox\coppe@lstarrow}}
2091 \lstnewenvironment{xml}[1][\{\xmlstyle\lstset{#1}\}}{}
2092 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
2093 numberstyle=\small\ttfamily,numbersep=0.5em,%
2094 keywordstyle=\ttb\color[rgb]{0,0,0.5},emphstyle=\ttb\color[rgb]{0.6,0,0},%
2095 stringstyle=\color[rgb]{0,0.5,0},frame=tb,showstringspaces=false,%
2096 extendedchars=true,inputencoding=utf8,breaklines=true,%
2097 postbreak=\usebox\coppe@lstarrow}}
2098 \lstnewenvironment{html}[1][\{\htmlstyle\lstset{#1}\}}{}
2099 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
2100 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
2101 keywordstyle=\ttb\color[rgb]{0,0,0.5},emph={MyClass,__init__},%
2102 emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
2103 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
2104 breaklines=true,escapechar=',%
2105 postbreak=\usebox\coppe@lstarrow}}
2106 \lstnewenvironment{prolog}[1][\{\prologstyle\lstset{#1}%
2107 \cscname\@lst @SetFirstNumber\endcscname}{\cscname\@lst @SaveFirstNumber\endcscname}]
2108 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
2109 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
2110 morekeywords={self,as},keywordstyle=\ttb\color[rgb]{0,0,0.5},%
2111 emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
2112 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
2113 breaklines=true,escapechar=',%
2114 postbreak=\usebox\coppe@lstarrow}}
2115 \lstnewenvironment{java}[1][\{\javastyle\lstset{#1}%
2116 \cscname\@lst @SetFirstNumber\endcscname}{\cscname\@lst @SaveFirstNumber\endcscname}]
2117 \newcommand\javainline[1]{\{\javastyle\lstinline!#1!}}
2118 \newcommand\javaexternal[2][\{\javastyle\lstinputlisting[1]{#2}\}}
2119 % fancyvrb's Verbatim environment is available for program output; the
2120 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
2121 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
2122 % example,
2123 % \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
2124 % ...
2125 % \end{Verbatim}
2126 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
2127 \renewcommand\theFancyVerbLine{\small\ttfamily\arabic{FancyVerbLine}}
2128 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
2129 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
2130 % language (English) -- needed because algorithm2e errors on an empty option.
2131 \edef\coppe@@algotlang{\coppemainstring{algorithm2eopt}}
2132 \ifx\coppe@@algotlang\empty
2133 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
2134 \else
2135 \expandafter\RequirePackage\expandafter[\coppe@@algotlang,linesnumbered,lined,algochapter,ruled]{algorithm2e}
2136 \fi
2137 % algorithm2e composes its own caption and therefore ignores the class-wide
2138 % \captionsetup{font=footnotesize,labelsep=endash}. Left alone it printed
2139 % "Algoritmo 6.1:Titulo" in bold at body size, while every other illustration

```

```

2140 % printed "Figura 6.1 -- Titulo" in footnotesize -- 2.10 asks for the
2141 % designating word, the number, a TRAVESSAO and the title, uniformly.
2142 %
2143 % The separator and the caption fonts are therefore set to match. The 'ruled'
2144 % layout (rules above and below) is kept: 2.10 prescribes where the caption and
2145 % the source go, not whether the float is delimited, and the rules read here
2146 % like the top-and-bottom delimitation a table carries.
2147 \SetAlgoCaptionSeparator{\ \textendash}
2148 \SetAlCapFnt{\footnotesize}
2149 \SetAlCapNameFnt{\footnotesize}
2150 \SetAlCapSty{textnormal}
2151 \SetAlCapNameSty{textnormal}
2152 \SetKwFor{Para}{para}{faça}{fim do para}
2153 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
2154 % setspace + hyperref + biblatex are loaded at the very end of the class
2155 % (see the block just before %</class>), so their hooks see every macro the
2156 % class defines, and the load order setspace -> hyperref -> biblatex is
2157 % respected.

\department This macro is used to set the author's affiliation. There are twelve options which
correspond to all academic units at COPPE/UFRJ. It defines the current and the
foreign names of these units.

2158 \newcommand\coppe@deptcode{}
2159 \newcommand\meta@deptname{}
2160 \newcommand\department[1]{%
2161 % The programme CODE is kept alongside the resolved name, and each name is
2162 % declared TWICE. \local@deptname keeps the TeX accent escapes with braces
2163 % (Computa{\c c}{\~ a}o) that the cover and the folha de rosto typeset;
2164 % \meta@deptname is the same name in literal UTF-8, and is the one that goes
2165 % into the XMP. pdfx copies whatever it is given straight through, so feeding it
2166 % the escaped form produced "Computa{c}o" in dc:description -- which is why the
2167 % subject line used to fall back to the bare programme code. Both forms are
2168 % written out here rather than derived from one another because there is no
2169 % reliable way to strip TeX accent syntax at the point where the .xmpdata is
2170 % written, and a table of thirteen entries is cheaper than that machinery.
2171 \gdef\coppe@deptcode{#1}%
2172 \ifthenelse{\equal{#1}{PEB}}{
2173   {\global\def\local@deptname{Engenharia Biom{\` e}dica}
2174     \global\def\meta@deptname{Engenharia Biomédica}
2175     \global\def\foreign@deptname{Biomedical Engineering}}{}
2176 \ifthenelse{\equal{#1}{PEC}}{
2177   {\global\def\local@deptname{Engenharia Civil}
2178     \global\def\meta@deptname{Engenharia Civil}
2179     \global\def\foreign@deptname{Civil Engineering}}{}
2180 \ifthenelse{\equal{#1}{PEE}}{
2181   {\global\def\local@deptname{Engenharia El{\` e}trica}
2182     \global\def\meta@deptname{Engenharia Elétrica}
2183     \global\def\foreign@deptname{Electrical Engineering}}{}
2184 \ifthenelse{\equal{#1}{PEM}}{
2185   {\global\def\local@deptname{Engenharia Mec{\` a}nica}
2186     \global\def\meta@deptname{Engenharia Mecânica}
2187     \global\def\foreign@deptname{Mechanical Engineering}}{}
2188 \ifthenelse{\equal{#1}{PEMM}}{
2189   {\global\def\local@deptname{Engenharia Metal{\` u}rgica e de Materiais}

```

```

2190 \global\def\meta@deptname{Engenharia Metalúrgica e de Materiais}
2191 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
2192 \ifthenelse{\equal{#1}{PEN}}{
2193   {\global\def\local@deptname{Engenharia Nuclear}
2194     \global\def\meta@deptname{Engenharia Nuclear}
2195     \global\def\foreign@deptname{Nuclear Engineering}}{}
2196 \ifthenelse{\equal{#1}{PEN0}}{
2197   {\global\def\local@deptname{Engenharia Oce{\'a}nica}
2198     \global\def\meta@deptname{Engenharia Oceânica}
2199     \global\def\foreign@deptname{Ocean Engineering}}{}
2200 \ifthenelse{\equal{#1}{PPE}}{
2201   {\global\def\local@deptname{Planejamento Energ{\'e}tico}
2202     \global\def\meta@deptname{Planejamento Energético}
2203     \global\def\foreign@deptname{Energy Planning}}{}
2204 \ifthenelse{\equal{#1}{PEP}}{
2205   {\global\def\local@deptname{Engenharia de Produ{\'c}o{~}o}
2206     \global\def\meta@deptname{Engenharia de Produção}
2207     \global\def\foreign@deptname{Production Engineering}}{}
2208 \ifthenelse{\equal{#1}{PEQ}}{
2209   {\global\def\local@deptname{Engenharia Qu{\'i}mica}
2210     \global\def\meta@deptname{Engenharia Química}
2211     \global\def\foreign@deptname{Chemical Engineering}}{}
2212 \ifthenelse{\equal{#1}{PESC}}{
2213   {\global\def\local@deptname{Engenharia de Sistemas e Computa{\'c}o{~}o}
2214     \global\def\meta@deptname{Engenharia de Sistemas e Computação}
2215     \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
2216 \ifthenelse{\equal{#1}{PET}}{
2217   {\global\def\local@deptname{Engenharia de Transportes}
2218     \global\def\meta@deptname{Engenharia de Transportes}
2219     \global\def\foreign@deptname{Transportation Engineering}}{}
2220 \ifthenelse{\equal{#1}{PENT}}{
2221   {\global\def\local@deptname{Engenharia de Nanotecnologia}
2222     \global\def\meta@deptname{Engenharia de Nanotecnologia}
2223     \global\def\foreign@deptname{Nanotechnology Engineering}}{}
2224 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```

2225 \renewcommand\title[1]{%
2226   \global\def\local@title{#1}%
2227   \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
2228 }

```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```

2229 \newcommand\foreigntitle[1]{%
2230   \global\def\foreign@title{#1}%
2231   \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
2232 }

```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```

2233 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}
2234 % Subtitle, mirroring the title machinery slot for slot: \subtitle feeds the
2235 % brazilian slot, \foreignsubtitle the english one, \subtitlein any other.
2236 % 3.1.1 and 3.1.2.1.1(c) require the subtitle on the capa and on the folha de
2237 % rosto, subordinated to the title by a colon.
2238 \newcommand\subtitle[1]{%
2239   \expandafter\gdef\csname coppe@subtitle@brazilian\endcsname{#1}}
2240 \newcommand\foreignsubtitle[1]{%
2241   \expandafter\gdef\csname coppe@subtitle@english\endcsname{#1}}
2242 \newcommand\subtitlein[2]{%
2243   \expandafter\gdef\csname coppe@subtitle@#1\endcsname{#2}}
2244 \newcommand\coppe@selectsubtitle{\@nameuse{coppe@subtitle@\coppe@mainlang}}
2245 % Title as the capa and the folha de rosto print it. Annexes A and B show the
2246 % title in caps and the subtitle in ordinary case after a colon, so the case
2247 % change is applied to the title alone.
2248 \newcommand\coppe@titleblock{%
2249   \MakeUppercase{\coppe@selecttitle}%
2250   \@ifundefined{coppe@subtitle@\coppe@mainlang}{\coppe@selectsubtitle}}
2251 % Number of volumes (3.1.1 and 3.1.2.1.1(d)): printed only when there is more
2252 % than one, since a single-volume work says nothing about volumes.
2253 % Coorientador (3.1.2.1.1(f)). Same shape as \advisor, including the optional
2254 % treatment; the label switches to the plural on the second call, as the
2255 % advisor label already did.
2256 \newcount\@coadvisor\@coadvisor0
2257 \newcommand\coadvisor[5][[]]{%
2258   \global\@namedef{CoppeCoadvisorTitle:\expandafter\the\@coadvisor}{#1}
2259   \global\@namedef{CoppeCoadvisorName:\expandafter\the\@coadvisor}{#2}
2260   \global\@namedef{CoppeCoadvisorSurname:\expandafter\the\@coadvisor}{#3}
2261   \global\@namedef{CoppeCoadvisorDegree:\expandafter\the\@coadvisor}{#4}
2262   \global\@namedef{CoppeCoadvisorInst:\expandafter\the\@coadvisor}{#5}
2263   \global\advance\@coadvisor by 1
2264 }
2265 \newcommand\local@coadvisorstring{%
2266   \ifnum\@coadvisor>1
2267     \coppemainstring{coadvisors}%
2268   \else
2269     \coppemainstring{coadvisor}%
2270   \fi}
2271 % Mesma coisa no idioma estrangeiro, para a pagina de abstract.
2272 \newcommand\foreign@coadvisorstring{%
2273   \ifnum\@coadvisor>1
2274     \coppeforeignstring{coadvisors}%
2275   \else
2276     \coppeforeignstring{coadvisor}%
2277   \fi}
2278 \newcommand\coppe@nvolumes{1}
2279 \newcommand\coppe@thisvolume{1}
2280 \newcommand\volumes[1]{\gdef\coppe@nvolumes{#1}}
2281 \newcommand\volume[1]{\gdef\coppe@thisvolume{#1}}
2282 \newcommand\coppe@volumeline{%
2283   \ifnum\coppe@nvolumes>1
2284     \par\vspace*{4mm}%
2285     \coppemainstring{volumename}\space\coppe@thisvolume\space
2286     \coppemainstring{ofname}\space\coppe@nvolumes

```

```

2287 % The closing \par matters: on the capa the next thing is \vspace{\stretch{3}},
2288 % which does not end a paragraph, so without it the city ran on into the volume
2289 % line ("Volume 2 de 3 RIO DE JANEIRO").
2290 \par
2291 \fi}

```

\advisor Defines globally the title, name and academic degree of the advisors.

```

2292 \newcount\@advisor\@advisor0
2293 % Argument order since v4.1: the TREATMENT ("Prof.", "Prof.a") became the
2294 % optional slot, and comes empty by default -- no norm asks for it and it only
2295 % clutters the folha de aprovacao. The INSTITUTION, required by 3.1.2.1.3(e),
2296 % became the last mandatory argument; it may be left empty ({} without error,
2297 % in which case nothing is printed for it. A pre-4.1 call has to be rewritten:
2298 % \advisor[UFRJ]{Prof.}{Nome}{Sobrenome}{D.Sc.} becomes
2299 % \advisor{Nome}{Sobrenome}{D.Sc.}{UFRJ}.
2300 \newcommand\advisor[5][]{\%
2301 \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
2302 \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
2303 \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
2304 \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
2305 \global\@namedef{CoppeAdvisorInst:\expandafter\the\@advisor}{#5}
2306 \global\advance\@advisor by 1
2307 \ifnum\@advisor>1
2308 \renewcommand\local@advisorstring{Orientadores}
2309 \renewcommand\foreign@advisorstring{Advisors}
2310 \fi
2311 }

```

\examiner

```

2312 \newcount\@examiner\@examiner0
2313 % Until v4.1 this stored only "#1 #2" and DISCARDED the third argument: the
2314 % titulacao every document passed was silently thrown away and never printed,
2315 % though 3.1.2.1.3(e) requires it. Each part is now kept separately, in the same
2316 % order \advisor uses: the optional slot is the TREATMENT, and the institution
2317 % is the last mandatory argument, which may be left empty.
2318 % CoppeExaminer: is retained with its old meaning for anything reading it.
2319 \newcommand\examiner[4][]{\%
2320 \global\@namedef{CoppeExaminerTitle:\expandafter\the\@examiner}{#1}
2321 \global\@namedef{CoppeExaminerName:\expandafter\the\@examiner}{#2}
2322 \global\@namedef{CoppeExaminerDegree:\expandafter\the\@examiner}{#3}
2323 \global\@namedef{CoppeExaminerInst:\expandafter\the\@examiner}{#4}
2324 \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1 #2}
2325 \global\advance\@examiner by 1
2326 }

```

\author It was redefined to allow the identification of the author's first names and surname.

```

2327 \renewcommand\author[2]{\%
2328 \global\def\@authname{#1}
2329 \global\def\@authsurn{#2}
2330 }

```

\date Sets the *document's* nominal month and year (used by the cover, the folha de rosto, and the catalog). Stores them in private slots (\coppe@docmonth /

\coppe@docyear) rather than mutating TeX's \month / \year primitives, which would clobber \today.

```
2331 \renewcommand\date[2]{%
2332   \def\coppe@docmonth{#1}%
2333   \def\coppe@docyear{#2}}
```

\local@monthname

```
2334 \newcommand\local@monthname{\ifcase\coppe@docmonth\or
2335   Janeiro\or Fevereiro\or Mar{\c c}\o\or Abril\or Maio\or Junho\or
2336   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}
```

\foreign@monthname

```
2337 \newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
2338   January\or February\or March\or April\or May\or June\or
2339   July\or August\or September\or October\or November\or December\fi}
```

\keyword

```
2340 \newcounter{keywords}
2341 \newcommand\keyword[1]{%
2342   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
2343   \global\addtocounter{keywords}{1}
2344 }
2345 % Keywords for the foreign abstract. Annex F of the manual shows the English
2346 % abstract closing with English keywords, and \keyword alone cannot serve both
2347 % pages. When no \foreignkeyword is given the foreign abstract falls back to
2348 % the \keyword list, which is what a document written before v4.1 expects.
2349 \newcounter{foreignkeywords}
2350 \newcommand\foreignkeyword[1]{%
2351   \global\@namedef{CoppeForeignKeyword:\expandafter\the\c@foreignkeywords}{#1}
2352   \global\addtocounter{foreignkeywords}{1}
2353 }
2354 % Keywords for the optional third abstract (Portuguese, for the banca). Same
2355 % shape and same fallback as \foreignkeyword: without it the Portuguese page
2356 % would close with the main-language keywords under a Portuguese label, which
2357 % is what a Spanish-, French- or Italian-main work would produce.
2358 \newcounter{braziliankeywords}
2359 \newcommand\braziliankeyword[1]{%
2360   \global\@namedef{CoppeBrazilianKeyword:\expandafter\the\c@braziliankeywords}{#1}
2361   \global\addtocounter{braziliankeywords}{1}
2362 }
2363 % 3.1.2.1.4: the keywords close the resumo, preceded by the term, a colon, and
2364 % separated from one another by semicolons.
2365 % #1 label, #2 csname prefix, #3 how many.
2366 \newcommand\coppe@printkeywords[3]{%
2367   \ifnum#3>0
2368     \par\vspace*{7mm}%
2369     \noindent{\singlespacing\textbf{#1:}}\enspace
2370     \count1=0
2371     \@whilenum\count1<#3\do{%
2372       \ifnum\count1>0 ;\space\fi
2373       \csname #2\the\count1\endcsname
2374       \advance\count1 by 1}%
2375     .\par}%
2376 \fi}
```

`\areaconcentracao` Data for the folha adicional (Coleta CAPES) and for the natureza block of the
`\linhapesquisa` folha de rosto and folha de aprovacao. Every one of them defaults to empty, and
`\tipoproducao` each page says what it does with a missing value: the folha adicional leaves the
`\projetovinculado` cell blank, and the folha de aprovacao writes the date as “a ser determinada”.
`\nomeprojeto` 2377 `\newcommand\coppe@areaconc{}`
`\agenciafomento` 2378 `\newcommand\areaconcentracao[1]{\gdef\coppe@areaconc{#1}}`
2379 `\newcommand\coppe@dataaprov{}`
2380 `\newcommand\dataaprovacao[1]{\gdef\coppe@dataaprov{#1}}`
2381 `\newcommand\coppe@linhapesq{}`
2382 `\newcommand\linhapesquisa[1]{\gdef\coppe@linhapesq{#1}}`
2383 `\newcommand\coppe@tipoproducao{}`
2384 `\newcommand\tipoproducao[1]{\gdef\coppe@tipoproducao{#1}}`
2385 `\newcommand\coppe@projetovinc{}`
2386 `\newcommand\projetovinculado[1]{\gdef\coppe@projetovinc{#1}}`
2387 `\newcommand\coppe@nomeprojeto{}`
2388 `\newcommand\nomeprojeto[1]{\gdef\coppe@nomeprojeto{#1}}`
2389 `\newcounter{coppeagencias}`
2390 `\newcommand\agenciafomento[2]{%`
2391 `\global\@namedef{CoppeAgencia:\expandafter\the\c@coppeagencias}{#1}%`
2392 `\global\@namedef{CoppeAgenciaSigla:\expandafter\the\c@coppeagencias}{#2}%`
2393 `\global\addtocounter{coppeagencias}{1}}`

`\freeconfig` This command allows easy changing of core class parameters.

2394 `\newcommand\freeconfig[6]{%`
2395 `\providecommand\@degree{}`
2396 `\renewcommand\@degree{#1}`
2397 `\providecommand\@degreename{}`
2398 `\renewcommand\@degreename{#2}`
2399 `\providecommand\local@degname{}`
2400 `\renewcommand\local@degname{#3}`
2401 `\providecommand\foreign@degname{}`
2402 `\renewcommand\foreign@degname{#4}`
2403 `\providecommand\local@doctype{}`
2404 `\renewcommand\local@doctype{#5}`
2405 `\providecommand\foreign@doctype{}`
2406 `\renewcommand\foreign@doctype{#6}%`
2407 `}`
2408 `% \end{macrocode}`
2409 `% \end{macro}`
2410 `%`
2411 `% \begin{macro}{\frontmatter}`
2412 `% The number of pages for both frontmatter and mainmatter printed`
2413 `% in the cataloging details page is computed by means of simple`
2414 `% LaTeX labels.`
2415 `% \begin{macrocode}`
2416 `\renewcommand\frontmatter{%`
2417 `\cleardoublepage`
2418 `\@mainmatterfalse`
2419 `% The count already started at the folha de rosto (see \maketitle) and simply`
2420 `% continues here: 2.7 requires every folha from the folha de rosto onwards to`
2421 `% be counted sequentially, printing arabic numerals only from the first textual`
2422 `% folha. Resetting the counter here -- as this class did until v4.1 -- made the`
2423 `% Introducao come out as page 1, which the norm does not allow.`
2424 `%`


```

2425 % The numeraisromanos option keeps the legacy behaviour (lowercase roman in the
2426 % pre-textual part, arabic restarting at 1 in the textual part). That departs
2427 % from 2.7, which has the pre-textual folhas counted but NOT numbered, so the
2428 % option warns.
2429 \if@copperoman
2430   \ClassWarningNoLine{coppe}{Option ‘numeraisromanos’ prints roman numerals
2431   on the pre-textual folhas and restarts the arabic count at 1 in the
2432   textual part. Section 2.7 of the SiBI manual (9th rev. ed., 2026) has
2433   those folhas counted but not numbered, and the arabic numbering
2434   continuing the count. Drop the option to comply}%
2435   \pagenumbering{roman}%
2436 \fi
2437 % \thispagestyle vale por UMA pagina. Se a folha de aprovacao transbordar --
2438 % uma banca grande com titulo longo consegue --, a pagina seguinte pegava o
2439 % estilo corrente e saia com folio, na parte pre-textual, onde a 2.7 nao
2440 % admite. O grupo com \pagestyle{empty} cobre quantas paginas forem, e o
2441 % \endgroup devolve o estilo anterior.
2442 \begingroup
2443   \pagestyle{empty}\thispagestyle{empty}
2444   \singlespacing
2445   \makefrontpage
2446   \clearpage
2447 \endgroup
2448 \if@copperoman
2449   \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}%
2450   \renewcommand{\headrulewidth}{0pt}}%
2451   \pagestyle{coppe}%
2452 \else
2453   \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
2454   \pagestyle{empty}%
2455 \fi
2456 }

\mainmatter

2457 \renewcommand\mainmatter{%
2458   \coppe@mainBegin
2459   \cleardoublepage
2460   \@mainmattertrue
2461   \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}%
2462   \renewcommand{\headrulewidth}{0pt}}%
2463   \pagestyle{coppe}%
2464 % No \pagenumbering{arabic} in the default path: it would reset the counter to
2465 % 1 and make the Introducao folha 1, contrary to 2.7. The counter has been
2466 % arabic and running since the folha de rosto; all that changes here is that
2467 % the folio starts being PRINTED. Only the legacy numeraisromanos path, which
2468 % switched to roman in the front matter, has to switch back -- and there the
2469 % restart at 1 is intrinsic to that (non-conforming) scheme.
2470 \if@copperoman\pagenumbering{arabic}\fi

\backmatter

2471 \renewcommand\backmatter{%
2472   \if@openright
2473     \cleardoublepage
2474   \else

```

```

2475 \clearpage
2476 \fi}
2477 %

```

\coppetexfinalpage Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with `\vfill`; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overridable: `\coppefinalengine` (engine name; auto-detected via `iftex`), `\coppefinalfont` (text body font; defaults to Computer Modern, the L^AT_EX default), and `\coppefinalmanual` (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before `\coppetexfinalpage` with `\renewcommand` if the document uses a non-default font or follows an earlier edition of the norm.

```

2478 \providecommand{\coppefinalengine}{%
2479   \ifPDFTeX\hologo{pdfLaTeX}\fi
2480   \ifLuaTeX\hologo{LuaLaTeX}\fi
2481   \ifXeTeX\hologo{XeLaTeX}\fi}
2482 % A classe carrega lmodern (as bitmap da Computer Modern saiam em Type 3 e
2483 % inviabilizavam o PDF/A), entao o colofao diz Latin Modern -- quem trocar a
2484 % fonte redefine este comando.
2485 \providecommand{\coppefinalfont}{%
2486   \if@coppesansserif
2487     Latin Modern Sans (\texttt{lmodern})%
2488   \else
2489     Latin Modern (\texttt{lmodern})%
2490   \fi}
2491 \providecommand{\coppefinalmanual}{%
2492   \emph{Norma para a Elaborac~ao Gr'afica de Teses e Dissertac~oes
2493     da COPPE/UFRJ}}
2494 \newcommand{\coppetexfinalpage}{%
2495   \clearpage
2496   % Force the colophon onto a verso (even) page so it faces the rear cover.
2497   % If the current page is odd (recto), fill it with an empty verso first.
2498   \if@twoside
2499     \ifodd\c@page
2500       \hbox{}\thispagestyle{empty}\newpage
2501     \fi
2502   \fi
2503   \thispagestyle{empty}%
2504   \null\vfill
2505   \begin{otherlanguage}{brazilian}%
2506     \begin{center}
2507       {\footnotesize\setstretch{1.0}}%
2508       \textsc{Colof\~ao}\par
2509       \vspace{0.7em}
2510       \begin{minipage}{0.75\textwidth}\centering
2511         Este documento foi composto em \LaTeX{} com \coppefinalengine{}
2512         em \today, na fonte \coppefinalfont. A bibliografia foi processada
2513         por Bib\LaTeX{} e Biber. Foi utilizada a classe \CopeTeX,
2514         vers\~ao\coppe@version, dispon\~vel no reposit\~orio GitHub

```

```

2515     \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em
2516     conformidade com a \coppefinalmanual.
2517     \end{minipage}\par}
2518     \end{center}
2519 \end{otherlanguage}%
2520 \vspace*{2cm}%
2521 \clearpage}

```

`\makecover` The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```

2522 \newcommand\makecover{%
2523   \begin{titlepage}%
2524   % A capa e um modelo institucional, e sai em espaco simples qualquer que seja o
2525   % espacamento do corpo. Sob a opcao doublespacing ela transbordava para uma
2526   % segunda e ate uma terceira folha -- que ainda por cima levavam folio, porque
2527   % o \thispagestyle do titlepage vale por uma pagina so. 0 \pagestyle{empty} e
2528   % local ao ambiente e cobre quantas paginas forem.
2529   \pagestyle{empty}%
2530   \singlespacing
2531   \centering
2532   \makebox[\linewidth]{%
2533     \coppe@logo{ufrj-logo}\hfill
2534     \coppe@logo{coppe-logo}}\par
2535   \vspace{1.5cm}
2536   {\singlespacing
2537   \MakeUppercase{\local@universityname}\par
2538     \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de \
2539       P{\ ' o}s-Gradua{\c c}{\~ a}o e Pesquisa de Engenharia}}\par
2540     \MakeUppercase{Programa de \local@deptname}\par}
2541   \vspace{\stretch{2}}
2542   \textbf{\MakeUppercase{\textbf{\@authname\ \@authsur}}}\par
2543   \vspace{\stretch{2}}
2544   \nohyphens{\coppe@titleblock}\par
2545   \coppe@volumeline
2546   \vspace{\stretch{3}}
2547   \MakeUppercase{\local@cityname}\par
2548   \coppe@docyear
2549   \end{titlepage}%
2550 }

```

`\maketitle`

```

2551 \renewcommand\maketitle{%
2552 % The capa is neither counted nor numbered (2.7), so it lives in its own
2553 % throwaway numbering scheme; \pagenumbering{arabic} right after it restarts
2554 % the counter, making the FOLHA DE ROSTO page 1 -- which is where 2.7 says the
2555 % count begins. Nothing is printed here: \frontmatter keeps \pagestyle{empty}
2556 % until the textual part starts.
2557 \pagenumbering{alph}
2558 \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%

```

```

2559 \makecover
2560 \pagenumbering{arabic}
2561 \begin{titlepage}
2562 % Mesma razao da capa: modelo institucional, espaco simples, e sem folio ainda
2563 % que transborde.
2564 \pagestyle{empty}%
2565 \singlespacing
2566 \begin{flushleft}
2567 \vspace*{1.5mm}
2568 \setlength\baselineskip{0pt}
2569 \setlength\parskip{1mm}
2570 \makebox[\linewidth]{%
2571     \coppe@logo{ufrj-logo}\hfill
2572     \coppe@logo{coppe-logo}}
2573 \end{flushleft}
2574 \vspace{1.05cm}
2575 \begin{center}
2576 \nohyphens{\@authname\ \@authsur}\par
2577 \vspace*{3cm}
2578 \nohyphens{\coppe@titleblock}\par
2579 \coppe@volumeline
2580 \end{center}
2581 \vspace*{2.1cm}
2582 \begin{flushright}
2583 % 2.4: on the folha de rosto the natureza/objetivo/instituicao block is set
2584 % SINGLE spaced, and aligned from the middle of the mancha grafica to the right
2585 % margin. 0.5\textwidth tracks the mancha, so the block follows automatically if
2586 % the margins ever change -- the old fixed 8.45cm did not.
2587 \begin{minipage}{0.5\textwidth}
2588 \singlespacing\frontcover@maintext
2589 \end{minipage}\par
2590 \end{flushright}
2591 % Only the natureza block is indented to the right (2.4). The linha de pesquisa
2592 % and the orientadores return to the LEFT MARGIN, as the CAPES model of the
2593 % folha de rosto and Annex B of the manual show. Until v4.1 they sat inside the
2594 % right-hand block, which is what a recent review flagged.
2595 \vspace*{12mm}
2596 {\singlespacing
2597 \ifthenelse{\equal{\coppe@linhapesq}{}}{\%
2598     \noindent\coppemainstring{researchline}: \coppe@linhapesq\par
2599     \vspace*{5mm}}{\%
2600 \nohyphens{\%
2601 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2602     \local@advisorstring: &
2603     \count1=0
2604     \toks@={}
2605     \@whilenum \count1<\@advisor \do{\%
2606     \ifcase\count1 % same as \ifnum0=\count1
2607         \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2608             \expandafter\endcsname\expandafter\space%
2609             \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2610     \else
2611         \toks@=\expandafter\expandafter\expandafter{\%
2612             \expandafter\the\expandafter\toks@%

```

```

2613         \expandafter&\expandafter\space%
2614         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2615         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2616     }%
2617 \fi
2618 \advance\count1 by 1}
2619 \the\toks@
2620 \end{tabularx}}%
2621 \ifnum\@coadvisor>0
2622     \par\vspace*{2mm}%
2623     \nohyphens{%
2624     \noindent\begin{tabularx}{\textwidth}[b]{\@{}}l@{ }>{\raggedright\arraybackslash}X@{}
2625     \local@coadvisorstring: &
2626     \count1=0
2627     \toks@={}
2628     \@whilenum \count1<\@coadvisor \do{%
2629     \ifcase\count1
2630         \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2631         \expandafter\endcsname\expandafter\space%
2632         \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2633     \else
2634         \toks@=\expandafter\expandafter\expandafter{%
2635         \expandafter\the\expandafter\toks@%
2636         \expandafter&\expandafter\space%
2637         \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2638         \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\
2639     }%
2640 \fi
2641 \advance\count1 by 1}
2642 \the\toks@
2643 \end{tabularx}}%
2644 \fi
2645 \par}
2646 \vspace*{\fill}
2647 % 3.1.2.1.1(g): the folha de rosto carries the ANO DE DEPOSITO alone -- the year
2648 % the revised final version is handed to the Programa. The month was never asked
2649 % for here (the capa, Annex A, also shows only the year), and printing it
2650 % invited confusion with the defence date. \date still takes month and year:
2651 % the month is what the abstract pages and the colophon use.
2652 \begin{center}
2653 \local@cityname\par
2654 \coppe@docyear
2655 \end{center}
2656 \end{titlepage}
2657 % Folha adicional (3.1.2.1.2): a sheet of its own right after the folha de
2658 % rosto, ahead of the approval sheet produced by \frontmatter. Until the 2026
2659 % edition this content lived on the VERSO of the folha de rosto (NBR 14724,
2660 % R57); the verso is gone with the printed copy.
2661 % LaTeX's titlepage environment ends with \setcounter{page}{1} whenever the
2662 % document is one-sided (see \endtitlepage in classes.dtx). That silently ate
2663 % the folha de rosto from the count, putting the folha de aprovacao at 1 instead
2664 % of 2. The count is therefore restated here: the folha de rosto WAS folha 1, so
2665 % whatever follows it is folha 2.
2666 \setcounter{page}{2}%

```

2667 % The page carrying the ficha catalografica is explicitly excluded from the
 2668 % count -- 2.7 of the 2026 manual ("a folha adicional que contem a ficha
 2669 % catalografica nao pode ser contada ou numerada"), and the 2025 edition said
 2670 % the same of the verso of the folha de rosto. Hence the counter is stepped
 2671 % back after it.
 2672 % An exame de qualificacao is not deposited with the library: no ficha
 2673 % catalografica, and no Coleta CAPES sheet. Everything else gets both.
 2674 \if@coppeexame\else
 2675 \makefolhaadicional\clearpage\addtocounter{page}{-1}%
 2676 \fi
 2677 \global\let\maketitle\relax%
 2678 \global\let\and\relax}

2679 % 3.1.2.1.3 lists the elements of the folha de aprovacao in order: (a) the
 2680 % AUTHOR, (b) the title in full, (c) the natureza/objetivo/instituicao/area de
 2681 % concentracao, (d) the date of approval, (e) the examining board. Until v4.1
 2682 % this page opened with the title in caps and the author under it, and set the
 2683 % natureza as a wall of uppercase (\frontpage@maintext). Annex D shows neither.
 2684 % One board member. \coppe@membroid is the identification line required by
 2685 % 3.1.2.1.3(e): name, titulacao and institution. Since v4.1 the TREATMENT
 2686 % ("Prof.", "Prof.a") is the optional slot and comes empty by default -- no norm
 2687 % asks for it -- while titulacao and institution are what actually identify the
 2688 % member and are shown whenever they are given. Each of the three is dropped
 2689 % silently when empty, so nothing breaks if the author leaves one blank.
 2690 \newcommand\coppe@membroid[4]{%
 2691 \ifthenelse{\equal{#1}{}}{\{#1\}}{#1}%
 2692 \ifthenelse{\equal{#3}{}}{\{#3\}}{#3}%
 2693 \ifthenelse{\equal{#4}{}}{\{#4\}}{#4}%
 2694 % Until v4.1 the class could draw a rule above each name, for a hand signature,
 2695 % and \coppe@membrogap held the gap above that rule, shrinking as the board grew
 2696 % so seven members still fitted on one sheet. The deposit has been digital only
 2697 % since Resolucao CEPG n. 246/2023, so there is nothing left to sign by hand and
 2698 % the rules are gone. The length stays declared because the seven-member sizing
 2699 % below still uses it as the gap between entries.
 2700 \newlength\coppe@membrogap
 2701 \setlength\coppe@membrogap{9mm minus 2mm}
 2702 \newcount\coppe@bancasize
 2703 \newcommand\coppe@membro[4]{%
 2704 \noindent\hspace*{8mm}%
 2705 \begin{minipage}{\dimexpr\textwidth-8mm\relax}
 2706 \coppe@membroid{#1}{#2}{#3}{#4}
 2707 \end{minipage}\par\vspace{2mm}}
 2708 \newcommand\makefrontpage{%
 2709 \begin{center}
 2710 \nohyphens{\@authname\ \@authsurn}\par
 2711 \vspace*{7mm}
 2712 \sloppy\nohyphens{\coppe@selecttitle
 2713 \@ifundefined{coppe@subtitle@\coppe@mainlang}{:}
 2714 \coppe@selectsubtitle}}\par
 2715 \end{center}\par
 2716 \vspace*{8mm}
 2717 % Same block, same indentation and same single spacing as the folha de rosto:
 2718 % 2.4 names the two pages together.
 2719 \begin{flushright}

```

2720 \begin{minipage}{0.5\textwidth}
2721 \singlespacing\frontcover@maintext
2722 \end{minipage}
2723 \end{flushright}
2724 \vspace*{12mm}
2725 % (d) date of approval, from \dataaprovacao. With no date given the page used to
2726 % print a rule to fill in by hand, which the digital-only deposit made pointless
2727 % -- nobody writes on a PDF. It now says so in words, in the main language:
2728 % "a ser determinada". The line is never blank and never a rule.
2729 \noindent\local@approvedonname:\enspace
2730 \ifthenelse{\equal{\coppe@dataaprov}{}}{
2731   {\coppemainstring{tbddate}}{\coppe@dataaprov}\par
2732   \vspace*{12mm}
2733 % (e) the examining board. By DEFAULT only the examiners are listed: the folha
2734 % de aprovacao records who examined the work, and the advisor conducted it. With
2735 % the 'orientadorexamina' option the advisors and coadvisors go back in, ahead
2736 % of the examiners, as president of the board (3.1.2.1.3(e)).
2737 %
2738 % The list is assembled into \coppe@bancalist with \g@addto@macro instead of
2739 % being built inside a tabularx: each entry now carries four fields (treatment,
2740 % name, titulacao, institution) and the old \toks@/\expandafter accumulation
2741 % could not carry them without becoming unreadable. \protected@edef keeps
2742 % accents and other protected commands in the names intact.
2743 % Size of the board decides the spacing above. The thresholds are measured, not
2744 % guessed: with the 9mm gap a seven-member board pushed three members onto a
2745 % second page, 7mm brought six back onto one sheet, 5mm did the same for seven,
2746 % and eight needed 3.5mm -- eight came up in the adversarial suite, in a
2747 % Spanish thesis whose longer natureza block leaves less room than the
2748 % Portuguese one. Boards of five or fewer keep the original 9mm, so every
2749 % document written before this change renders exactly as it did.
2750 % The count is of what actually gets PRINTED, so a document that leaves the
2751 % advisors out does not get the tight spacing of a board it does not show.
2752 \coppe@bancasize=\@examiner
2753 \if@coppeorientexam
2754   \advance\coppe@bancasize by \@advisor
2755   \advance\coppe@bancasize by \@coadvisor
2756 \fi
2757 \ifnum\coppe@bancasize>7
2758   \setlength\coppe@membrogap{3.5mm minus 1.5mm}%
2759 \else\ifnum\coppe@bancasize>6
2760   \setlength\coppe@membrogap{5mm minus 2mm}%
2761 \else\ifnum\coppe@bancasize>5
2762   \setlength\coppe@membrogap{7mm minus 2mm}%
2763 \fi\fi\fi
2764 \gdef\coppe@bancalist{}%
2765 % Orientadores e coorientadores so entram sob 'orientadorexamina'. Fora dela a
2766 % folha lista apenas quem examinou.
2767 \if@coppeorientexam
2768   \count1=0
2769   \@whilenum\count1<\@advisor\do{%
2770     \protected@edef\coppe@tmp{\noexpand\coppe@membro
2771       {\csname CoppeAdvisorTitle:\the\count1\endcsname}%
2772       {\csname CoppeAdvisorName:\the\count1\endcsname}\space
2773       {\csname CoppeAdvisorSurname:\the\count1\endcsname}%

```

```

2774         {\csname CoppeAdvisorDegree:\the\count1\endcsname}%
2775         {\csname CoppeAdvisorInst:\the\count1\endcsname}}%
2776         \expandafter\g@addto@macro\expandafter\coppe@bancalist
2777         \expandafter{\coppe@tmp}%
2778         \advance\count1 by 1}%
2779 % Coorientadores come right after the orientadores and before the rest of the
2780 % board, keeping the order 3.1.2.1.3(e) prescribes.
2781     \count1=0
2782     \@whilenum\count1<\@coadvisor\do{%
2783         \protected@edef\coppe@tmp{\noexpand\coppe@membro
2784             {\csname CoppeCoadvisorTitle:\the\count1\endcsname}%
2785             {\csname CoppeCoadvisorName:\the\count1\endcsname}\space
2786             \csname CoppeCoadvisorSurname:\the\count1\endcsname}%
2787             {\csname CoppeCoadvisorDegree:\the\count1\endcsname}%
2788             {\csname CoppeCoadvisorInst:\the\count1\endcsname}}%
2789         \expandafter\g@addto@macro\expandafter\coppe@bancalist
2790         \expandafter{\coppe@tmp}%
2791         \advance\count1 by 1}%
2792 \fi
2793 \count1=0
2794 \@whilenum\count1<\@examiner\do{%
2795     \protected@edef\coppe@tmp{\noexpand\coppe@membro
2796         {\csname CoppeExaminerTitle:\the\count1\endcsname}%
2797         {\csname CoppeExaminerName:\the\count1\endcsname}%
2798         {\csname CoppeExaminerDegree:\the\count1\endcsname}%
2799         {\csname CoppeExaminerInst:\the\count1\endcsname}}%
2800     \expandafter\g@addto@macro\expandafter\coppe@bancalist
2801     \expandafter{\coppe@tmp}%
2802     \advance\count1 by 1}%
2803 \noindent\local@approvedname:\par
2804 \nohyphens{\singlespacing\coppe@bancalist}\par
2805 % No city/date block at the foot: 3.1.2.1.3 ends the folha de aprovacao at the
2806 % examining board, and Annex D shows nothing below the signature lines. The date
2807 % that belongs on this page is the date of APPROVAL, printed above.
2808 % \frontpage@bottomtext stays defined for anyone calling it directly.
2809 \vspace*{\fill}}

2810 \newcommand\coppe@hypersetup{%
2811 \begingroup
2812 % changes to \toks@ and \count@ are kept local;
2813 % it's not necessary for them, but it is usually the case
2814 % for \count1, because the first ten counters are written
2815 % to the DVI file, thus you got lucky because of PDF output
2816 \toks@={}% in this special case not necessary
2817 \count@=0 %
2818 \@whilenum\count@<\value{keywords}\do{%
2819     % * a keyword separator is not necessary,
2820     %     if there is just one keyword
2821     % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
2822     %     at least once, to get rid of the loop depended \count@
2823     \ifcase\count@ % same as \ifnum0=\count@
2824         \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
2825     \else
2826         \toks@=\expandafter\expandafter\expandafter{%

```



```

2827         \expandafter\the\expandafter\toks@
2828         \expandafter;\expandafter\space
2829         \csname CoppeKeyword:\the\count@\endcsname
2830     }%
2831 \fi
2832 \advance\count@ by 1 %
2833 }%
2834 \edef\x{\endgroup
2835     \noexpand\hypersetup{%
2836         pdfkeywords={\the\toks@}%
2837     }%
2838 }%
2839 \x
2840 \hypersetup{%
2841     pdfauthor={\@authname\ \@authsur},
2842     pdftitle={\local@title},
2843     pdfsubject={\local@doctype\ de \@degree\ em \local@deptname\ da COPPE/UFRJ},
2844     pdfcreator={LaTeX with CoppeTeX toolkit},
2845     breaklinks={true},
2846     raiselinks={true},
2847     pageanchor={true},
2848 }%
2849 % 'semlinks' vai por ultimo, para vencer o que vier antes, e usa hidelinks em
2850 % vez de colorlinks=false: hidelinks tira a cor E a moldura, que sao os dois
2851 % jeitos de o hyperref marcar um link, e nao mexe em mais nada. Os links
2852 % continuam existindo, clicaveis, e os bookmarks continuam no painel.
2853 \if@coppesemlinks\hypersetup{hidelinks}\fi}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il;” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain T_EX command `\isundefined` (cf. the T_EX FAQ).

```

2854 \newcommand\makecatalog{%
2855     \vspace*{\fill}
2856     \begin{center}
2857         \setlength{\fboxsep}{5mm}
2858         \framebox[120mm][c]{\makebox[5mm][c]{}%
2859             \begin{minipage}[c]{105mm}
2860                 \setlength{\parindent}{5mm}
2861                 \noindent\sloppy\nohyphens\@authsur,
2862                 \nohyphens\@authname\par
2863                 \nohyphens{%
2864                     \coppe@selecttitle/\@authname\ \@authsur. -- \local@cityname:
2865                     UFRJ/COPPE, \coppe@docyear.}\par
2866                 \pageref{front:pageno},
2867                 \pageref{LastPage}
2868                 p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm}.\par
2869                 % There is an issue here. When the last entry must be split between lines,
2870                 % the spacing between it and the next paragraph becomes smaller.
2871                 % Should we manually introduce a fixed space? But how could we know that
2872                 % a name was split? Is this happening yet?
2873                 \nohyphens{%
2874                     \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2875                         \local@advisorstring: &

```

```

2876 \count1=0
2877 \toks@={}
2878 \@whilenum \count1<\@advisor \do{%
2879 \ifcase\count1 % same as \ifnum0=\count1
2880 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2881 \expandafter\endcsname\expandafter\space%
2882 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2883 \else
2884 \toks@=\expandafter\expandafter\expandafter{%
2885 \expandafter\the\expandafter\toks@
2886 \expandafter&\expandafter\space
2887 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2888 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2889 }%
2890 \fi
2891 \advance\count1 by 1}
2892 \the\toks@
2893 \end{tabularx}}\par
2894 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
2895 UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
2896 Refer{\^ e}ncias Bibliogr{\` a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
2897 \count1=0
2898 \count2=1
2899 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
2900 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
2901 \advance\count1 by 1
2902 \advance\count2 by 1}
2903 I. \csname CoppeAdvisorSurname:0\endcsname,%
2904 \ \csname CoppeAdvisorName:0\endcsname%
2905 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
2906 II. \local@universityname, COPPE, Programa de \local@deptname.
2907 III. T{\` i}tulo.}
2908 \end{minipage}}
2909 \end{center}
2910 \vspace*{\fill}}

```

`\fichacatalografica` The folha adicional required from August 2026 (SiBI manual, 9th rev. ed., `\makefolhaadicional` 3.1.2.1.2 and Annex H): the CAPES collection fields plus the ficha catalografica, sitting immediately after the folha de rosto and counted by nobody. `\fichacatalografica` takes the PDF produced by the SiBI generator at <http://fichacatalografica.sibi.ufrj.br/>, optionally with a page number.

The sheet is institutional, so it is always in Portuguese whatever the main language of the work – and deliberately without `\selectlanguage`, since the brazilian babel language need not be loaded in, say, a Spanish-main document.

```

2911 \newcommand\coppe@fichafile{}
2912 \newcommand\coppe@fichapage{1}
2913 \newcommand\fichacatalografica[2][1]{%
2914 \gdef\coppe@fichapage{#1}\gdef\coppe@fichafile{#2}}
2915 \newcommand\coppe@marcatipo[1]{%
2916 \ifthenelse{\equal{\coppe@tipoproducao}{#1}}{$\times$}{\phantom{$\times$}}}
2917 \newcommand\coppe@marcavinc[1]{%
2918 \ifthenelse{\equal{\coppe@projetovinc}{#1}}{$\times$}{\phantom{$\times$}}}
2919 % Annex H prints the five fields inside a CLOSED FRAME, one cell each, with a

```

```

2920 % horizontal rule between cells and NOTHING to write on: the space to fill in is
2921 % the blank space of the cell. Until v4.1 the class laid the fields out as loose
2922 % paragraphs and drew a dotted \hrulefill to write on, which is not what the
2923 % model shows. \coppe@campo now prints the value when there is one and leaves
2924 % empty vertical space when there is not.
2925 \newlength\coppe@capesgap
2926 \setlength\coppe@capesgap{11mm}
2927 \newcommand\coppe@campo[1]{%
2928   \ifthenelse{\equal{#1}{}}{\vspace*{\coppe@capesgap}}{\par#1}}
2929 % One cell of the frame: the label, then whatever the field holds.
2930 \newcommand\coppe@capescelula[2]{%
2931   \begin{minipage}[t]{\dimexpr\linewidth-2\tabcolsep\relax}
2932     \vspace{1mm}\#1\par#2\vspace{1.5mm}
2933   \end{minipage}\\hline}
2934 \newcommand\coppe@checatipo{%
2935   \ifthenelse{\equal{\coppe@tipoproducao}{}}{\}%
2936   \ifthenelse{\equal{\coppe@tipoproducao}{bibliografica}\OR
2937     \equal{\coppe@tipoproducao}{artistica}\OR
2938     \equal{\coppe@tipoproducao}{tecnologica}\OR
2939     \equal{\coppe@tipoproducao}{tecnica}}{\}%
2940     \ClassWarning{coppe}{Unknown \string\coppe@tipoproducao value
2941       '\coppe@tipoproducao'. Annex H of the SiBI manual allows only
2942       bibliografica, artistica, tecnologica or tecnica; no box will be
2943       ticked}}}%
2944 \newcommand\makefolhaadicional{%
2945   \clearpage
2946   \thispagestyle{empty}%
2947   \coppe@checatipo
2948   {\singlespacing
2949   \begin{center}
2950     {\large\bfseries Informa\c c\~oes Coleta CAPES}
2951   \end{center}
2952   \vspace{7mm}
2953   \noindent\begin{tabular}{|@{\hspace{\tabcolsep}}p{\dimexpr\textwidth-2\tabcolsep-2\arrayrulewidth}\hline
2954     \coppe@capescelula{Tipo de produ\c c\~ao intelectual:}{%
2955       \vspace{1mm}
2956       \hspace*{10mm}(\coppe@marcatipo{bibliografica})\enspace
2957       Bibliogr\'afica\par
2958       \hspace*{10mm}(\coppe@marcatipo{artistica})\enspace Art\'istica\par
2959       \hspace*{10mm}(\coppe@marcatipo{tecnologica})\enspace Tecnol\'ogica\par
2960       \hspace*{10mm}(\coppe@marcatipo{tecnica})\enspace T\'ecnica}
2961     \coppe@capescelula{Projeto de pesquisa vinculado \a produ\c c\~ao
2962       intelectual?}{%
2963       \vspace{1mm}
2964       \hspace*{10mm}(\coppe@marcavinc{sim})\enspace Sim\par
2965       \hspace*{10mm}(\coppe@marcavinc{nao})\enspace N\~ao}
2966     \coppe@capescelula{Nome do projeto de pesquisa:}{%
2967       \coppe@campo{\coppe@nomeprojeto}}
2968     \coppe@capescelula{\Area de concentra\c c\~ao da produ\c c\~ao
2969       intelectual:}{\coppe@campo{\coppe@areaconc}}
2970     \coppe@capescelula{Ag\~encia(s) de fomento financiadora(s) da produ\c c\~ao
2971       intelectual:}{%
2972       \ifthenelse{value{coppe@agencias}=0}{\vspace*{\coppe@capesgap}}{\%
2973

```

```

2974         \count1=0
2975         \@whilenum\count1<\value{coppeagencias}\do{%
2976             \hspace*{10mm}\csname CoppeAgencia:\the\count1\endcsname
2977             \ (\csname CoppeAgenciaSigla:\the\count1\endcsname)\par
2978             \advance\count1 by 1}}
2979 \end{tabular}\par
2980 \vspace{9mm}
2981 \begin{center}
2982     {\large\bfseries Ficha catalogr\'afica}\par
2983     \vspace{4mm}
2984     \ifthenelse{\equal{\coppe@fichafile}{}}{%
2985         \if@copperascunho
2986             \makecatalog
2987         \else
2988             \framebox[125mm][c]{\parbox[c][62mm][c]{115mm}{\centering\footnotesize
2989                 Espa\c co reservado \'a ficha catalogr\'afica.\\[1ex]
2990                 Gere-a em \texttt{fichacatalografica.sibi.ufrj.br} e inclua o
2991                 arquivo com \texttt{\string\fichacatalografica\{arquivo.pdf\}},
2992                 ou solicite-a \'a biblioteca do seu Programa.}}}%
2993     \fi}{%
2994         \includegraphics[page=\coppe@fichapage,width=125mm]{\coppe@fichafile}}%
2995 \end{center}
2996 \par}}

```

\dedication

```

2997 \newcommand\dedication[1]{
2998     \gdef\@dedic{#1}
2999     \cleardoublepage
3000     \vspace*{\fill}
3001     \begin{flushright}
3002         \begin{minipage}{60mm}
3003             \raggedleft \it \normalsize \@dedic
3004         \end{minipage}
3005     \end{flushright}}

```

\coppe@coadvisorrows The coadvisor block of the abstract pages, in the shape the folha de rosto already uses. It prints nothing at all unless the `coorientador` class option is given AND the document actually declared a `\coadvisor`, so the default output is byte for byte what it was. The argument is the label macro, which differs per page: the main-language pages pass `\local@coadvisorstring` and the foreign one passes `\foreign@coadvisorstring`.

```

3006 \newcommand\coppe@coadvisorrows[1]{%
3007     \if@coppecoadvresumo
3008         \ifnum\@coadvisor>0
3009             \par\vspace*{2mm}%
3010             \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
3011                 #1: &
3012             \count1=0
3013             \toks@={}
3014             \@whilenum \count1<\@coadvisor \do{%
3015                 \ifcase\count1
3016                     \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
3017                         \expandafter\endcsname\expandafter\space%
3018                         \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}

```

```

3019         \else
3020         \toks@=\expandafter\expandafter\expandafter{%
3021         \expandafter\the\expandafter\toks@
3022         \expandafter&\expandafter\space%
3023         \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
3024         \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\
3025         }%
3026         \fi
3027         \advance\count1 by 1}
3028         \the\toks@
3029     \end{tabularx}%
3030 \fi
3031 \fi}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

3032 \newenvironment{abstract}{%
3033     \cleardoublepage
3034     \thispagestyle{plain}
3035     \abstract@toptext\par
3036     \vspace*{8.6mm}
3037     \begin{center}
3038         \sloppy\nohyphens{\MakeUppercase\local@title}\par
3039         \vspace*{13.2mm}
3040         \@authname\ \@authsur \par
3041         \vspace*{7mm}
3042         \local@monthname/\coppe@docyear
3043     \end{center}\par
3044     \vspace*{1.5cm}
3045     \noindent%
3046     \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
3047         \local@advisorstring: &
3048         \count1=0
3049         \toks@={}
3050         \@whilenum \count1<\@advisor \do{%
3051             \ifcase\count1 % same as \ifnum0=\count1
3052                 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
3053                 \expandafter\endcsname\expandafter\space%
3054                 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
3055             \else
3056                 \toks@=\expandafter\expandafter\expandafter{%
3057                 \expandafter\the\expandafter\toks@
3058                 \expandafter&\expandafter\space
3059                 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
3060                 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
3061                 }%
3062             \fi
3063             \advance\count1 by 1}
3064             \the\toks@
3065         \end{tabularx}%
3066     \coppe@coadvisorrows{\local@coadvisorstring}\par
3067     \vspace*{2mm}
3068     \noindent\local@deptstring: \local@deptname\par
3069     \vspace*{7mm}}{%
3070     \coppe@printkeywords{\coppemainstring{keywordsname}}%

```

```

3071 {CoppeKeyword:}{\value{keywords}}%
3072 \vspace*{\fill}}

```

foreignabstract (*env.*)

```

3073 \newenvironment{foreignabstract}{%
3074 \cleardoublepage
3075 \thispagestyle{plain}
3076 \begin{otherlanguage}{\coppe@foreignlang}
3077 \foreignabstract@toptext\par
3078 \vspace*{8.6mm}
3079 \begin{center}
3080 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
3081 \vspace*{13.2mm}
3082 \@authname\ \@authsurn \par
3083 \vspace*{7mm}
3084 \foreign@monthname/\coppe@docyear
3085 \end{center}\par
3086 \vspace*{1.5cm}
3087 \noindent%
3088 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
3089 \foreign@advisorstring: &
3090 \count1=0
3091 \toks@={}
3092 \@whilenum \count1<\@advisor \do{%
3093 \ifcase\count1 % same as \ifnum0=\count1
3094 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
3095 \expandafter\endcsname\expandafter\space%
3096 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
3097 \else
3098 \toks@=\expandafter\expandafter\expandafter{%
3099 \expandafter\the\expandafter\toks@
3100 \expandafter&\expandafter\space
3101 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
3102 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
3103 }%
3104 \fi
3105 \advance\count1 by 1}
3106 \the\toks@
3107 \end{tabularx}%
3108 \coppe@coadvisorrows{\foreign@coadvisorstring}\par
3109 \vspace*{2mm}
3110 \noindent\foreign@deptstring: \foreign@deptname\par
3111 \vspace*{7mm}}{%
3112 \ifnum\value{foreignkeywords}>0
3113 \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
3114 {CoppeForeignKeyword:}{\value{foreignkeywords}}%
3115 \else
3116 \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
3117 {CoppeKeyword:}{\value{keywords}}%
3118 \fi
3119 \end{otherlanguage}
3120 \vspace*{\fill}
3121 \global\let\@author\@empty
3122 \global\let\@date\@empty

```

```

3123 \global\let\foreign@title\@empty
3124 \global\let\foreign@title\relax
3125 \global\let\local@title\@empty
3126 \global\let\local@title\relax
3127 \global\let\author\relax
3128 \global\let\author\relax
3129 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor **english**; harmless otherwise (the document just won't include it).

```

3130 \newenvironment{brazilianabstract}{%
3131   \cleardoublepage
3132   \thispagestyle{plain}
3133   \begin{otherlanguage}{brazilian}
3134   \abstract@top\text\par
3135   \vspace*{8.6mm}
3136   \begin{center}
3137     \sloppy\nohyphens{\MakeUppercase\local@title}\par
3138     \vspace*{13.2mm}
3139     \@authname\ \@authsur \par
3140     \vspace*{7mm}
3141     \local@monthname/\coppe@docyear
3142   \end{center}\par
3143   \vspace*{1.5cm}
3144   \noindent%
3145   \begin{tabularx}{\textwidth}{b}{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
3146     \local@advisorstring: &
3147     \count1=0
3148     \toks@={}
3149     \@whilenum \count1<\@advisor \do{%
3150       \ifcase\count1
3151         \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
3152           \expandafter\endcsname\expandafter\space%
3153           \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
3154       \else
3155         \toks@=\expandafter\expandafter\expandafter{%
3156           \expandafter\the\expandafter\toks@
3157           \expandafter&\expandafter\space
3158           \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
3159           \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
3160         }%
3161       \fi
3162       \advance\count1 by 1}
3163     \the\toks@
3164   \end{tabularx}%
3165   \coppe@coadvisorrows{\local@coadvisorstring}\par
3166   \vspace*{2mm}
3167   \noindent\local@deptstring: \local@deptname\par

```

```

3168 \vspace*{7mm}}{%
3169 % \coppestring takes ONE argument and resolves against the current language;
3170 % this runs before \end{otherlanguage}, so \language is already brazilian.
3171 \ifnum\value{braziliankeywords}>0
3172   \coppe@printkeywords{\coppestring{keywordsname}}%
3173   {CoppeBrazilianKeyword:}{\value{braziliankeywords}}%
3174 \else
3175   \coppe@printkeywords{\coppestring{keywordsname}}%
3176   {CoppeKeyword:}{\value{keywords}}%
3177 \fi
3178 \end{otherlanguage}
3179 \vspace*{\fill}}

```

\listoffigures

```

3180 \renewcommand\listoffigures{%
3181   \coppe@hasLof
3182   \if@twocolumn
3183     \@restonecoltrue\onecolumn
3184   \else
3185     \@restonecolfalse
3186   \fi
3187   \chapter*{\listfigurename}%
3188   \coppe@listtoc{\listfigurename}%
3189   \@mkboth{\MakeUppercase\listfigurename}%
3190           {\MakeUppercase\listfigurename}%
3191   \@starttoc{lof}%
3192   \if@restonecol\twocolumn\fi
3193 }

```

\listoftables

```

3194 \renewcommand\listoftables{%
3195   \if@twocolumn
3196     \@restonecoltrue\onecolumn
3197   \else
3198     \@restonecolfalse
3199   \fi
3200   \chapter*{\listtablename}%
3201   \coppe@listtoc{\listtablename}%
3202   \@mkboth{%
3203     \MakeUppercase\listtablename}%
3204     {\MakeUppercase\listtablename}%
3205   \@starttoc{lot}%
3206   \if@restonecol\twocolumn\fi
3207 }

```

\printlosymbols

```

3208 % 0 \renewcommand ficava FORA do grupo e vazava: depois de imprimir a lista de
3209 % simbolos, \glossaryname continuava valendo "Lista de Simbolos" pelo resto do
3210 % documento, e o Glossario pos-textual da 3.1.4.2 saia com o titulo errado --
3211 % no sumario tambem. Dentro do grupo, a troca dura o que tem de durar.
3212 \newcommand\printlosymbols{%
3213 \begingroup\@coppeglosspretrue
3214 \renewcommand\glossaryname{\listsymbolname}%
3215 \@input@{\jobname.los}\endgroup}

```


\makelosymbols

```
3216 \def\makelosymbols{%
3217   \newwrite\@losfile
3218   \immediate\openout\@losfile=\jobname.syx
3219   \newcommand\symb1[3][]{\@bsphack\begingroup
3220     \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
3221     \@sanitize%
3222     \@wrls{\@tempsymb1}{##2}{##3}}\typeout%
3223   {Writing index of symbols file \jobname.syx}%
3224   \let\makelosymbols\@empty%
3225 }%
3226 \@onlypreamble\makelosymbols

3227 % \coppe@semfolio engole o numero de pagina que o makeindex insiste em escrever.
3228 % As duas listas nao levam pagina (ver o comentario no modulo 'glossary'), e o
3229 % makeindex nao tem como nao escrever o numero -- entao ele escreve, e este
3230 % comando o descarta na composicao. E tambem o que substitui \hyperpage: sem
3231 % numero nao ha para onde o link levar.
3232 \newcommand\coppe@semfolio[1]{%
3233 \newcommand\@wrls[3]{%
3234   \protected@write\@losfile{%
3235     {\string\indexentry{#1[#2] #3|coppe@semfolio}{\thepage}}}%
3236   \endgroup%
3237   \@esphack}
```

\printloabbreviations

```
3238 \newcommand\printloabbreviations{%
3239 \begingroup\@coppeglosspretrue
3240 \renewcommand\glossaryname{\listabbreviationname}%
3241 \@input@{\jobname.lab}\endgroup}
```

\makeloabbreviations

```
3242 \def\makeloabbreviations{%
3243   \newwrite\@labfile
3244   \immediate\openout\@labfile=\jobname.abx
3245   \newcommand\abbrev[3][]{\@bsphack\begingroup
3246     \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
3247     \@sanitize
3248     \@wrlab{\@tempsymb1}{##2}{##3}}\typeout
3249   {Writing index of abbreviations file \jobname.abx}%
3250   \let\makeloabbreviations\@empty
3251 }
3252 \@onlypreamble\makeloabbreviations

3253 \newcommand\@wrlab[3]{%
3254   \protected@write\@labfile{%
3255     {\string\indexentry{#1[#2] #3|coppe@semfolio}{\thepage}}}%
3256   \endgroup%
3257   \@esphack}
```

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
\sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in
the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations

must be active), then just “SIGLA” on later uses. `\sigla*{key}` forces the full form. The first-use flag is set globally so it survives groups (floats, footnotes).

```

3258 \newcommand\newsigla[3]{%
3259   \@namedef{coppe@sigla@s@#1}{#2}%
3260   \@namedef{coppe@sigla@l@#1}{#3}}
3261 \newcommand\coppe@sigla@reg[1]{%
3262   \@ifundefined{abbrev}{}%
3263     {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}%
3264 \newcommand\coppe@sigla@full[1]{%
3265   \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}
3266 \newcommand\coppe@sigla@undef[1]{%
3267   \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
3268   \textbf{??#1??}}
3269 \newcommand\coppe@sigla@first[1]{%
3270   \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
3271 \newcommand\coppe@sigla@norm[1]{%
3272   \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
3273   {\@ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
3274     {\@nameuse{coppe@sigla@s@#1}}}%
3275 \newcommand\coppe@sigla@star[1]{%
3276   \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
3277   {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}%
3278 \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

3279 %%% \AtBeginDocument{%
3280 %%% \ifpackageloaded{hyperref}{%
3281 %%%   \def\@wrlab#1#2{%
3282 %%%     \protected@write\@labfile{%
3283 %%%       {\string\indexentry{[#1] #2|hyperpage}{\thepage}}}%
3284 %%%     \endgroup
3285 %%%     \@esphack}}}%
3286 %%%   \def\@wrlab#1#2{%
3287 %%%     \protected@write\@labfile{%
3288 %%%       {\string\indexentry{[#1] #2}{\arabic{page}}}}}%
3289 %%%     \endgroup
3290 %%%     \@esphack}}}%

3291 % Catalog-label writers. Two language-related traps must be avoided here:
3292 % 1. The label NAMES contain ':' and '.', which babel makes active in
3293 %    some languages (e.g. Brazilian's "shorthand" set). If the write
3294 %    argument is tokenised while one of those is active, pdfTeX aborts
3295 %    with "<inserted text> \endwrite" because the shorthand expansion
3296 %    injects unbalanced braces. \detokenize emits its argument as
3297 %    catcode-12 characters regardless of the current activation, so the
3298 %    label NAMES survive whatever language is current at writer time.
3299 % 2. Spanish babel redefines \@roman (and therefore \roman{...}) to
3300 %    produce small-caps roman numerals via \es@scroman, whose expansion
3301 %    contains an \uppercase group that is unbalanced inside \write. The
3302 %    same crash signature appears. The fix is to bypass LaTeX's
3303 %    \roman/\arabic counter wrappers entirely and use the raw e-TeX
3304 %    primitives \romannumeral and \number directly on the \c@page count
3305 %    register --- both are fully expandable, language-agnostic, and
3306 %    always produce plain ASCII digits or lowercase roman letters.
3307 %    \Roman{page} is left in place because babel-spanish only redefines
3308 %    \@Roman under the (non-default) ucroman option.

```

```

3309 \AtBeginDocument{%
3310   \ltx@ifpackageloaded{hyperref}{%
3311     \def\coppe@bibEnd{%
3312       \immediate\write\@auxout{%
3313         \string\newlabel{\detokenize{bib:end}}%
3314         {{}\number\c@page}{\detokenize{page.}\number\c@page}{}}}%
3315     \def\coppe@bibBegin{%
3316       \immediate\write\@auxout{%
3317         \string\newlabel{\detokenize{bib:begin}}%
3318         {{}\number\c@page}{\detokenize{page.}\number\c@page}{}}}%
3319     \def\coppe@mainBegin{%
3320       \immediate\write\@auxout{%
3321         \string\newlabel{\detokenize{front:pageno}}%
3322         {{}\Roman{page}}{\detokenize{page.}\romannumeral\c@page}{}}}%
3323     \def\coppe@hasLof{%
3324       \immediate\write\@auxout{%
3325         \string\newlabel{\detokenize{cat:lofflag}}%
3326         {{}\detokenize{:~il.};}{\detokenize{page.}\romannumeral\c@page}{}}}%
3327   }%
3328   \def\coppe@bibEnd{%
3329     \immediate\write\@auxout{%
3330       \string\newlabel{\detokenize{bib:end}}{{}\number\c@page}{}}}%
3331   \def\coppe@bibBegin{%
3332     \immediate\write\@auxout{%
3333       \string\newlabel{\detokenize{bib:begin}}{{}\number\c@page}{}}}%
3334   \def\coppe@mainBegin{%
3335     \immediate\write\@auxout{%
3336       \string\newlabel{\detokenize{front:pageno}}{{}\Roman{page}}{}}}%
3337   \def\coppe@hasLof{%
3338     \immediate\write\@auxout{%
3339       \string\newlabel{\detokenize{cat:lofflag}}{{}\detokenize{:~il.};}{}}}%
3340   }%
3341 }
3342 % The reference list is now produced by biblatex's \printbibliography (see the
3343 % bibliography setup after babel, above). The former thebibliography
3344 % redefinition and its \bibindent are no longer used and have been removed; the
3345 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
3346 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (*env.*)

```

3347 \newlength{\recuolongquote}
3348 \setlength{\recuolongquote}{4cm}
3349
3350 \newenvironment*{longquote}[1][default]{%
3351   \begin{list}{}{%
3352     \setlength{\leftmargin}{\recuolongquote}%
3353     \setlength{\rightmargin}{0pt}%
3354     \setlength{\itemindent}{0pt}%
3355     \setlength{\listparindent}{0pt}%
3356     \setlength{\parsep}{0pt}%
3357     \setlength{\topsep}{\baselineskip}%
3358   }%
3359   \item\relax
3360   \footnotesize

```

```

3361 \singlespacing
3362 \ifthenelse{\not\equal{#1}{default}}{%
3363 {\itshape\selectlanguage{#1}}%
3364 }%
3365 \ignorespaces
3366 }{%
3367 \end{list}%
3368 \ignorespacesafterend
3369 }

3370 \newenvironment{theglossary}{%
3371 \if@twocolumn%
3372 \@restonecoltrue\onecolumn%
3373 \else%
3374 \@restonecolfalse%
3375 \fi%
3376 \@mkboth{\MakeUppercase\glossaryname}%
3377 {\MakeUppercase\glossaryname}%
3378 \chapter*{\glossaryname}%
3379 \if@coppeglosspre
3380 \coppe@listtoc{\glossaryname}%
3381 \else
3382 \addcontentsline{toc}{chapter}{\glossaryname}%
3383 \fi
3384 \list{}
3385 {\setlength{\listparindent}{0in}%
3386 \setlength{\labelwidth}{1.0in}%
3387 \setlength{\leftmargin}{1.5in}%
3388 \setlength{\labelsep}{0.5in}%
3389 \setlength{\itemindent}{0in}}%
3390 \sloppy}%
3391 {\if@restonecol\twocolumn\fi%
3392 \endlist}
3393 %
3394 \renewenvironment{theindex}{%
3395 \if@twocolumn
3396 \@restonecolfalse
3397 \else
3398 \@restonecoltrue
3399 \fi
3400 \twocolumn[\@makeschapterhead{\indexname}]%
3401 \@mkboth{\MakeUppercase\indexname}%
3402 {\MakeUppercase\indexname}%
3403 \thispagestyle{plain}\parindent\z@
3404 \addcontentsline{toc}{chapter}{\indexname}
3405 \parskip\z@ \@plus .3\p@\relax
3406 \columnseprule \z@
3407 \columnsep 35\p@
3408 \let\item\@idxitem}
3409 {\if@restonecol\onecolumn\else\clearpage\fi}
3410 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
3411 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
3412 \newcommand\glossaryname{\coppemainstring{glossary}}
3413 %

```

```

3414 \newcommand\local@advisorstring{Orientador}
3415 \newcommand\foreign@advisorstring{Advisor}
3416 % Fed by the string dispatcher instead of a hard-coded Portuguese literal.
3417 % Expansion is lazy (it happens inside \makefrontpage), so \coppe@mainlang is
3418 % already set by the time this is used.
3419 \ifthenelse{\boolean{maledoc}}{%
3420   \newcommand\local@approvedname{\coppemainstring{approvedmale}}%
3421 }{%
3422   \newcommand\local@approvedname{\coppemainstring{approvedfemale}}%
3423 }
3424 \ifthenelse{\boolean{maledoc}}{%
3425   \newcommand\local@approvedonname{\coppemainstring{approvedonmale}}%
3426 }{%
3427   \newcommand\local@approvedonname{\coppemainstring{approvedonfemale}}%
3428 }
3429 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
3430 \newcommand\local@deptstring{Programa}
3431 \newcommand\foreign@deptstring{Department}
3432 \newcommand\local@cityname{Rio de Janeiro}
3433 \newcommand\local@statename{RJ}
3434 \newcommand\local@countryname{Brasil}
3435 %
3436 % 3.1.2.1.1(e) of the 2026 manual requires the natureza, the objetivo, the
3437 % name of the institution AND the area de concentracao in this block. The area
3438 % was missing; it is appended as its own sentence when \areaconcentracao was
3439 % given, and simply omitted otherwise.
3440 \newcommand\frontcover@maintext{
3441   \sloppy\nohyphens{\local@doctype\ de \@degreename\
3442   \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3443   ao Programa de P{' o}s-gradua{\c c}{\~ a}o em \local@deptname,
3444   COPPE, da \local@universityname, como parte dos requisitos
3445   necess{' a}rios {' a} obten{\c c}{\~ a}o do t{' i}tulo de
3446   \local@degname\ em \local@deptname.}%
3447   \ifthenelse{equal{\coppe@areaconc}}{}{}{}%
3448   \par\vspace{1ex}%
3449   \sloppy\nohyphens{'Area de concentra{\c c}{\~ a}o: \coppe@areaconc.}}%
3450 }
3451 %
3452 \newcommand\frontpage@maintext{
3453   \noindent {\MakeUppercase\local@doctype}
3454   \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
3455   \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
3456   DE P{' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
3457   UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
3458   NECESS{' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
3459   {\MakeUppercase\local@degname} EM CI{'E}NCIAS EM
3460   {\MakeUppercase\local@deptname.\par}}%
3461 }
3462 %
3463 \newcommand\frontpage@bottomtext{%
3464   \begin{center}
3465     {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
3466     {\MakeUppercase\local@monthname\ DE \coppe@docyear}
3467   \end{center}}%

```

```

3468 }
3469 %
3470 \newcommand\abstract@toptext{%
3471   \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
3472   \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3473   \sloppy\nohyphens{{\` a} COPPE/UFRJ como parte dos requisitos
3474   necess{\` a}rios para a obten{\c c}{\~ a}o do grau de
3475   \local@degname\ em Ci{\^ e}ncias (\@degree)}
3476 }
3477 \newcommand\foreignabstract@toptext{%
3478   \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
3479   COPPE/UFRJ as a partial fulfillment of the requirements for the
3480   degree of \foreign@degname\ of Science (\@degree)}
3481 }
3482 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- **setspace** – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before **hyperref** patches all the cross-reference and TOC plumbing.
- **hyperref** – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, list-of-* commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the **coppe** class redefines (e.g. `\@chapter`, `\listofquadros`).
- **biblatex** – must come after **hyperref** so its citation links and back-references hook into **hyperref** correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```

3483 \RequirePackage{setspace}
3484 % Apply the default line spacing now that setspace is available. The flag was
3485 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
3486 \if@coppedoublespace\doublespacing\else\onehalfspacing\fi
3487 % pdfx MUST precede hyperref -- it loads hyperref itself, with the options
3488 % PDF/A needs, and cannot fix up an already-loaded one. Level a-2b rather than
3489 % a-1b (the "ISO 19005-1" of Annex I): a-1b forbids transparency, which
3490 % tcolorbox and several graphics packages produce, so a-1b would fail on
3491 % documents that are otherwise perfectly fine. a-2b is ISO 19005-2, accepted by
3492 % Pantheon and by the SiBI's own instructions, which also list PDF/A-2b.
3493 \if@coppepdfa
3494   \RequirePackage[a-2b]{pdfx}
3495 % pdfTeX stamps /PTEX.Fullbanner into the document information dictionary.
3496 % It is not one of the keys ISO 19005-2 6.6.2.3.2 maps to an XMP property, and
3497 % no extension schema declares it, so it is a DocInfo entry with no XMP
3498 % counterpart -- a finding a PDF/A checker has every reason to raise, and one
3499 % that costs nothing to remove. -1 sets every suppression bit.
3500 %
3501 % The two engines spell this differently, and not just in the name. pdfTeX has

```

```

3502 % \pdfsuppressptexinfo, whose bits only ever govern PTEX.* entries, so -1 (all
3503 % of them) is safe. LuaTeX moved the pdfTeX integer parameters behind
3504 % \pdfvariable in 0.85 AND renamed this one to suppressoptionalinfo, whose bits
3505 % reach further: besides the four PTEX.* entries (1, 2, 4, 8) it can also
3506 % suppress Creator, the dates, Producer, Trapped and -- the dangerous one --
3507 % the trailer /ID, which PDF/A requires. So -1 is exactly what must NOT be
3508 % written there; 15 is the four PTEX.* bits and nothing else. Checked on a real
3509 % LuaTeX run: /ID survives.
3510 %
3511 % Getting the name wrong is not a quiet failure. \pdfvariable with an unknown
3512 % variable name makes LuaTeX typeset the name, and the document stops with
3513 % "Missing \begin{document}" -- which is how the first attempt at this was
3514 % caught.
3515 \ifdefined\pdfsuppressptexinfo
3516 \pdfsuppressptexinfo=-1
3517 \else
3518 \ifdefined\pdfvariable
3519 \pdfvariable suppressoptionalinfo=15
3520 \fi
3521 \fi
3522 % pdfx takes the XMP metadata from \jobname.xmpdata, which it reads when it is
3523 % LOADED -- before the preamble has run, so before \title, \author and
3524 % \keyword exist. The file is therefore WRITTEN by the class and picked up on
3525 % the NEXT run. The build already runs pdflatex three times for cross-
3526 % references, so the deposited PDF always carries current metadata; only a
3527 % first-ever compilation of a brand-new file emits pdfx's "no \jobname.xmpdata"
3528 % warning and lacks dc:title.
3529 %
3530 % The write happens at the END of the document, not at \begin{document}. This
3531 % class has always documented -- and every example follows it -- that \title,
3532 % \author, \department and \keyword go AFTER \begin{document}, next to
3533 % \maketitle. A write at \begin{document} therefore ran before any of them
3534 % existed and died with "Undefined control sequence" on \@authname, so the
3535 % pdfa option was unusable in exactly the style the manual teaches. It only
3536 % ever worked for a document that set its metadata in the preamble. At the end
3537 % everything is set whichever style the author chose, and \immediate still puts
3538 % the file on disk in time for the next run.
3539 %
3540 % The hook is enddocument/afteraux, and the stream is \@mainaux rather than one
3541 % of our own. TeX has sixteen output streams and example.tex was already using
3542 % all of them (toc, lof, lot, loa, lol, loq, los, idx, glo, aux, bcf, ...), so
3543 % \newwrite here failed with "No room for a new \write" on precisely the
3544 % document that most needs validating. enddocument/afteraux fires after LaTeX
3545 % has closed the .aux, which frees that stream for the one \immediate\openout
3546 % we need and costs the document nothing. Kernels older than 2020-10-01 have no
3547 % such hook; there the class falls back to \AtEndDocument with a stream of its
3548 % own, which is the pre-existing behaviour.
3549 %
3550 % Every field is guarded, and the guards are not cosmetic. \coppe@selecttitle
3551 % is \@nameuse{coppe@title@<lang>}, and \csname on a name that does not exist
3552 % yet MAKES it \relax: unexpandable, so \write puts the control sequence NAME
3553 % into the file instead of the title. The .xmpdata then carried the literal
3554 % \coppe@title@brazilian, pdfx read it back on the following run, and the
3555 % document stopped compiling at all -- a single bad run bricked the file until

```

```

3556 % someone deleted the .xmpdata by hand. A field that is not set is now simply
3557 % omitted, which pdfx handles, instead of poisoning the next run.
3558 %
3559 % The writes must be \immediate. \protected@write defers to the shipout, and
3560 % the stream is closed here and now, so deferred writes would land nowhere --
3561 % which is exactly what happened first time round: a 0-byte .xmpdata.
3562 % \let\protect\string inside the group gives the protection that
3563 % \protected@write would have given: accents and other robust commands are
3564 % written literally instead of being expanded. pdfx reads the .xmpdata as
3565 % LaTeX, so \a arrives there intact.
3566 \ifdefined\AddToHook
3567   \let\coppe@xmpout\@mainaux
3568   \def\coppe@xmphook{\AddToHook{enddocument/afteraux}}%
3569 \else
3570   \newwrite\coppe@xmpout
3571   \let\coppe@xmphook\AtEndDocument
3572 \fi
3573 \coppe@xmphook{%
3574   \gdef\coppe@xmpkeys{%
3575     \count@=0
3576     \@whilenum\count@<\value{keywords}\do{%
3577       \ifnum\count@>0
3578         \g@addto@macro\coppe@xmpkeys{\string\sep\space}%
3579       \fi
3580       \protected@edef\coppe@tmp{\csname CoppeKeyword:\the\count@\endcsname}%
3581       \expandafter\g@addto@macro\expandafter\coppe@xmpkeys
3582       \expandafter{\coppe@tmp}%
3583       \advance\count@ by 1}%
3584   \immediate\openout\coppe@xmpout=\jobname.xmpdata
3585   \begingroup
3586     \let\protect\string
3587     \@ifundefined{coppe@title@\coppe@mainlang}{%
3588       \immediate\write\coppe@xmpout{\string\Title{\coppe@selecttitle}}}%
3589     \@ifundefined{@authname}{%
3590       \immediate\write\coppe@xmpout{%
3591         \string\Author{\@authname\space\@authsurn}}}%
3592     \immediate\write\coppe@xmpout{\string\Keywords{\coppe@xmpkeys}}%
3593     \ifthenelse{\equal{\coppe@deptcode}{}}{%
3594       \immediate\write\coppe@xmpout{\string\Subject{\local@doctype\space de
3595         \@degree name. Programa de \meta@deptname\space(\coppe@deptcode),
3596         COPPE/UFRJ}}}%
3597     \immediate\write\coppe@xmpout{\string\Publisher{\local@universityname}}%
3598     \@ifundefined{coppe@docyear}{%
3599       \immediate\write\coppe@xmpout{\string\Date{\coppe@docyear}}}%
3600   \endgroup
3601   \immediate\closeout\coppe@xmpout}%
3602 \fi
3603 \RequirePackage{hyperref}
3604 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
3605 % former natbib+bibtex setup. The numbers class option selects the numeric
3606 % citation style; otherwise the author-date default is used. natbib=true keeps
3607 % \citet/\citep available for users migrating from natbib.
3608 \if@coppenumeric
3609   \RequirePackage[backend=biber,damodel=coppe,%

```



```

3610     bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
3611 \else
3612   \RequirePackage[backend=biber,datamodel=coppe,%
3613     bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
3614 \fi
3615 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
3616 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
3617 % "Bibliografia" (babel's default \bibname). The title is rendered through
3618 % \coppestring so it follows the current main language in BOTH the chapter
3619 % heading and the .toc entry (biblatex's \bibstring proved unreliable across
3620 % the .toc write, and babel's \bibname caption override was equally fragile).
3621 \defbibheading{bibliography}{%
3622   \chapter*{\coppestring{references}}%
3623   \addcontentsline{toc}{chapter}{\coppestring{references}}}
3624 % The catalog page range (bib:begin/bib:end) was written by the old
3625 % thebibliography redefinition, which biblatex's \printbibliography does not
3626 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
3627 % begin label uses that hook and the end label is emitted by wrapping
3628 % \printbibliography to write it just after the reference list.
3629 \AtBeginBibliography{\coppe@bibBegin}
3630 \let\coppe@orig@printbibliography\printbibliography
3631 \renewcommand\printbibliography[1][]{%
3632   \coppe@orig@printbibliography[#1]%
3633   \coppe@bibEnd}
3634 \</class>
3635 \<*glossary>
3636 actual '='
3637 quote '!'
3638 level '>'
3639 %%% Sem numero de pagina, e portanto sem pontilhado.
3640 %%%
3641 %%% A 4.1.1 do Manual descreve a lista de abreviaturas e siglas como uma
3642 %%% relacao alfabetica das siglas com o SIGNIFICADO de cada uma, e a de
3643 %%% simbolos do mesmo modo. Nenhuma das duas e lista de ilustracoes, e nenhuma
3644 %%% pede localizacao. O numero de pagina era heranca de as duas listas terem
3645 %%% sido construidas com a maquina do indice remissivo, onde a pagina e o
3646 %%% ponto.
3647 %%%
3648 %%% O makeindex sempre escreve o numero; quem o descarta e o encapsulador
3649 %%% \coppe@semfolio, definido na classe, que engole o argumento e nao imprime
3650 %%% nada. Por isso todos os delimitadores ficam vazios: sem numero, o
3651 %%% pontilhado levaria o olho ate coisa nenhuma, e a descricao longa passa a
3652 %%% quebrar como texto comum, alinhada sob a primeira linha -- que era o outro
3653 %%% pedido do issue #60.
3654 delim_0 ""
3655 delim_1 ""
3656 delim_2 ""
3657 delim_n ""
3658 delim_r ""
3659 lethead_flag 0
3660 headings_flag 0
3661 preamble
3662 "\n\begin{theglossary}\n \\\makeatletter"

```

```

3663 postamble
3664 "\n \end{theglossary}\n"
3665 </glossary>

3666 <*class>
3667 \newcommand{\annex}{
3668     \renewcommand{\appendixname}{\coppestring{annex}}%
3669     \appendix
3670     \@coppeannextrue
3671 % \appendix has already re-issued the centred chapter format, but it baked in
3672 % the \appendixname of that moment. Re-issue it now that the name is "Anexo".
3673     \coppe@appendixheadings
3674 }
3675 </class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

3676 <*dbx>
3677 \ProvidesFile{coppe.dbx}[2026/09/09 v4.1 CoppeTeX ABNT datamodel]
3678 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
3679 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
3680 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
3681 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
3682 % performance) are already in biblatex's standard datamodel.
3683 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
3684 % Custom fields.
3685 \DeclareDatamodelFields[type=field,datatype=literal]{course} % "... em <course>"
3686 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
3687 \DeclareDatamodelFields[type=field,datatype=literal]{platform} % videogame platform
3688 \DeclareDatamodelFields[type=field,datatype=literal]{scale} % map scale (escala)
3689 \DeclareDatamodelFields[type=list,datatype=literal]{artist} % game/illustration art
3690 \DeclareDatamodelFields[type=list,datatype=literal]{director} % movie/tvshow/video
3691 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
3692 \DeclareDatamodelFields[type=field,datatype=literal]{caderno} % newspaper section
3693 \DeclareDatamodelFields[type=field,datatype=literal]{relator} % jurisprudence rapporteur
3694 \DeclareDatamodelFields[type=field,datatype=literal]{filingdate} % patente: deposito
3695 \DeclareDatamodelFields[type=field,datatype=literal]{grantdate} % patente: concessao
3696 \DeclareDatamodelEntryfields{course,coppedegree}
3697 \DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
3698 {artist,platform,director,producer,version}
3699 \DeclareDatamodelEntryfields[map]{scale}
3700 \DeclareDatamodelEntryfields[article]{caderno}
3701 \DeclareDatamodelEntryfields[jurisdiction]{relator}
3702 \DeclareDatamodelEntryfields[patent]{filingdate,grantdate}
3703 </dbx>
3704 <*bbx>
3705 \ProvidesFile{coppe.bbx}[2026/09/09 v4.1 CoppeTeX ABNT bibliography style]
3706 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in
3707 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
3708 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
3709 % so load setspace here as well (idempotent).
3710 \RequirePackage{setspace}
3711 % ABNT reference list keeps the date at the end (imprenta), same for author-date

```

```

3712 % and numeric modes, so we derive from the standard style, not authoryear.
3713 \RequireBibliographyStyle{standard}
3714
3715 % Note: standard style never dashes repeated authors, which already matches the
3716 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
3717 \ExecuteBibliographyOptions{%
3718   maxbibnames=99,      % ABNT 4.3.1.3: list all authors by default
3719   giveninits=false,
3720   uniquename=false,
3721   uniquelist=false,
3722   labeldateparts=true, % needed for author-date extradate (1999a/1999b)
3723   sorting=nyt,
3724 }
3725
3726 % Language strings (Disponível em / Acesso em / In, etc.)
3727 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
3728 \DeclareLanguageMapping{english}{english-coppe}
3729
3730 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
3731 \DeclareNameAlias{author}{family-given}
3732 \DeclareNameAlias{editor}{family-given}
3733 \DeclareNameAlias{translator}{family-given}
3734 \DeclareNameAlias{default}{family-given}
3735 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
3736 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
3737
3738 % ABNT (NBR 6023 8.1.1.4): a collection entered under the name of its
3739 % organizer prints the kind of responsibility in parentheses after the
3740 % name, abbreviated, lowercase and in the singular even for several
3741 % organizers -- e.g. "SOUSA, A. M. de; OLIVEIRA, L. da C. (org.)". The
3742 % standard style prints ", Org." (comma, no parentheses) when the editor
3743 % takes the author slot, so override editor+others to wrap the role string
3744 % in "(...)". The host position ("In: NAME (org.)") already parentheses
3745 % the role via coppe:bookeditor below. The role word itself is the
3746 % lowercase, singular "org." (and "ed.", "cur.", ...) set in the language
3747 % packs (brazilian-coppe.lbx et al.).
3748 \renewbibmacro*{editor+others}{%
3749   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
3750     {\printnames{editor}%
3751       \setunit*{\addspace}%
3752       \printtext{\mkbibparens{\usebibmacro{editor+othersstrg}}}%
3753       \clearname{editor}}
3754   {}
3755 % Same parenthesised role for drivers that take the editor through the plain
3756 % editor macro instead of editor+others (e.g. @manual).
3757 \renewbibmacro*{editor}{%
3758   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
3759     {\printnames{editor}%
3760       \setunit*{\addspace}%
3761       \printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3762       \clearname{editor}}
3763   {}
3764
3765 % Uppercase the surname (family + prefix) only inside the bibliography list;

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```

3766 % citations keep normal caps (post-2020 rule).
3767 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
3768 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
3769 % field names; the month stays abbreviated via \mkbibmonth.
3770 % Autor-entidade com subordinacao: a 4.3.2.13 poe em caixa alta a jurisdicao ou
3771 % a entidade superior, e deixa em caixa normal o que vem depois -- o orgao
3772 % subordinado ("BRASIL. Supremo Tribunal Federal.") e o qualificador entre
3773 % parenteses ("SAO PAULO (Estado)."). \mkbibnamefamily so conhecia nome de
3774 % pessoa e punha tudo em caixa alta. Corta-se no primeiro " (" e, dentro do que
3775 % sobra a esquerda, no primeiro ". ": o que estiver antes vai para caixa alta,
3776 % o resto sai como foi digitado.
3777 % O biber entrega o nome literal ENTRE CHAVES ("{Brasil. Supremo Tribunal
3778 % Federal}"), e as chaves escondem o ponto do casamento de padrao abaixo: sem
3779 % desembrulhar, o nome inteiro caia no \MakeUppercase. \coppe@ucstrip tira a
3780 % chave externa -- um argumento delimitado que seja um grupo unico perde as
3781 % chaves na captura.
3782 % Autor-entidade: a 4.3.2.13 poe em caixa alta a jurisdicao ou a entidade
3783 % superior e deixa o orgao subordinado em caixa normal -- "BRASIL. Supremo
3784 % Tribunal Federal." Qual das duas coisas o nome e, so quem escreve sabe:
3785 % "Associacao Brasileira de Normas Tecnicas" e uma entidade so, e vai inteira
3786 % para versal; "Brasil. Supremo Tribunal Federal" tem subordinacao, e a parte
3787 % depois do ponto fica como esta.
3788 %
3789 % A regra, entao, e mecanica e previsivel: um nome que TENHA ponto sai
3790 % exatamente como o autor digitou; um nome sem ponto vai para caixa alta. O
3791 % teste e feito sobre a forma detokenizada, porque entre as palavras de um
3792 % nome o biblatex nao poe espaco -- poe \bibnamedelim e \bibnamedelimi --, e
3793 % procurar ". " no fluxo de tokens nunca encontra nada.
3794 % O parentese conta junto com o ponto: "Sao Paulo (Estado)" leva o
3795 % qualificador em caixa normal, e so quem escreve sabe disso.
3796 % O argumento chega como \namepartfamily -- uma sequencia de controle, nao o
3797 % texto. Detokenizar sem expandir antes devolve a string "\namepartfamily",
3798 % que nao tem ponto nenhum, e por isso TODO nome caia no \MakeUppercase.
3799 \protected\def\coppe@ucfamily#1{%
3800   \protected@edef\coppe@uctmp{#1}%
3801   \edef\coppe@ucstr{\detokenize\expandafter{\coppe@uctmp}}%
3802   \expandafter\coppe@ucscan\coppe@ucstr.\coppe@ucstop{#1}%
3803 \protected\def\coppe@ucscan#1.#2\coppe@ucstop#3{%
3804   \ifblank{#2}
3805     {\expandafter\coppe@ucscanp\coppe@ucstr\coppe@ucstop{#3}}
3806     {#3}}
3807 \protected\def\coppe@ucscanp#1(#2\coppe@ucstop#3{%
3808   \ifblank{#2}{\MakeUppercase{#3}}{#3}}
3809 % A 4.3 separa os meses de um intervalo por barra -- "jan./jun. 1991" --, e o
3810 % biblatex usa travessao.
3811 \renewcommand*{\bibdaterangesep}{/}
3812 \protected\def\coppe@abntdate#1#2#3{%
3813   \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
3814   \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
3815   \iffieldundef{#1}{\stripzeros{\thefield{#1}}}}
3816 \AtBeginBibliography{%
3817   \singlespacing % entries single-spaced internally (2.4)
3818   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
3819   \renewcommand*{\mkbibnamefamily}[1]{\coppe@ucfamily{#1}}%

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```

3820 \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
3821 \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
3822 \let\mkbibdateshort\coppe@abntdate
3823 }
3824
3825 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
3826 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
3827 \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
3828 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
3829 % article AND the next word are uppercased; otherwise just the first word. The
3830 % delimited macros split off words; \coppeUCkeep strips the trailing-space
3831 % sentinel so no spurious space is left before the following punctuation.
3832 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
3833 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
3834 \coppeUCifart{#1}
3835 {\MakeUppercase{#1}\ifblank{#2}{ }\{\space\coppeUCtitleB#2\coppeUCstop}}%
3836 {\MakeUppercase{#1}\ifblank{#2}{ }\{\space\coppeUCkeep#2\coppeUCstop}}
3837 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
3838 \MakeUppercase{#1}\ifblank{#2}{ }\{\space\coppeUCkeep#2\coppeUCstop}}
3839 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
3840 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
3841 \protected\def\coppeUCifart#1{%
3842 \begingroup
3843 \lowercase{\def\coppeUC@w{#1}}%
3844 \edef\coppeUC@w{\coppeUC@w,%}
3845 \expandafter\in@\expandafter{\coppeUC@w}{,a,o,as,os,um,uma,uns,umas,the,an,%}
3846 \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
3847 \endgroup}
3848 \DeclareFieldFormat{title}{%
3849 \ifboolexpr{ test {\ifnameundef{labelname}} }
3850 {#1}%
3851 {\mkbibbold{#1}}}
3852 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
3853 % format receives the RAW title text (the title format above only sees the
3854 % field-printing commands), so \coppeUCtitle can split off and uppercase the
3855 % first word here.
3856 \DeclareFieldFormat{titlecase}{%
3857 \ifboolexpr{ test {\ifnameundef{labelname}} }
3858 {\coppeUCtitle{#1}}%
3859 {#1}}
3860 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%
3861 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
3862 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
3863 % the title of a whole document, so override those back to bold.
3864 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
3865 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
3866 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
3867 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
3868 \DeclareFieldFormat[legislation,jurisdiction,legal]{volume}{v\adddot\addnbspace#1}
3869 \DeclareFieldFormat[legislation,jurisdiction,legal]{number}{n\adddot\addnbspace#1}
3870 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
3871 \DeclareFieldFormat[booktitle]{\mkbibbold{#1}}
3872 \DeclareFieldFormat[maintitle]{\mkbibbold{#1}}
3873

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3874 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
3875 \DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
3876 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
3877 \DeclareFieldFormat{pages}{p\adddot\addnbspace#1}
3878
3879 % --- Periodical volume / number prefixes ---
3880 \DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
3881 \DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}
3882
3883 % --- "In:" before host title (incollection/inproceedings/inbook) ---
3884 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
3885
3886 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
3887 \DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}
3888 % 0 4.2 do manual fecha toda entrada eletronica assim, sem execucao:
3889 %   Disponível em: <url>. Acesso em: <data>.
3890 % 0 biblatex, por padrao, poe a data de acesso entre parenteses e sem
3891 % dois-pontos -- "<url> (Acesso em <data>)" --, que e a convencao dele, nao a
3892 % da norma.
3893 \DeclareFieldFormat{urldate}{\bibstring{urlseen}\addcolon\space#1}
3894 % A descricao fisica ("1 carta", "3 fotografias", "1 partitura") vem ANTES do
3895 % bloco eletronico em todos os exemplos da 4.2. Os drivers padrao do biblatex
3896 % a imprimem DEPOIS, entao ela e impressa aqui e o campo e limpo, para nao
3897 % sair duas vezes.
3898 \renewbibmacro*{url+urldate}{%
3899   \iffielddundef{addendum}
3900   {}
3901   {\printfield{addendum}\clearfield{addendum}%
3902     \setunit{\adddot\space}}%
3903   \iffielddundef{url}
3904   {}
3905   {\printfield{url}%
3906     \iffielddundef{urlyear}{\setunit{\adddot\space}\printurldate}}%
3907
3908 % --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL.
3909 %   Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
3910 %   differ: the newspaper name (journaltitle) and city, then either
3911 %   "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
3912 %   "p. <pages>, <date>" (page before date, when none),
3913 %   per NBR 6023 §4.2.3.6. ---
3914 \DeclareBibliographyDriver{article}{%
3915   \usebibmacro{bibindex}%
3916   \usebibmacro{begentry}%
3917   \usebibmacro{author/translator+others}%
3918   \setunit{\labelnamepunct}\newblock
3919   \usebibmacro{title}%
3920   \newunit\newblock
3921   \printfield{journaltitle}%
3922 % "J. Manag. Stud., Oxford": a virgula depois de um titulo abreviado terminado
3923 % em ponto e engolida pelo rastreador de pontuacao, e a 4.2.3.4 a exige.
3924   \iflistundef{location}{\printtext{\setunit{\space}\printlist{location}}%
3925   \iffieldequalstr{entrysubtype}{newspaper}
3926     {\iffielddundef{caderno}
3927       {\setunit{\addcomma\space}\printfield{pages}}%

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```

3928     \setunit{\addcomma\space}\printdate}%
3929     {\setunit{\addcomma\space}\printdate%
3930     \setunit{\adddot\space}\printfield{caderno}%
3931     \setunit{\addcomma\space}\printfield{pages}}}%
3932     {\setunit{\addcomma\space}\printfield{volume}%
3933     \setunit{\addcomma\space}\printfield{number}%
3934     \setunit{\addcomma\space}\printfield{pages}%
3935     \setunit{\addcomma\space}\printdate}%
3936 \newunit\newblock
3937 \printfield{doi}%
3938 \newunit\newblock
3939 \usebibmacro{url+urldate}%
3940 \usebibmacro{finentry}}
3941
3942 % --- incollection: ABNT puts the host organizer before the booktitle, as
3943 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
3944 % p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
3945 \newbibmacro*{coppe:bookeditor}{%
3946   \ifnameundef{editor}
3947   {}
3948   {\printnames{editor}%
3949     \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3950     \clearname{editor}%
3951     \setunit{\adddot\space}}}
3952 \newbibmacro*{coppe:partdriver}{%
3953   \usebibmacro{bibindex}\usebibmacro{begentry}%
3954   \usebibmacro{author/translator+others}%
3955   \setunit{\labelnamepunct}\newblock
3956   \usebibmacro{title}%
3957   \newunit\newblock
3958   \usebibmacro{in:}%
3959   \usebibmacro{coppe:bookeditor}%
3960   \printfield{booktitle}%
3961   \newunit\newblock
3962   \printfield{edition}%
3963   \newunit\newblock
3964   \printlist{location}%
3965   \setunit*{\addcolon\space}\printlist{publisher}%
3966   \setunit*{\addcomma\space}\printdate%
3967   \newunit\newblock
3968   \printfield{pages}%
3969   \setunit{\addcomma\space}\printfield{chapter}%
3970   \newunit\newblock
3971   \usebibmacro{url+urldate}%
3972   \usebibmacro{finentry}}
3973 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
3974 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
3975
3976 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
3977 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
3978
3979 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
3980 % "(Mestrado em Memoria Social e Documento)": a 4.2.1.1 poe o grau dentro dos
3981 % parenteses, antes do curso, e deixa o substantivo do documento fora deles --

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3982 % "Dissertacao (Mestrado em ...)". O grau vem do campo coppedegree, que o
3983 % sourcemap preenche junto com o tipo.
3984 % O grau sai do proprio tipo do trabalho, que ja e uma chave de string; passar
3985 % a chave por um campo e depois resolve-la com \ifbibstring nao funciona,
3986 % porque \thefield nao se expande onde \ifbibstring precisa. Um campo
3987 % coppedegree preenchido a mao pelo autor tem precedencia.
3988 \newcommand*\coppe@degword{%
3989   \iffieldundef{coppedegree}
3990     {\iffieldequalstr{type}{mscdiss}{\bibstring{coppemscdeg}}{%
3991       \iffieldequalstr{type}{dscthesis}{\bibstring{coppedscdeg}}{%
3992         \iffieldequalstr{type}{phdthesis}{\bibstring{coppephddeg}}{%
3993           \iffieldequalstr{type}{tcc}{\bibstring{coppetccdeg}}{}}}}}%
3994   {\printfield{coppedegree}}}%
3995 \DeclareFieldFormat{course}{%
3996   \mkbibparens{\coppe@degword\addspace\coppestring{in:}\addspace#1}%
3997 % A ordem da 4.2.1.1 para tese e dissertacao:
3998 %   AUTOR. Titulo. ano. NNN f. Tipo (Grau em Curso) -- Instituicao, Local, ano.
3999 % A data aparece duas vezes de proposito, como nos exemplos do manual: uma
4000 % depois do titulo, outra fechando a nota de tese.
4001 \DeclareBibliographyDriver{thesis}{%
4002   \usebibmacro{bibindex}\usebibmacro{begentry}%
4003   \usebibmacro{author}%
4004   \setunit{\labelnamepunct}\newblock
4005   \usebibmacro{title}%
4006   \newunit\newblock
4007   \printdate%
4008   \newunit\newblock
4009   \printfield{pagetotal}%
4010   \newunit\newblock
4011   \printfield{type}\setunit*{\addspace}\printfield{course}%
4012   \iflistundef{institution}
4013     {}
4014     {\setunit*{\addspace\textendash\addspace}\printlist{institution}}%
4015   \setunit*{\addcomma\space}\printlist{location}%
4016   \setunit*{\addcomma\space}\printdate%
4017   \newunit\newblock
4018   \printfield{note}%
4019   \newunit\newblock
4020   \usebibmacro{url+urldate}%
4021   \usebibmacro{finentry}}%
4022 % --- Games / audiovisual: render the custom fields (artist, platform,
4023 %   director, producer, version) the generic misc driver ignores. ---
4024 \newbibmacro*{coppe:media}{%
4025   \iflistundef{artist}{ }\setunit{\addperiod\space}%
4026   \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
4027   \iffieldundef{platform}{ }\setunit{\addperiod\space}%
4028   \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}%
4029   \iflistundef{director}{ }\setunit{\addperiod\space}%
4030   \printtext{\coppestring{directedby}\addspace}\printlist{director}}%
4031   \iflistundef{producer}{ }\setunit{\addperiod\space}%
4032   \printtext{\coppestring{producedby}\addspace}\printlist{producer}}%
4033 \newbibmacro*{coppe:mediadriver}{%
4034   \usebibmacro{bibindex}\usebibmacro{begentry}%
4035   \usebibmacro{author/translator+others}%

```



```

4036 \setunit{\labelnamepunct}\newblock
4037 \usebibmacro{title}%
4038 \newunit\newblock
4039 \usebibmacro{coppe:media}%
4040 \newunit\newblock
4041 \usebibmacro{coppe:imprenta}%
4042 \newunit\newblock
4043 \printfield{version}%
4044 \newunit\newblock
4045 \printfield{note}%
4046 \newunit\newblock
4047 \usebibmacro{url+urldate}%
4048 \usebibmacro{finentry}}
4049 % --- Publicacao periodica (4.2.3.1 a 4.2.3.3) -----
4050 % O titulo do periodico vai INTEIRO em caixa alta, e nao so a primeira palavra
4051 % como numa obra sem autor: e a forma dos exemplos do manual. Sem driver
4052 % proprio saia o default do biblatex -- "SAO Paulo Medical Journal (1941)." --,
4053 % sem local, sem editora e com o ano entre parenteses.
4054 \DeclareFieldFormat[periodical,suppperiodical]{title}{\MakeUppercase{#1}}
4055 \DeclareFieldFormat[periodical,suppperiodical]{volume}{v\addot\addnbspace#1}
4056 \DeclareFieldFormat[periodical,suppperiodical]{number}{n\addot\addnbspace#1}
4057 \DeclareBibliographyDriver{periodical}{%
4058 \usebibmacro{bibindex}\usebibmacro{begentry}%
4059 % Pelo campo, e nao pelo bibmacro: o bibmacro 'title' passa pelo formato
4060 % 'titlecase', e num periodico -- que nao tem autor -- ele nao chegava a
4061 % imprimir coisa nenhuma. A entrada saia começando por um ponto solto.
4062 \printfield{title}%
4063 \newunit\newblock
4064 \printlist{location}%
4065 \iflistundef{publisher}{\setunit*{\addcolon\space}\printlist{publisher}}%
4066 \setunit*{\addcomma\space}\printfield{volume}%
4067 \setunit*{\addcomma\space}\printfield{number}%
4068 \setunit*{\addcomma\space}\printdate%
4069 \newunit\newblock
4070 \printfield{note}%
4071 \newunit\newblock
4072 \usebibmacro{url+urldate}%
4073 \usebibmacro{finentry}}
4074 \DeclareBibliographyDriver{suppperiodical}{%
4075 \usedriver{}{periodical}\usebibmacro{finentry}}
4076 % --- Eventos (4.2.4) -----
4077 % O nome do evento vai em caixa alta, seguido do numero da edicao, do ano e da
4078 % cidade -- "ENCONTRO ANUAL DA ANPAD, 14., 1982, Florianopolis." --, e so
4079 % depois vem o titulo dos anais, que e o elemento em destaque. Por isso o nome
4080 % do evento tem campo proprio (|eventtitle|, sinonimo |evento|) em vez de vir
4081 % empacotado no titulo: nada mais na entrada vai para caixa alta.
4082 \DeclareFieldFormat{eventtitle}{\MakeUppercase{#1}}
4083 % "Anais [...]" nao e uma obra entrada pelo titulo: quem faz as vezes de
4084 % cabeca da entrada e o nome do evento, ja em caixa alta. Sem esta declaracao
4085 % o titulo dos anais saia "ANAIS [...]" e ainda perdia o ponto que o separa
4086 % da imprenta.
4087 \DeclareFieldFormat[proceedings]{title}{#1}
4088 \DeclareFieldFormat[proceedings,inproceedings]{number}{#1\addot}
4089 \DeclareFieldFormat[proceedings,inproceedings]{volume}{v\addot\addnbspace#1}

```

```

4090 \newbibmacro*{coppe:evento}{%
4091   \printfield{eventtitle}%
4092   \setunit*{\addcomma\space}\printfield{number}%
4093   \setunit*{\addcomma\space}\printeventdate%
4094   \setunit*{\addcomma\space}\printfield{venue}}
4095 \DeclareBibliographyDriver{proceedings}{%
4096   \usebibmacro{bibindex}\usebibmacro{begentry}%
4097   \usebibmacro{coppe:evento}%
4098   \newunit\newblock
4099 % O ponto que separa o titulo dos anais da imprensa e literal: o rastreador de
4100 % pontuacao do biblatex nao reconhece o fim de um titulo terminado em "]", e
4101 % nem \newunit nem \addot produzia alguma coisa ali.
4102   \iffielddundef{title}{ }\{\printfield{title}\printtext{.}\}%
4103   \setunit{\space}\newblock
4104   \usebibmacro{coppe:imprensa}%
4105   \newunit\newblock
4106   \usebibmacro{url+urldate}%
4107   \usebibmacro{finentry}}
4108 \DeclareBibliographyDriver{inproceedings}{%
4109   \usebibmacro{bibindex}\usebibmacro{begentry}%
4110   \usebibmacro{author/translator+others}%
4111   \setunit{\labelnamepunct}\newblock
4112   \usebibmacro{title}%
4113   \newunit\newblock
4114   \usebibmacro{in:}%
4115   \usebibmacro{coppe:evento}%
4116   \newunit\newblock
4117   \iffielddundef{booktitle}{ }\{\printfield{booktitle}\printtext{.}\}%
4118   \setunit{\space}\newblock
4119   \usebibmacro{coppe:imprensa}%
4120   \newunit\newblock
4121   \printfield{volume}%
4122   \setunit*{\addcomma\space}\printfield{pages}%
4123   \newunit\newblock
4124   \usebibmacro{url+urldate}%
4125   \usebibmacro{finentry}}
4126 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadrivere}}
4127 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadrivere}}
4128 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadrivere}}
4129 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadrivere}}
4130 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadrivere}}
4131 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
4132 % imprensa) plus the mandatory scale ("Escala 1:N").
4133 \DeclareFieldFormat{scale}{\bibstring{coppescale}\adnbspace#1}
4134 % O dois-pontos separa local e editora; sem editora ele vazava para a data
4135 % ("Recife: 1976"), e a 4.2.12 quer "Recife, 1976". Por isso a guarda.
4136 \newbibmacro*{coppe:imprensa}{%
4137   \printlist{location}%
4138   \iflistundef{publisher}
4139     {}
4140     {\setunit*{\addcolon\space}\printlist{publisher}}%
4141   \setunit*{\addcomma\space}\printdate}
4142 \DeclareBibliographyDriver{map}{%
4143   \usebibmacro{bibindex}\usebibmacro{begentry}%

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```

4144 \usebibmacro{author/translator+others}%
4145 \setunit{\labelnamepunct}\newblock
4146 \usebibmacro{title}%
4147 \newunit\newblock
4148 \usebibmacro{coppe:imprenta}%
4149 \newunit\newblock
4150 % "Recife, 1976. 71 fotografias. Escala 1: 20.000." -- a descricao fisica vem
4151 % ANTES da escala, e por isso ela nao pode esperar pelo bloco eletronico.
4152 \iffieldundef{addendum}{\printfield{addendum}\clearfield{addendum}}%
4153 \newunit\newblock
4154 \printfield{scale}%
4155 \newunit\newblock
4156 \printfield{note}%
4157 \newunit\newblock
4158 \usebibmacro{url+urldate}%
4159 \usebibmacro{finentry}}
4160 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
4161 % number. date. The holder (depositante) is printed in normal case, not the
4162 % ALL-CAPS family treatment the bibliography gives the leading author.
4163 \newbibmacro*{coppe:depositor}{%
4164 \ifnameundef{holder}
4165 {}
4166 {\printtext{\bibstring{depositor}\addcolon\space}%
4167 \begingroup
4168 \def\mkbibnamefamily##1{##1}%
4169 \def\mkbibnameprefix##1{##1}%
4170 \printnames{holder}%
4171 \endgroup}}
4172 \DeclareBibliographyDriver{patent}{%
4173 \usebibmacro{bibindex}%
4174 \usebibmacro{begentry}%
4175 \usebibmacro{author/translator+others}%
4176 \setunit{\labelnamepunct}\newblock
4177 \usebibmacro{title}%
4178 \newunit\newblock
4179 \usebibmacro{coppe:depositor}%
4180 \newunit\newblock
4181 \printfield{type}%
4182 \setunit*{\addspace}\printfield{number}%
4183 \newunit\newblock
4184 % A 4.2.5 fecha a patente com as duas datas, nomeadas: "Deposito: 27 jan. 2006.
4185 % Concessao: 25 mar. 2008." Sem elas so saia o ano.
4186 \iffieldundef{filingdate}
4187 {}
4188 {\printtext{\bibstring{coppefiling}\addcolon\space}\printfield{filingdate}%
4189 \setunit{\addot\space}}%
4190 \iffieldundef{grantdate}
4191 {}
4192 {\printtext{\bibstring{coppegrant}\addcolon\space}\printfield{grantdate}}%
4193 \ifbool{test {\iffieldundef{filingdate}} and test {\iffieldundef{grantdate}}} {
4194 {\printdate}
4195 {}%
4196 \newunit\newblock
4197 \usebibmacro{url+urldate}%

```

```

4198 \usebibmacro{finentry}}
4199 % Exotic ABNT types that biblatex's standard style leaves without a driver:
4200 % give each a baseline (book-like for published docs: norma/partitura; misc for
4201 % the rest) so they print. Per-type ABNT refinements can follow later.
4202 % \usedriver desliga o \finentry do driver chamado, porque quem chama e que
4203 % deve fechar a entrada -- e aqui ninguem fechava. Resultado: norma, partitura,
4204 % audiovisual, iconografico, obra de arte, tridimensional e conjunto de dados
4205 % saiam sem o ponto final, que a 4.2 poe em toda referencia.
4206 \DeclareBibliographyDriver{standard}{\usedriver{}{book}\usebibmacro{finentry}}
4207 \DeclareBibliographyDriver{music}{\usedriver{}{book}\usebibmacro{finentry}}
4208 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
4209 % the law id + ementa (plain, not bold), then location and date.
4210 \newbibmacro*{coppe:legaldriver}{%
4211 \usebibmacro{bibindex}\usebibmacro{begentry}%
4212 \usebibmacro{author/translator+others}%
4213 \setunit{\labelnamepunct}\newblock
4214 \usebibmacro{title}%
4215 \newunit\newblock
4216 % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
4217 \iffieldundef{relator}{\setunit{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\newblock}}{\printfield{relator}\newunit\newblock}
4218 \printfield{edition}%
4219 \newunit\newblock
4220 \printlist{location}%
4221 % A 4.2.6 traz o veiculo de publicacao com volume, numero e pagina -- "Sao
4222 % Paulo, v. 62, n. 3, p. 217-220, 1998." --, que se perdiam por completo.
4223 \iflistundef{publisher}{\setunit*\addcolon\space\printlist{publisher}}{\setunit*\addcolon\space\printfield{volume}%
4224 \setunit*\addcomma\space\printfield{number}%
4225 \setunit*\addcomma\space\printfield{pages}%
4226 \setunit*\addcomma\space\printdate%
4227 \newunit\newblock
4228 \printfield{note}%
4229 \newunit\newblock
4230 \usebibmacro{url+urldate}%
4231 \usebibmacro{finentry}}
4232 % 4.2.14: "UNIVERSIDADE ... Rio de Janeiro, 1998. Disponível em: ..." -- o
4233 % driver padrao do biblatex para @online nao imprime o local.
4234 \DeclareBibliographyDriver{online}{%
4235 \usebibmacro{bibindex}\usebibmacro{begentry}%
4236 \usebibmacro{author/translator+others}%
4237 \setunit{\labelnamepunct}\newblock
4238 \usebibmacro{title}%
4239 \newunit\newblock
4240 \usebibmacro{coppe:imprenta}%
4241 \newunit\newblock
4242 \printfield{note}%
4243 \newunit\newblock
4244 \usebibmacro{url+urldate}%
4245 \usebibmacro{finentry}}
4246 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
4247 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
4248 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
4249 % Audio e video vao pelo driver de midia, que imprime diretor e produtor --
4250 % "Producao de Jorge Ramos de Andrade." da 4.2.9. O driver misc os ignorava.

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4252 \DeclareBibliographyDriver{audio}{\usebibmacro{coppe:mediadriv}}
4253 \DeclareBibliographyDriver{video}{\usebibmacro{coppe:mediadriv}}
4254 \DeclareBibliographyDriver{software}{\usedriver{}{misc}\usebibmacro{finentry}}
4255 \DeclareBibliographyDriver{image}{\usedriver{}{misc}\usebibmacro{finentry}}
4256 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}\usebibmacro{finentry}}
4257 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}\usebibmacro{finentry}}
4258 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}\usebibmacro{finentry}}
4259
4260 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
4261 \DeclareSourcemap{%
4262   \maps[datatype=bibtex]{%
4263     % entry-type PT synonyms -> canonical English
4264     \map{\step[tourcesource=livro,typetarget=book]}
4265     \map{\step[tourcesource=artigo,typetarget=article]}
4266     \map{\step[tourcesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
4267     \map{\step[tourcesource=partedelivro,typetarget=incollection]}
4268     \map{\step[tourcesource=emlivro,typetarget=inbook]}
4269     \map{\step[tourcesource=verbete,typetarget=inreference]}
4270     \map{\step[tourcesource=anais,typetarget=proceedings]}
4271     \map{\step[tourcesource=trabalhodeevento,typetarget=inproceedings]}
4272     \map{\step[tourcesource=periodico,typetarget=periodical]}
4273     \map{\step[tourcesource=relatorio,typetarget=report]}
4274     \map{\step[tourcesource=relatoriotecnico,typetarget=report]}
4275     \map{\step[tourcesource>manual,typetarget>manual]}
4276     \map{\step[tourcesource=correspondencia,typetarget=letter]}
4277     \map{\step[tourcesource=patente,typetarget=patent]}
4278     \map{\step[tourcesource=diverso,typetarget=misc]}
4279     \map{\step[tourcesource=jogo,typetarget=game]}
4280     \map{\step[tourcesource=jogoeletronico,typetarget=videogame]}
4281     \map{\step[tourcesource=jogodetabuleiro,typetarget=boardgame]}
4282     \map{\step[tourcesource=filme,typetarget=movie]}
4283     \map{\step[tourcesource=serietv,typetarget=tvshow]}
4284     \map{\step[tourcesource=programadettv,typetarget=tvshow]}
4285     % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
4286     % except map (declared in coppe.dbx).
4287     \map{\step[tourcesource=legislacao,typetarget=legislation]}
4288     \map{\step[tourcesource=atonormativo,typetarget=legislation]}
4289     \map{\step[tourcesource=jurisprudencia,typetarget=jurisdiction]}
4290     \map{\step[tourcesource=documentojuridico,typetarget=legal]}
4291     \map{\step[tourcesource=documentocivil,typetarget=legal]}
4292     \map{\step[tourcesource=norma,typetarget=standard]}
4293     \map{\step[tourcesource=partitura,typetarget=music]}
4294     \map{\step[tourcesource=iconografico,typetarget=image]}
4295     \map{\step[tourcesource=obradearte,typetarget=artwork]}
4296     \map{\step[tourcesource=tridimensional,typetarget=artwork]}
4297     \map{\step[tourcesource=cartografico,typetarget=map]}
4298     \map{\step[tourcesource=mapa,typetarget=map]}
4299     \map{\step[tourcesource=audiovisual,typetarget=video]}
4300     \map{\step[tourcesource=conjuntodedados,typetarget=dataset]}
4301     \map{\step[tourcesource=fasciculo,typetarget=supperperiodical]}
4302     \map{\step[tourcesource=trabalhoapresentado,typetarget=unpublished]}
4303     % thesis family: retype to thesis and preset the (babel-aware) type word
4304     \map{\step[tourcesource=tese,typetarget=thesis]}
4305     \map{\step[tourcesource=dissertacao,typetarget=thesis]}

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4306 \map{\step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
4307 \map{\step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fieldv
4308 \map{\step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
4309 \map{\step[typesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
4310 \map{\step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
4311 \map{\step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
4312 \map{\step[typesource=tesephd,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
4313 \map{\step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
4314 % field PT synonyms (unaccented) -> canonical English
4315 \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
4316 \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
4317 \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
4318 \map{\step[fieldsource=titulo,fieldtarget=title]}
4319 \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
4320 \map{\step[fieldsource=autor,fieldtarget=author]}
4321 \map{\step[fieldsource=editora,fieldtarget=publisher]}
4322 \map{\step[fieldsource=local,fieldtarget=location]}
4323 \map{\step[fieldsource=ano,fieldtarget=year]}
4324 \map{\step[fieldsource=data,fieldtarget=date]}
4325 \map{\step[fieldsource=edicao,fieldtarget=edition]}
4326 \map{\step[fieldsource=paginas,fieldtarget=pages]}
4327 \map{\step[fieldsource=numero,fieldtarget=number]}
4328 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
4329 \map{\step[fieldsource=organizador,fieldtarget=editor]}
4330 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
4331 \map{\step[fieldsource=tipo,fieldtarget=type]}
4332 \map{\step[fieldsource=nota,fieldtarget=note]}
4333 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
4334 \map{\step[fieldsource=curso,fieldtarget=course]}
4335 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
4336 \map{\step[fieldsource=artista,fieldtarget=artist]}
4337 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
4338 \map{\step[fieldsource=diretor,fieldtarget=director]}
4339 \map{\step[fieldsource=produtor,fieldtarget=producer]}
4340 \map{\step[fieldsource=versao,fieldtarget=version]}
4341 \map{\step[fieldsource=escala,fieldtarget=scale]}
4342 \map{\step[fieldsource=evento,fieldtarget=eventtitle]}
4343 \map{\step[fieldsource=dataevento,fieldtarget=eventdate]}
4344 \map{\step[fieldsource=locaevento,fieldtarget=venue]}
4345 \map{\step[fieldsource=deposito,fieldtarget=filingdate]}
4346 \map{\step[fieldsource=concessao,fieldtarget=grantdate]}
4347 }%
4348 }
4349
4350 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
4351 \setlength{\bibhang}{0pt}
4352 \setlength{\bibitemsep}{\baselineskip}
4353
4354 </bbx>
4355 <*cbx>
4356 \ProvidesFile{coppe.cbx}[2026/09/09 v4.1 CoppeTeX ABNT author-date citations]
4357 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
4358 % author-year cites with normal-caps names (post-2020 rule) out of the box.
4359 \RequireCitationStyle{authoryear-comp}

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4360
4361 % --- NBR 10520:2023 author separator -----
4362 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
4363 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
4364 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
4365 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
4366 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
4367 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
4368 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
4369 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
4370
4371 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
4372 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
4373 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
4374 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
4375 % so it is never added to the bibliography.
4376 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
4377 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
4378 % \citeapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
4379 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
4380 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
4381 \DeclareRobustCommand{\citetapud}[4][]{%
4382   #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
4383 \DeclareRobustCommand{\citeapud}[4][]{%
4384   \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
4385     \citeauthor{#4}\addcomma\space\citeyear{#4}%
4386     \ifblank{#1}{ }\addcomma\space\mkpageprefix[pagination]{#1}}}%
4387 \end{cbx}
4388 \end{lbxbr}
4389 \ProvidesFile{brazilian-coppe.lbx}[2026/09/09 v4.1 CoppeTeX brazilian strings]
4390 \InheritBibliographyExtras{brazilian}
4391 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale,
4392   coppemscdeg,coppedscdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4393 \DeclareBibliographyStrings{%
4394   inherit = {brazilian},
4395   availablefrom = {{Dispon\'}{\i}vel em}{Dispon\'}{\i}vel em}},
4396   coppemscdeg = {{Mestrado}{Mestrado}},
4397   coppedscdeg = {{Doutorado}{Doutorado}},
4398   coppephddeg = {{Doutorado}{Doutorado}},
4399   coppetccdeg = {{Gradua\c{c}\~{a}o}{Gradua\c{c}\~{a}o}},
4400   coppefiling = {{Dep\'}osito}{Dep\'}osito}},
4401   coppegrant = {{Concess\~{a}o}{Concess\~{a}o}},
4402   urlseen = {{Acesso em}{Acesso em}},
4403 % A 4.3 abrevia os meses nas tres primeiras letras seguidas de ponto -- menos
4404 % maio, que se escreve por extenso. O biblalex-brazilian abrevia "mai.".
4405   may = {{maio}{maio}},
4406   in = {{In}{In}},
4407   editor = {{organizador}{org\addot}},
4408   editors = {{organizadores}{org\addot}},
4409   depositor = {{Depositante}{Depositante}},
4410   coppescale = {{Escala}{Escala}},
4411   mscdiss = {{Disserta\c{c}\~{a}o}{Disserta\c{c}\~{a}o}},
4412   dscthesi = {{Tese}{Tese}},
4413   phdthesis = {{Tese}{Tese}},

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4414 tcc          = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
4415 }
4416 \lbr/>
4417 \lbr/>
4418 \ProvidesFile{english-coppe.lbx}[2026/09/09 v4.1 CoppeTeX english strings]
4419 \InheritBibliographyExtras{english}
4420 \NewBibliographyString{availablefrom,mscdiss,dsctthesis,tcc,depositor,coppescale,
4421   coppemscdeg,coppedscdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4422 \DeclareBibliographyStrings{%
4423   inherit      = {english},
4424   availablefrom = {{Available from}{Available from}},
4425   coppemscdeg   = {{M\addot Sc\addot}{M\addot Sc\addot}},
4426   coppedscdeg   = {{D\addot Sc\addot}{D\addot Sc\addot}},
4427   coppephddeg   = {{Ph\addot D\addot}{Ph\addot D\addot}},
4428   coppetccdeg   = {{B\addot Sc\addot}{B\addot Sc\addot}},
4429   coppefiling   = {{Filed}{Filed}},
4430   coppegrant    = {{Granted}{Granted}},
4431   urlseen       = {{Access on}{Access on}},
4432   in            = {{In}{In}},
4433   editor        = {{editor}{ed\addot}},
4434   editors       = {{editors}{ed\addot}},
4435   depositor     = {{Depositor}{Depositor}},
4436   coppescale    = {{Scale}{Scale}},
4437   mscdiss       = {{M\addot Sc\addot\ Dissertation}{M\addot Sc\addot\ Diss\addot}},
4438   dsctthesis    = {{D\addot Sc\addot\ Thesis}{D\addot Sc\addot\ Thesis}},
4439   phdthesis     = {{Ph\addot D\addot\ Thesis}{Ph\addot D\addot\ Thesis}},
4440   tcc           = {{Undergraduate Thesis}{Undergraduate Thesis}},
4441 }
4442 \lbr/>
4443 \lbr/>
4444 \ProvidesFile{spanish-coppe.lbx}[2026/09/09 v4.1 CoppeTeX spanish strings]
4445 \InheritBibliographyExtras{spanish}
4446 \NewBibliographyString{availablefrom,mscdiss,dsctthesis,tcc,depositor,coppescale,
4447   coppemscdeg,coppedscdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4448 \DeclareBibliographyStrings{%
4449   inherit      = {spanish},
4450   availablefrom = {{Disponible en}{Disponible en}},
4451   coppemscdeg   = {{Maestr\~{a}}{Maestr\~{a}}},
4452   coppedscdeg   = {{Doctorado}{Doctorado}},
4453   coppephddeg   = {{Doctorado}{Doctorado}},
4454   coppetccdeg   = {{Grado}{Grado}},
4455   coppefiling   = {{Dep\~{o}sito}{Dep\~{o}sito}},
4456   coppegrant    = {{Concesi\~{o}n}{Concesi\~{o}n}},
4457   urlseen       = {{Consultado el}{Consultado el}},
4458   in            = {{En}{En}},
4459   editor        = {{editor}{ed\addot}},
4460   editors       = {{editores}{ed\addot}},
4461   depositor     = {{Solicitante}{Solicitante}},
4462   coppescale    = {{Escala}{Escala}},
4463   mscdiss       = {{Tesis de Maestr\~{a}}{Tesis de Maestr\~{a}}},
4464   dsctthesis    = {{Tesis Doctoral}{Tesis Doctoral}},
4465   phdthesis     = {{Tesis Doctoral}{Tesis Doctoral}},
4466   tcc           = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
4467 }

```



```

4468 </lbxes>
4469 <*lbxf>
4470 \ProvidesFile{french-coppe.lbx}[2026/09/09 v4.1 CoppeTeX french strings]
4471 \InheritBibliographyExtras{french}
4472 \NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale,
4473   coppemscdeg,coppedsdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4474 \DeclareBibliographyStrings{%
4475   inherit      = {french},
4476   availablefrom = {{Disponibile \a}{Disponibile \a}},
4477   coppemscdeg  = {{Master}{Master}},
4478   coppedsdeg   = {{Doctorat}{Doctorat}},
4479   coppephddeg  = {{Doctorat}{Doctorat}},
4480   coppetccdeg  = {{Licence}{Licence}},
4481   coppefiling  = {{D\ep\ot}{D\ep\ot}},
4482   coppegrant   = {{D\elivrance}{D\elivrance}},
4483   urlseen     = {{Consult\le}{Consult\le}},
4484   in           = {{Dans}{Dans}},
4485   editor       = {{\ed}{\ed\adddot}},
4486   editors      = {{\editeurs}{\ed\adddot}},
4487   depositor    = {{D\eposant}{D\eposant}},
4488   coppescale   = {{\Echelle}{\Echelle}},
4489   mscdiss      = {{M\emoire de Master}{M\emoire de Master}},
4490   dscthesis    = {{Th\ese de Doctorat}{Th\ese de Doctorat}},
4491   phdthesis    = {{Th\ese de Doctorat}{Th\ese de Doctorat}},
4492   tcc          = {{M\emoire de fin d\etudes}{M\emoire de fin d\etudes}},
4493 }
4494 </lbxf>
4495 <*lbxit>
4496 \ProvidesFile{italian-coppe.lbx}[2026/09/09 v4.1 CoppeTeX italian strings]
4497 \InheritBibliographyExtras{italian}
4498 \NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale,
4499   coppemscdeg,coppedsdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4500 \DeclareBibliographyStrings{%
4501   inherit      = {italian},
4502   availablefrom = {{Disponibile da}{Disponibile da}},
4503   urlseen      = {{Consultato il}{Consultato il}},
4504   in           = {{In}{In}},
4505   editor       = {{curatore}{cur\adddot}},
4506   editors      = {{curatori}{cur\adddot}},
4507   depositor    = {{Depositante}{Depositante}},
4508   coppescale   = {{Scala}{Scala}},
4509   mscdiss      = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
4510   dscthesis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
4511   phdthesis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
4512   tcc          = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
4513 }
4514 </lbxit>
4515 <*langes>
4516 %% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.
4517 %% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other
4518 %% spanish slot) is active. Populates the dispatcher table for the
4519 %% spanish language and registers the biblatex language mapping.
4520 %% Institutional names (universityname/cityname/statename/countryname)
4521 %% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de

```

```

4522 %% rosto are a Brazilian institutional template that does not translate.
4523 \ProvidesFile{coppe-lang-spanish.def}[2026/09/09 v4.1 CoppeTeX spanish class strings]
4524 \copperdefstring{spanish}{appendix}      {Ap\endice}
4525 \copperdefstring{spanish}{annex}         {Anexo}
4526 \copperdefstring{spanish}{frame}         {Cuadro}
4527 \copperdefstring{spanish}{listframe}     {Lista de Cuadros}
4528 \copperdefstring{spanish}{source}        {Fuente}
4529 \copperdefstring{spanish}{program}        {Programa}
4530 \copperdefstring{spanish}{listprogram}    {Lista de Programas}
4531 \copperdefstring{spanish}{references}     {Referencias}
4532 \copperdefstring{spanish}{in:}           {en}
4533 \copperdefstring{spanish}{art}            {Arte}
4534 \copperdefstring{spanish}{platform}       {Plataforma}
4535 \copperdefstring{spanish}{directedby}     {Direcci\on}
4536 \copperdefstring{spanish}{producedby}    {Producci\on}
4537 \copperdefstring{spanish}{listabbreviation}{Lista de Abreviaturas y Siglas}
4538 \copperdefstring{spanish}{listsymbol}     {Lista de S\imbolos}
4539 \copperdefstring{spanish}{glossary}       {Glosario}
4540 \copperdefstring{spanish}{advisor}        {Director}
4541 \copperdefstring{spanish}{advisors}       {Directores}
4542 \copperdefstring{spanish}{dept}           {Programa}
4543 \copperdefstring{spanish}{approvedmale}   {Aprobado por}
4544 \copperdefstring{spanish}{approvedfemale} {Aprobada por}
4545 \copperdefstring{spanish}{approvedonmale} {Aprobado el}
4546 \copperdefstring{spanish}{approvedonfemale}{Aprobada el}
4547 \copperdefstring{spanish}{keywordsname}  {Palabras clave}
4548 \copperdefstring{spanish}{volumename}    {Volumen}
4549 \copperdefstring{spanish}{coadvisor}     {Codirector}
4550 \copperdefstring{spanish}{coadvisors}    {Codirectores}
4551 \copperdefstring{spanish}{ofname}        {de}
4552 \copperdefstring{spanish}{researchline}{L\i nea de investigaci\on}
4553 \copperdefstring{spanish}{tbddate}       {por determinar}
4554 \copperdefstring{spanish}{universityname}{Universidade Federal do Rio de Janeiro}
4555 \copperdefstring{spanish}{cityname}      {Rio de Janeiro}
4556 \copperdefstring{spanish}{statename}     {RJ}
4557 \copperdefstring{spanish}{countryname}   {Brasil}
4558 \copperdefstring{spanish}{monthname1}    {enero}
4559 \copperdefstring{spanish}{monthname2}    {febrero}
4560 \copperdefstring{spanish}{monthname3}    {marzo}
4561 \copperdefstring{spanish}{monthname4}    {abril}
4562 \copperdefstring{spanish}{monthname5}    {mayo}
4563 \copperdefstring{spanish}{monthname6}    {junio}
4564 \copperdefstring{spanish}{monthname7}    {julio}
4565 \copperdefstring{spanish}{monthname8}    {agosto}
4566 \copperdefstring{spanish}{monthname9}    {septiembre}
4567 \copperdefstring{spanish}{monthname10}   {octubre}
4568 \copperdefstring{spanish}{monthname11}   {noviembre}
4569 \copperdefstring{spanish}{monthname12}   {diciembre}
4570 \copperdefstring{spanish}{algorithm2eopt}{spanish}
4571 \copperdefstring{spanish}{babelname}     {spanish}
4572 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
4573 %% language-mapping registration until \begin{document} so biblatex's
4574 %% \DeclareLanguageMapping is available.
4575 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}

```

```

4576 </langes>
4577 <*langfr>
4578 %% coppe-lang-french.def -- CoppeTeX 4.x French class-string pack.
4579 %% See coppe-lang-spanish.def for the structure. Institutional names
4580 %% remain Portuguese; only document-content strings translate.
4581 \ProvidesFile{coppe-lang-french.def}[2026/09/09 v4.1 CoppeTeX french class strings]
4582 \copperdefstring{french}{appendix}      {Annexe}
4583 \copperdefstring{french}{annex}         {Annexe}
4584 \copperdefstring{french}{frame}         {Encadr\'e}
4585 \copperdefstring{french}{listframe}     {Liste des encadr\'es}
4586 \copperdefstring{french}{source}        {Source}
4587 \copperdefstring{french}{program}       {Programme}
4588 \copperdefstring{french}{listprogram}   {Liste des programmes}
4589 \copperdefstring{french}{references}    {R\'ef\'erences}
4590 \copperdefstring{french}{in:}           {dans}
4591 \copperdefstring{french}{art}           {Illustration}
4592 \copperdefstring{french}{platform}      {Plateforme}
4593 \copperdefstring{french}{directedby}    {R\'ealis\'e par}
4594 \copperdefstring{french}{producedby}    {Produit par}
4595 \copperdefstring{french}{listabbreviation}{Liste des abr\'eviations et acronymes}
4596 \copperdefstring{french}{listsymbol}    {Liste des symboles}
4597 \copperdefstring{french}{glossary}      {Glossaire}
4598 \copperdefstring{french}{advisor}       {Directeur}
4599 \copperdefstring{french}{advisors}      {Directeurs}
4600 \copperdefstring{french}{dept}          {Programme}
4601 \copperdefstring{french}{approvedmale}  {Approuv\'e par}
4602 \copperdefstring{french}{approvedfemale}{Approuv\'ee par}
4603 \copperdefstring{french}{approvedonmale}{Approuv\'e le}
4604 \copperdefstring{french}{approvedonfemale}{Approuv\'ee le}
4605 \copperdefstring{french}{keywordsname} {Mots-cl\'es}
4606 \copperdefstring{french}{volumename}   {Volume}
4607 \copperdefstring{french}{coadvisor}    {Co-directeur}
4608 \copperdefstring{french}{coadvisors}   {Co-directeurs}
4609 \copperdefstring{french}{ofname}       {de}
4610 \copperdefstring{french}{researchline}{Axe de recherche}
4611 \copperdefstring{french}{tbddate}      {\`a d\'eterminer}
4612 \copperdefstring{french}{universityname}{Universidade Federal do Rio de Janeiro}
4613 \copperdefstring{french}{cityname}     {Rio de Janeiro}
4614 \copperdefstring{french}{statename}    {RJ}
4615 \copperdefstring{french}{countryname}  {Brasil}
4616 \copperdefstring{french}{monthname1}   {janvier}
4617 \copperdefstring{french}{monthname2}   {f\'evrier}
4618 \copperdefstring{french}{monthname3}   {mars}
4619 \copperdefstring{french}{monthname4}   {avril}
4620 \copperdefstring{french}{monthname5}   {mai}
4621 \copperdefstring{french}{monthname6}   {juin}
4622 \copperdefstring{french}{monthname7}   {juillet}
4623 \copperdefstring{french}{monthname8}   {ao\~ut}
4624 \copperdefstring{french}{monthname9}   {septembre}
4625 \copperdefstring{french}{monthname10} {octobre}
4626 \copperdefstring{french}{monthname11} {novembre}
4627 \copperdefstring{french}{monthname12} {d\'ecembre}
4628 \copperdefstring{french}{algorithm2eopt}{french}
4629 \copperdefstring{french}{babelname}    {french}

```

```

4630 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
4631 \</langfr>
4632 \<langit>
4633 %% coppe-lang-italian.def -- CoppeTeX 4.x Italian class-string pack.
4634 %% See coppe-lang-spanish.def for the structure. The algorithm2e
4635 %% language option for Italian is spelled "italiano" -- that's the
4636 %% spelling algorithm2e ships, even though babel calls the language
4637 %% "italian".
4638 \ProvidesFile{coppe-lang-italian.def}[2026/09/09 v4.1 CoppeTeX italian class strings]
4639 \copperdefstring{italian}{appendix}      {Appendice}
4640 \copperdefstring{italian}{annex}         {Allegato}
4641 \copperdefstring{italian}{frame}         {Riquadro}
4642 \copperdefstring{italian}{listframe}     {Elenco dei riquadri}
4643 \copperdefstring{italian}{source}        {Fonte}
4644 \copperdefstring{italian}{program}       {Programma}
4645 \copperdefstring{italian}{listprogram}   {Elenco dei programmi}
4646 \copperdefstring{italian}{references}    {Riferimenti}
4647 \copperdefstring{italian}{in:}          {in}
4648 \copperdefstring{italian}{art}           {Illustrazione}
4649 \copperdefstring{italian}{platform}      {Piattaforma}
4650 \copperdefstring{italian}{directedby}    {Regia di}
4651 \copperdefstring{italian}{producedby}    {Prodotto da}
4652 \copperdefstring{italian}{listabbreviation}{Elenco di abbreviazioni e sigle}
4653 \copperdefstring{italian}{listsymbol}    {Elenco dei simboli}
4654 \copperdefstring{italian}{glossary}      {Glossario}
4655 \copperdefstring{italian}{advisor}       {Relatore}
4656 \copperdefstring{italian}{advisors}      {Relatori}
4657 \copperdefstring{italian}{dept}          {Programma}
4658 \copperdefstring{italian}{approvedmale}  {Approvato da}
4659 \copperdefstring{italian}{approvedfemale}{Approvata da}
4660 \copperdefstring{italian}{approvedonmale}{Approvato il}
4661 \copperdefstring{italian}{approvedonfemale}{Approvata il}
4662 \copperdefstring{italian}{keywordsname}  {Parole chiave}
4663 \copperdefstring{italian}{volumename}    {Volume}
4664 \copperdefstring{italian}{coadvisor}     {Correlatore}
4665 \copperdefstring{italian}{coadvisors}    {Correlatori}
4666 \copperdefstring{italian}{ofname}        {di}
4667 \copperdefstring{italian}{researchline}{Linea di ricerca}
4668 \copperdefstring{italian}{tbddate}       {da determinare}
4669 \copperdefstring{italian}{universityname}{Universidade Federal do Rio de Janeiro}
4670 \copperdefstring{italian}{cityname}      {Rio de Janeiro}
4671 \copperdefstring{italian}{statename}     {RJ}
4672 \copperdefstring{italian}{countryname}   {Brasil}
4673 \copperdefstring{italian}{monthname1}    {gennaio}
4674 \copperdefstring{italian}{monthname2}    {febbraio}
4675 \copperdefstring{italian}{monthname3}    {marzo}
4676 \copperdefstring{italian}{monthname4}    {aprile}
4677 \copperdefstring{italian}{monthname5}    {maggio}
4678 \copperdefstring{italian}{monthname6}    {giugno}
4679 \copperdefstring{italian}{monthname7}    {luglio}
4680 \copperdefstring{italian}{monthname8}    {agosto}
4681 \copperdefstring{italian}{monthname9}    {settembre}
4682 \copperdefstring{italian}{monthname10}{ottobre}
4683 \copperdefstring{italian}{monthname11}{novembre}

```

```

4684 \copperdefstring{italian}{monthname12}{dicembre}
4685 \copperdefstring{italian}{algorithm2eopt}{italiano}
4686 \copperdefstring{italian}{babelname} {italian}
4687 \AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
4688 </langit>
4689 <*numbbx>
4690 \ProvidesFile{coppe-numeric.bbx}[2026/09/09 v4.1 CoppeTeX ABNT numeric bibliography]
4691 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
4692 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
4693 % the PT->EN source map and the language mappings) and only adds the [n] list
4694 % label and the numeric sorting on top.
4695 \RequireBibliographyStyle{coppe}
4696
4697 % ABNT numeric system lists references in the order they are first cited
4698 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
4699 % numbered list. 'labelnumber' generates the [n] field.
4700 \ExecuteBibliographyOptions{labelnumber,sorting=none}
4701
4702 % Bracketed labels, both in the list and for any shorthand entries.
4703 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
4704 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
4705
4706 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
4707 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
4708 \defbibenvironment{bibliography}
4709 {
4710   \list
4711     {\printtext[labelnumberwidth]{%
4712       \printfield{labelprefix}%
4713       \printfield{labelnumber}}}
4714     {\setlength{\labelwidth}{\labelnumberwidth}%
4715       \setlength{\leftmargin}{\labelwidth}%
4716       \setlength{\labelsep}{\biblabelsep}%
4717       \addtolength{\leftmargin}{\labelsep}%
4718       \setlength{\itemsep}{\bibitemsep}%
4719       \setlength{\parsep}{\bibparsep}}%
4720     {\renewcommand*{\makelabel}[1]{\hss#1}}
4721   {\item}
4722
4723 </numbbx>
4724 <*numcbx>
4725 \ProvidesFile{coppe-numeric.cbx}[2026/09/09 v4.1 CoppeTeX ABNT numeric citations]
4726 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
4727 % numbers [1], compressed and sorted within a single cite group. This is the
4728 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
4729 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
4730 % \citet -> Author [1].
4731 \RequireCitationStyle{numeric-comp}
4732 </numcbx>

```

10.2 The bibliographic databases

The two BibLaTeX/biber databases shipped with the class — `example.bib` (the sample database cited by the example monograph) and `coppe.bib` (the references

of this manual) — are generated from this source too, so the whole distribution still has a single editable origin. They live inside an `\iffalse` block below: the documentation driver never typesets them, and only `coppe.ins` extracts them, through the `examplebib` and `coppebib` docstrip targets. Edit the entries here, never the generated `.bib` files.

10.3 The generated companion files

Everything the distribution ships is extracted from this file: the class, the BibLaTeX styles, the language packs, the databases, the `.ist`, the demonstration documents and the `latexmkrc`. One `pdflatex coppe.ins` rebuilds all of it and there is no second place to edit. Each one sits in an `\iffalse` block: the documentation driver never typesets them, and only `coppe.ins` extracts them.

The line is drawn at what is DISTRIBUTED. What exists only to prove the class works stays outside, as ordinary files under version control:

`tests/` the regression suite. Nobody installing CoppeTeX needs it, and generating it from here would put proof and product in the same box.

`adversativa/` twelve documents — four kinds of work by three languages — each firing everything the class offers at once.

`tools/*.ps1` the build harness itself.

`NORMA_COPPE_2026.tex`, `futuremanual2026.tex` documents about the norm that live in this repository with their own life cycle.

Both suites are run by one command, `tools\prova.ps1`, and that is what has to come out clean before a version is tagged.

10.3.1 `example_pt.tex` -- pt-main demo

`example_pt.tex` — CoppeTeX 4.0 multilingual demo, main language: brazilian. Short demonstration document (cover + folha + abstracts + 1 chapter) used by the CPGP submission package. The full pt-main example with bibliography and lists is the historical `example.tex`.

10.3.2 `example_en.tex` -- en-main demo

`example_en.tex` — CoppeTeX 4.0 multilingual demo, main language: english.

10.3.3 `example_es.tex` -- es-main demo

`example_es.tex` — CoppeTeX 4.0 multilingual demo, main language: spanish. Requires the language pack files `coppe-lang-spanish.def` and `spanish-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.4 `example_fr.tex` -- fr-main demo

`example_fr.tex` — CoppeTeX 4.0 multilingual demo, main language: french. Requires the language pack files `coppe-lang-french.def` and `french-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.5 `example_it.tex` -- it-main demo

`example_it.tex` – CoppeTeX 4.0 multilingual demo, main language: italian. Requires the language pack files `coppe-lang-italian.def` and `italian-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.6 `example_pdfa.tex` -- the full example as PDF/A

`example_pdfa.tex` – o `example.tex` completo, compilado com a opção ‘pdfa’.

Existe para dar ao veraPDF o documento mais exigente que a classe produz: o exemplo inteiro, com bibliografia, listas, figuras, tabelas, algoritmos e as caixas do `tcolorbox` – ou seja, transparência, que é justamente o que o a-1b proibiria e o a-2b admite. Validar só um documento curto não diria nada sobre o que vai de fato para o depósito.

`\PassOptionsToClass` tem de vir ANTES do `\documentclass` que o `example.tex` traz dentro de si; é o único jeito de acrescentar uma opção de classe a um arquivo gerado pelo docstrip sem duplicá-lo aqui e deixar as duas cópias divergirem com o tempo.

10.3.7 `covers_5languages.tex` -- the five covers side by side

`covers_5languages.tex` – One-page visual demonstration of the CoppeTeX 4.0 multilingual model: the five built-in main languages side by side, all sharing the same Portuguese institutional template, differing only in the title (and the body language, not visible on the cover).

This file uses the article class deliberately – it is a meta-document about the coppe class, not a thesis. It expects the five cover PNGs `cover_pt.png`, `cover_en.png`, `cover_es.png`, `cover_fr.png`, `cover_it.png` to live in the same directory (generated by `pdftoppm` from the `example_<lang>.pdf` demos).

10.3.8 `latexmkrc` -- latexmk/Overleaf build configuration

`latexmkrc` – CoppeTeX build configuration (works locally and on Overleaf).

Run with: `latexmk example` (or `latexmk futuremanual2026`, your thesis...) `latexmk` reads this file automatically when it is in the build directory. On Overleaf, place this file at the project root; Overleaf runs `latexmk`.

Engine: pdfLaTeX by default (fast). For LuaLaTeX (recommended, full Unicode) set `$pdf_mode = 4` below, or use the `"% !TeX program = lualatex"` magic comment / the Overleaf "Menu > Compiler" setting.

Acknowledgments

Thanks to all COPPETEX users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

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Numbers written in *italic* refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in roman refer to the code lines where the entry is used.

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\, 918, 927	\@coppedoublespacefalse	\@coppetwosidetru 1752
\; ... 837, 839, 841,	 1685	\@date 3122
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Change History

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General: Creation Date.	1	<code>\department</code> : Included a sort key in <code>syml</code>	60
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General: Documentation:		General: Fixed version problem between <code>cls</code> and <code>dtx</code>	1
bibliography fixed, title translation.	1	v3.3	
Sourceforge submission.	1	General: Extend <code>abrev</code> to work like <code>syml</code> and accept a sorting key .	1
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v0.5		v4.0	
<code>abstract</code> : Changed from macro to environment.	78	General: Catalog-label writers (<code>coppe@mainBegin</code> , <code>coppe@bibBegin</code> , <code>coppe@bibEnd</code> , <code>coppe@hasLof</code>) made language-agnostic: <code>detokenize</code> wrap on label names protects ':' and '.' from babel shorthand activation, and the raw <code>etex</code> primitives <code>romannumeral/number</code> replace LaTeX's <code>roman/arabic</code> to bypass <code>spanish-babel</code> 's <code>small-caps @roman</code> redefinition that would otherwise crash <code>pdftex</code> with "Unbalanced write command" on every <code>es-main</code> document.	1
<code>\backmatter</code> : Added <code>mainmatter</code> pages counter.	66	Class-wide string dispatcher (<code>copperdefstring</code> , <code>coppestring</code> , <code>coppemainstring</code> , <code>coppeforeignstring</code>) replaces scattered <code>iflanguage</code> and <code>if</code> english branches; <code>pt/en</code> outputs remain byte-identical. .	1
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v2.2.2			
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<code>\examiner</code> : Examiners expansion without degree.	62		

spanish, french, italian); plug-in mechanism for any other Babel language via coppe-lang-lang.def plus lang-coppe.lbx.	1		
New user API: titlein, usecoppelanguage, brazilianabstract environment. .	1		
<code>\coppetexfinalpage</code> : Initial coppetexfinalpage colophon page.	67		
v4.1			
General: A missing cover logo now raises a class error naming the file and saying it ships with the class, instead of TeX's bare "File not found". No graphicspath is set: kpathsea finds a logo installed beside the class and does not report where it found anything, so the class cannot deduce its own directory.	1		
Breaking change in how the board is declared: advisor, coadvisor and examiner now take the TREATMENT as the optional argument, empty by default, and the INSTITUTION as the last mandatory one, which may be left empty. 3.1.2.1.3(e) asks for name, titulacao and institution and asks for no treatment, so that is what the folha de aprovacao prints.	1		
Conformance with the revised 9th edition (2026) of the UFRJ/SiBI manual: continuous pagination from the folha de rosto, folio in 10pt at 2cm from the top and right edges, one-sided layout with a 3cm left margin, additional sheet with the catalog card and the CAPES collection fields, approval sheet of 3.1.2.1.3, keywords closing the three abstracts, pre-textual lists out of the sumario, Apendice and Anexo headings centred, fifth heading level, and Latin Modern instead of the bitmap fonts.	1		
		Fixes found by the adversarial review: cover and folha de rosto compose in single spacing so they do not overflow under doublespacing, no folio is printed on a pre-textual sheet when something does overflow, the approval sheet holds a board of eight, and es-nosectiondot removes the period babel-spanish put in the sumario indicative.	1
		New option semlinks: hidelinks over everything else, so the links keep working and lose only the colour and the frame. For anyone printing.	1
		New options: pdfa (PDF/A-2b via pdfx, XMP built from the document), assinaturas, coorientador, rascunhoficha, listasnosumario, morewrites/semmorewrites. New command coadvisor. Under pdfTeX the class loads morewrites by itself when installed, because a work that asks for every list needs seventeen of TeX's sixteen output streams.	1
		newcoppesfloat fixed twice: newfloat was wrapped in a group, and since newfloat defines the environment locally the new float vanished when the group closed; and its list came out unnumbered and without leaders because l@figure was not inherited. Author-declared floats are now numbered within the chapter like every other illustration. . . .	1
		No signature rules on the folha de aprovacao: the deposit has been digital only since Resolucao CEPG n. 246/2023, so there is nothing to sign by hand. The assinaturas option is still accepted and only warns. .	1
		Single source: pdfatex coppe.ins generates everything distributed – class, biblatex styles, language packs, .bib bases, .ist, the five	

per-language examples, example_pdfa, the cover montage and latexmkrc. The regression suite, the adversarial documents and the harness prove the class and are NOT distributed.	1
The class sets the document in the sans-serif family by DEFAULT; comserifa restores the serif one. Neither the manual nor the COPPE norm prescribes a family, and the SiBI manual that carries the rule is itself set in Arial. semserifa is still accepted and now does nothing.	1
The folha adicional draws the Coleta CAPES fields as the closed frame of Annex H – one cell per field, a rule between cells, and no rules to write on – instead of loose paragraphs with dotted fill rules.	1
The folha de aprovacao lists only the EXAMINERS. The sheet records who examined the work, and the advisor conducted it; the new orientadorexamina option puts advisors and coadvisors back, ahead of the examiners, for the Programas that seat the advisor on the board. Before	
this there was no choice, and the only way out was not to declare the advisor at all, which also removed them from the capa and the folha de rosto.	1
The lists of abbreviations and of symbols no longer carry a page number, nor the dot leaders that led to it. 4.1.1 describes both as an alphabetical list of the term and its meaning and asks for no location; the folio was inherited from building them with the index machinery. makeindex always writes the number, so the class discards it at composition through the coppe@semfolio encapsulator, and every delimiter in coppe.ist is now empty.	1
The research-line label on the folha de rosto follows the main language of the work instead of being hard-coded in Portuguese.	1
With no dataaprovacao given, the folha de aprovacao writes the date as "a ser determinada" in the main language instead of drawing a rule to fill in by hand: the deposit is digital and nobody writes on a PDF.	1